

ECON 3818: Introduction to Statistics with Computer Application

Sungyoun Chung



Part I

Exploring Data

- Variables and their distributions (Ch1,2 and 3)
- Relationships among variables (Ch 4, 5 and 6)

Ch1. Picturing Distributions with Graphs

Statistics?

- Statistics
 - the science of learning from data
- Data
 - Data are numbers with a *context*. And the context makes the number informative
 - **Example.** A new baby weighed 8.5 pounds at birth.

Individuals and Variables

- Individuals
 - the objects described by a set of data. Individuals may be people, but they may also be animals or things
- Variable
 - any characteristic of an individual. A variable can take different values for different individuals
- Exploring Data
 - (1) Who?: what individuals (observations) do the data describe?
 - (2) What?: what are the *exact definitions* of these variables? In what *unit of measurement* is each variable recorded?
 - (3) Where?: where were the data collected (location, setting, or population)?
 - (4) When?: Observations change from year to year
 - (5) Why?: what purpose do the data have? Do we want answers for just these individuals or for some larger group that these individuals are supposed to represent?

- Categorical and Quantitative Variables
 1. **Categorical Variable:** places an individual into one of several groups or categories
 - e.g., education level, gender, marital status, zip code, type of residence, ...
 2. **Quantitative Variable:** takes numerical values for which arithmetic operations such as adding and averaging make sense
 - e.g., height, weight, age, income, distance, test score, ...

Figure 1: ACS PUMS Data

	A	B	C	D	E	F	G	H	I	J	K
1	RT	SERIALNO	DIVISION	PUMA	REGION	STATE	ADJHSG	ADJINC	WGTP	NP	TYPEHUGQ
2	H	2024GQ0000080	9	400	4	2	1000000	1015250	0	1	3
3	H	2024GQ0000715	9	300	4	2	1000000	1015250	0	1	3
4	H	2024GQ0000753	9	101	4	2	1000000	1015250	0	1	3
5	H	2024GQ0001209	9	300	4	2	1000000	1015250	0	1	3
6	H	2024GQ0001307	9	400	4	2	1000000	1015250	0	1	3
7	H	2024GQ0001380	9	200	4	2	1000000	1015250	0	1	3
8	H	2024GQ0001482	9	200	4	2	1000000	1015250	0	1	2
9	H	2024GQ0002208	9	300	4	2	1000000	1015250	0	1	3
10	H	2024GQ0002244	9	300	4	2	1000000	1015250	0	1	3
11	H	2024GQ0002486	9	400	4	2	1000000	1015250	0	1	3
12	H	2024GQ0003127	9	300	4	2	1000000	1015250	0	1	3
13	H	2024GQ0003186	9	400	4	2	1000000	1015250	0	1	2
14	H	2024GQ0003412	9	300	4	2	1000000	1015250	0	1	2
15	H	2024GQ0003451	9	400	4	2	1000000	1015250	0	1	3
16	H	2024GQ0003843	9	400	4	2	1000000	1015250	0	1	3
17	H	2024GQ0003889	9	200	4	2	1000000	1015250	0	1	3
18	H	2024GQ0003959	9	300	4	2	1000000	1015250	0	1	3
19	H	2024GQ0004105	9	400	4	2	1000000	1015250	0	1	2

- RT: Record type
- SERIALNO: Household serial number
- DIVISION: Census division
- PUMA: Public Use Microdata Area
- REGION: Census region
- STATE: FIPS code

Practice Question 1-1

- (1) What are the individuals (observations) in this data set?
- (2) Which of these variables are categorical, and which are quantitative?

Table 1: Vehicle characteristics and fuel economy variables

Make and Model	Vehicle Class	Transmission Type	Number of Cylinders	City mpg	Highway mpg	Annual Fuel Cost
Subaru Impreza	Midsize	Manual	4	24	32	\$1,450
Nissan Rogue Sport	Small	Automatic	4	25	32	\$1,400
Hyundai Elantra	Midsize	Automatic	4	28	37	\$1,200
Chevrolet Impala	Large	Automatic	6	19	28	\$1,750

Categorical Variables: Pie Charts and Bar Graphs

- Exploratory Data Analysis
 - Use of graphs and numerical summaries to describe the variables in a data set and the relationships among them
- Two principles for organizing the exploration of a set of data
 1. Begin by examining each variable by itself. Then move on to study the **relationships** among the variables
 2. Begin with a **graph** or graphs. Then add **numerical summaries** of specific aspects of the data
- Distribution of a variable (single variable)
 - The distribution of a variable tells us what values it takes and how often it takes these values
 - The distribution of a categorical variable lists the categories and gives either the count or the percentage of individuals who fall in each category
 - Pie chart or a bar graph to display the distribution of a categorical variable more vividly
 - (1) Pie Charts: show the distribution of a categorical variable as a “pie” whose slices are sized by the counts or percentages for the categories (Figure 2)
 - (2) Bar Graph: shows the distribution of a categorical variable by representing each category as a bar, with the height of each bar representing the count or percentage for the corresponding category (Figure 3)

Figure 2: Pie Chart

Share of Oscar 2024 Nominations

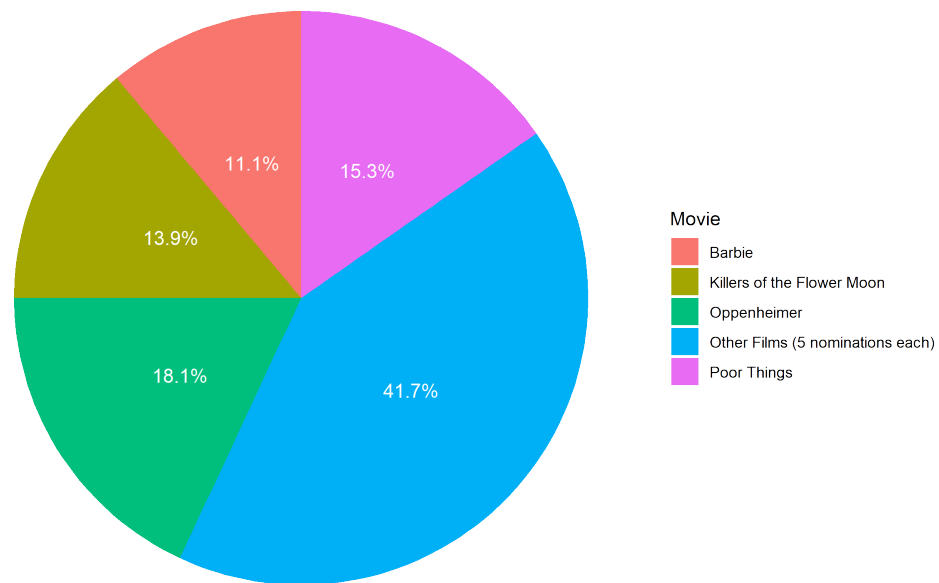
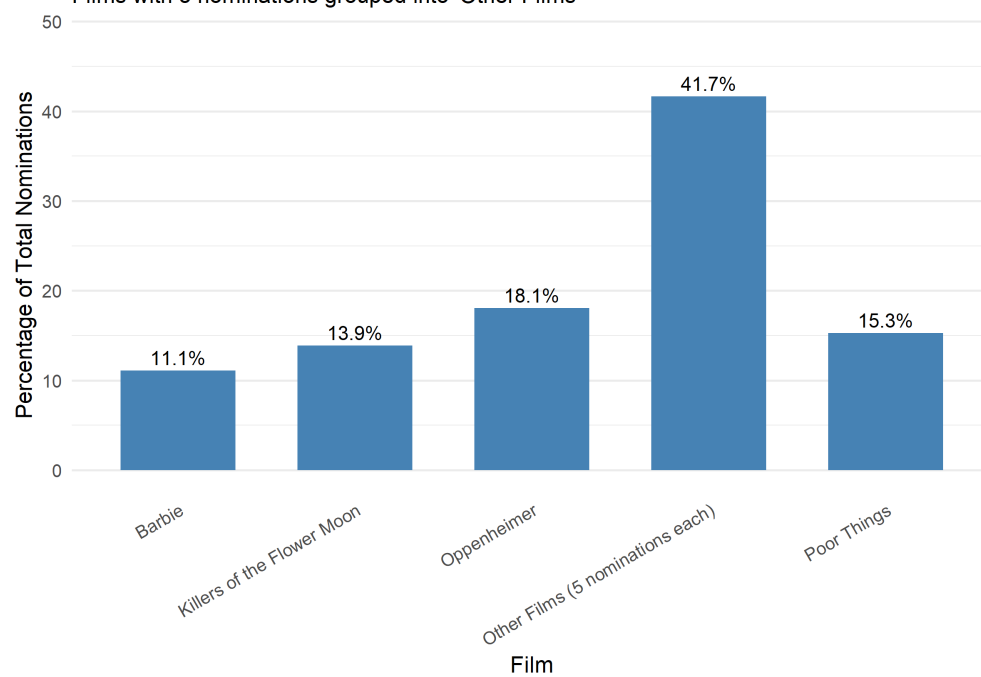


Figure 3: Bar Graph

Share of Oscar 2024 Nominations by Film
Films with 5 nominations grouped into 'Other Films'



Quantitative Variables: Histograms

- Histogram
 - A graph of the distribution of one **quantitative** variable. Bars on the graph each represent a class, with the base of each bar covering the class on the horizontal axis.
 - The **horizontal** axis is marked in the units of measurement for the variable. The range of the data is divided into classes of equal width
 - The **vertical** axis displays the scale of counts, and each bar's height is the class count.

Figure 4: Adjusted Cohort Graduation Rate (ACGR)

state	2010-2011	2021-2022
ALABAMA	0.720	0.882
ALASKA	0.680	0.778
ARIZONA	0.779	0.773
ARKANSAS	0.807	0.882
CALIFORNIA	0.763	0.870
COLORADO	0.739	0.823
CONNECTICUT	0.830	0.889
DELAWARE	0.785	0.878
DISTRICT OF COLUMBIA	0.586	0.764
FLORIDA	0.706	0.873
GEORGIA	0.675	0.841
HAWAII	0.800	0.860
IDAHO	0.773	0.799
ILLINOIS	0.838	0.873
INDIANA	0.857	0.877
IOWA	0.883	0.899
KANSAS	0.830	0.891
KENTUCKY	0.861	0.901

state	2010-2011	2021-2022
LOUISIANA	0.7090	0.831
MAINE	0.8379	0.861
MARYLAND	0.8282	0.863
MASSACHUSETTS	0.8340	0.901
MICHIGAN	0.7433	0.810
MINNESOTA	0.7686	0.836
MISSISSIPPI	0.7370	0.889
MISSOURI	0.8125	0.898
MONTANA	0.8220	0.858
NEBRASKA	0.8600	0.871
NEVADA	0.6200	0.817
NEW HAMPSHIRE	0.8609	0.877
NEW JERSEY	0.8317	0.852
NEW MEXICO	0.6300	0.766
NEW YORK	0.7680	0.867
NORTH CAROLINA	0.7790	0.864
NORTH DAKOTA	0.8625	0.851
OHIO	0.8000	0.862

state	2010-2011	2021-2022
OKLAHOMA	0.8480	0.801
OREGON	0.6765	0.813
PENNSYLVANIA	0.8260	0.870
RHODE ISLAND	0.7730	0.833
SOUTH CAROLINA	0.7360	0.838
SOUTH DAKOTA	0.8339	0.821
TENNESSEE	0.8550	0.904
TEXAS	0.8590	0.897
UTAH	0.7600	0.882
VERMONT	0.8746	0.828
VIRGINIA	0.8200	0.891
WASHINGTON	0.7660	0.836
WEST VIRGINIA	0.7650	0.912
WISCONSIN	0.8700	0.903
WYOMING	0.7970	0.818
United States	0.7900	0.866

Figure 5: 2010-11

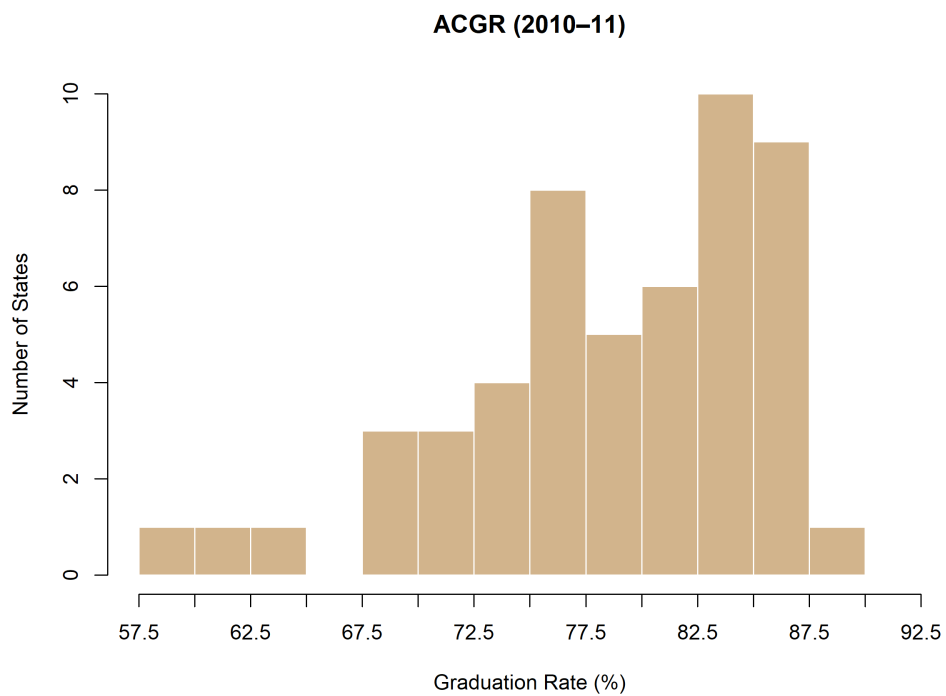
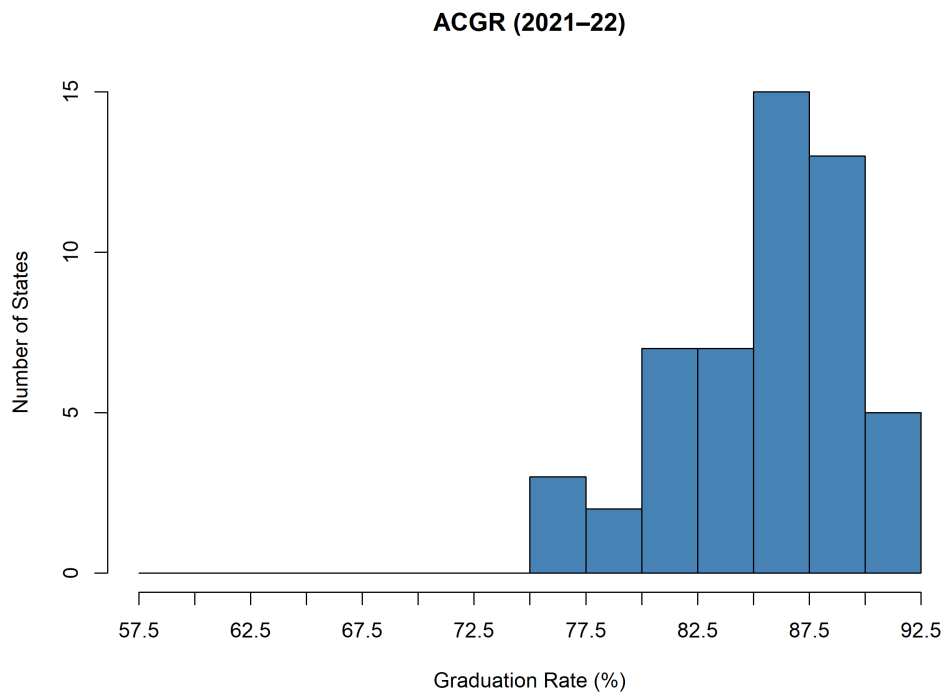


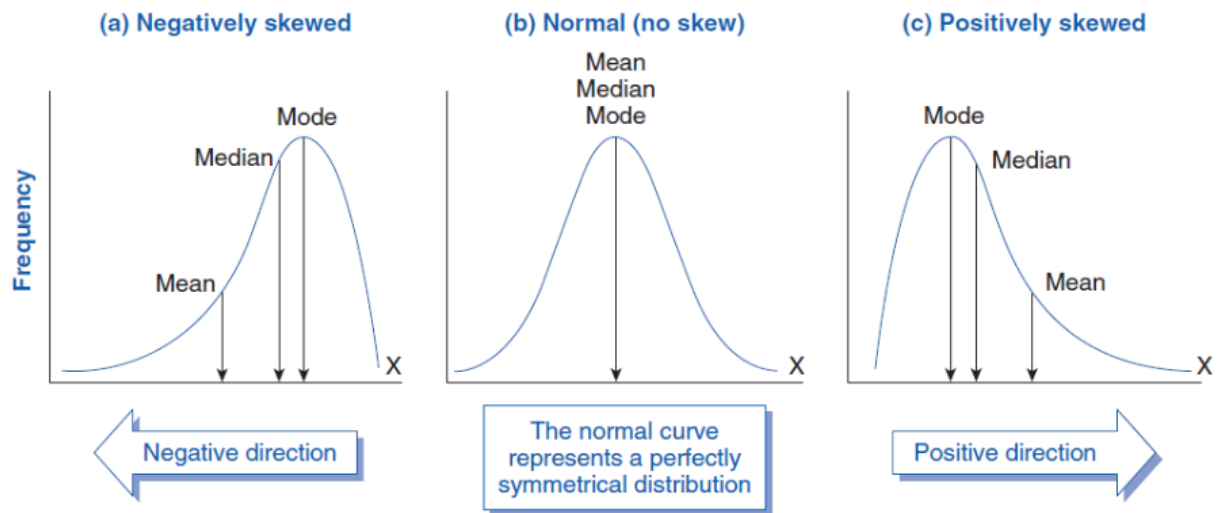
Figure 6: 2021-22



Interpreting Histograms

- The purpose of graphs is to help us understand how the data help us answer some real-world question.
- Examining a Histogram
 - Look for the overall pattern and for striking deviations from that pattern
 - You can describe the overall pattern of a histogram by its *shape*, *center*, and *variability*
 - *Outlier* is an individual (observation) value that falls outside the overall pattern
- Symmetric and Skewed Distributions
 - **Symmetric**: if the right and left sides of the histogram are approximately mirror images of each other
 - **Skewed to the right (positive)**: if the right side of the histogram extends much farther out than the left side (long right tail)
 - **Skewed to the left (negative)**: if the left side of the histogram extends much farther out than the right side (long left tail)

Figure 7: Distributions of Histograms



Quantitative Variables: Stemplots

- Stemplot
 - A graph of the distribution of a quantitative variable. Each observation is shown as a stem consisting of all but the final digit, and a leaf, the final digit
 - However, stemplots do not work well for large data sets, where each stem must hold a large number of leaves.

Example

Consider the following data on graduation rates (in percent):

72, 74, 75, 77, 78, 80, 81, 83, 85

A stem-and-leaf plot separates each observation into a *stem* (the leading digit) and a *leaf* (the trailing digit):

Stem		Leaf
7		2 4 5 7 8
8		0 1 3 5

Time Plots

- Time Plot
 - A time plot of a variable plots each observation against the time at which it was measured.
 - Put time on the horizontal axis of your plot and the variable you are measuring on the vertical axis.
 - Connecting the data points by lines helps emphasize any change over time
- Time Series Data
 - Measurement of a variable taken repeatedly, recording the time as well as the value of each measurement
 - For example, monthly unemployment rate, daily stock prices, annual GDP, weekly COVID-19 cases, ...
- Patterns in a time plot
 - (1) **Trend**: a long-term upward or downward movement in a time series over time (Figure 8)
 - (2) **Cycle**: regular up-and-down movements in time series data (Figure 9)

Figure 8: Time Plot of Unemployment Rate

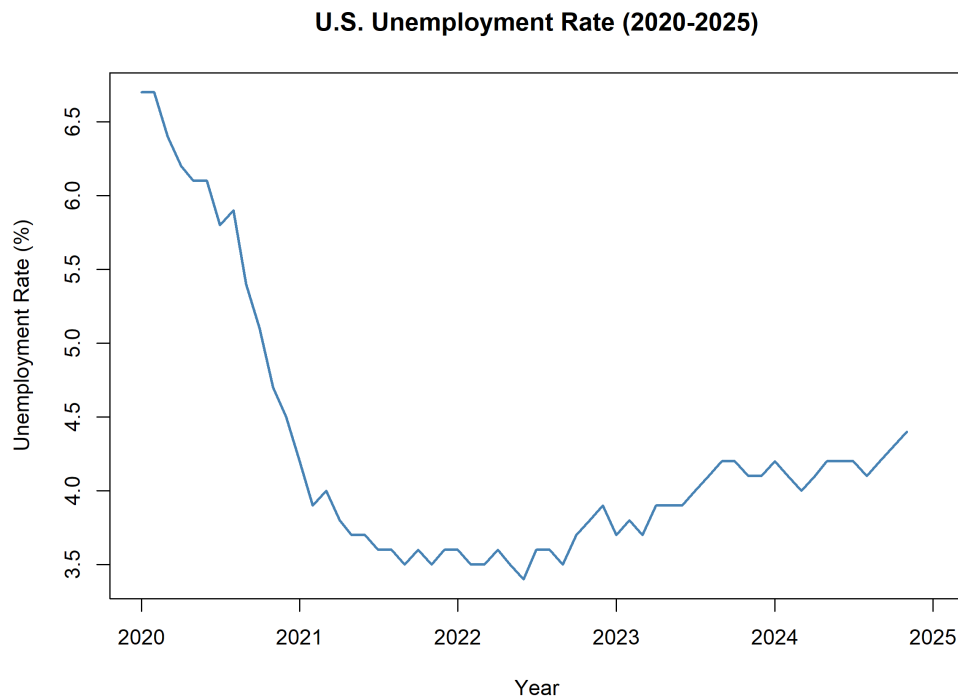
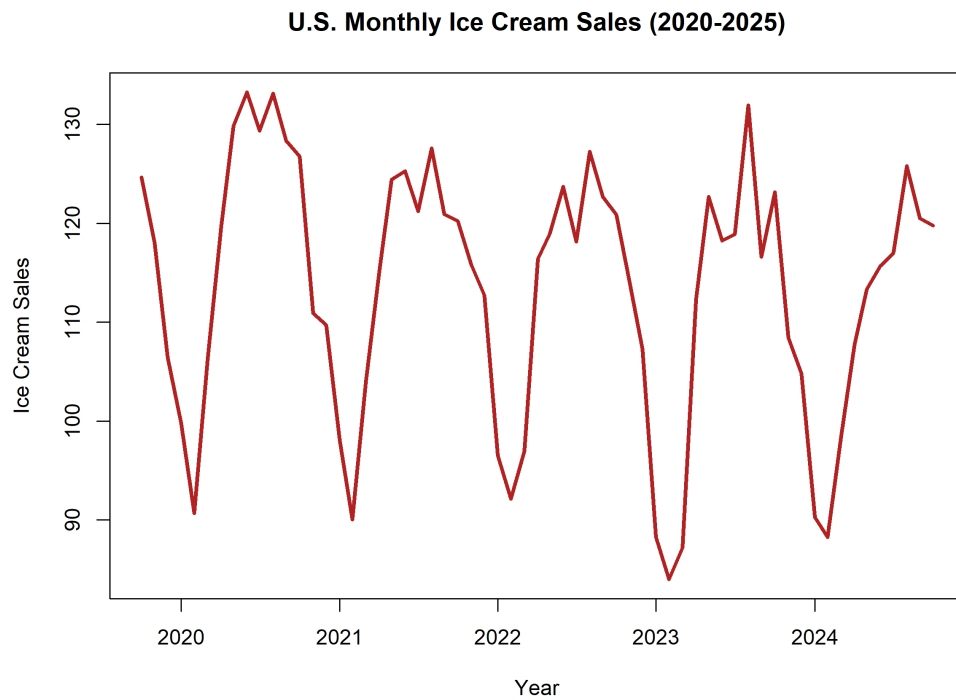


Figure 9: Time Plot of Ice Cream Sales



CH 2. Describing Distributions with Numbers

- The overall pattern in a histogram illustrates some of the important features of the distribution of a variable
 - The center of the histogram tells us about the value of a “typical” observation on this variable
 - Whereas the variability gives us a sense of how close most of the observations are to this value

Measuring Center: The Mean

- The Mean $\bar{x}$
 - To most common measure of center is the ordinary arithmetic average, or *mean*
 - To find the mean of a set of observations, add their values and divide by the number of observations. If the n observations are $x_1, x_2, \dots, x_n$, their mean is

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \cdots + x_n}{n}$$

or, in more compact notation,

$$\bar{x} = \frac{1}{n} \sum x_i$$

- Resistant Measure
 - A resistant measure is a statistical measure that is relatively unaffected by large changes in numerical values of a small proportion of the observations in the distribution that the measure describes
 - Because the *mean* cannot resist the influence of extreme observations (*outliers*), we say that it is ***not a resistant measure of center***

Practice Question 2-1

- Question: The following data represent the number of hours a student studied over 7 days:

2, 3, 4, 5, 6, 4, 6

Find the mean number of hours studied per day.

- Solutions: There are $n = 7$ observations. The mean is

$$\bar{x} = \frac{30}{7} \approx 4.29$$

Thus, the student studied an average of approximately 4.29 hours per day.

- Question: What if a student who studied 0.5 hours or a student who studied 20 hours were added to this data set? How would each new value affect the mean?

- Solutions:

- * **Adding a student who studied 0.5 hours:**

The new total is

$$30 + 0.5 = 30.5$$

and the new number of observations is $n = 8$.

The new mean is

$$\bar{x} = \frac{30.5}{8} = 3.8125$$

- * **Adding a student who studied 20 hours:**

The new total is

$$30 + 20 = 50$$

and the new number of observations is $n = 8$.

The new mean is

$$\bar{x} = \frac{50}{8} = 6.25$$

Measuring Center: The Median

- Median
 - The **median** M is the midpoint of a distribution, the number such that half the observations are smaller and the other half are larger. To find the median of a distribution
 - (1) Arrange all observations in order of size, from smallest to largest
 - (2) If the number of observations n is odd, the median M is the center observation in the ordered list.
 - (3) If the number of observations n is even, the median M is midway between the two center observations in the ordered list
 - (4) You can always locate the median in the ordered list of observations by counting up $(n + 1)/2$ observations from the start of the list

Practice Question 2-2

- Question: The following data represent the number of hours a student studied over 7 days:

2, 3, 4, 5, 6, 4, 6

- Find the median number of hours studied per day.
- What happens to the median if a student who studied 0.5 hours is added?
- What happens to the median if a student who studied 20 hours is added?

Comparing the Mean and the Median

- The mean ($\bar{x}$) and median (M) of a roughly symmetric distribution are close together
- If the distribution is exactly symmetric, the mean and median are exactly the same.
- In a skewed distribution, the mean is usually farther out in the long tail than the median
- Don't confuse the "average" value of a variable (the mean) with its "middle" value, which we might describe by the median, or with its "typical" value, which we might describe by the *mode*
 - Mode: the mode of a set of values is the most frequently occurring value

Measuring Variability: The Quartiles

- The simplest useful numerical description of a distribution requires both a measure of *center* and a measure of *variability*

Example

The U.S. Census Bureau reports that in 2017 the median income of American households was \$61,372. The mean was much higher, \$86,220, because the distribution of incomes is skewed to the right. The bottom 20% of households had income less than \$24,638, and households in the top 5% took in more than \$237,034.

- The Quartiles
 - One way to measure the variability is to give the smallest and largest observations and look at the variability of the middle half of the data.
 - (1) First quartile (Q_1): 25th percentile
 - (2) Second quartile (Q_2): 50th percentile (the median)
 - (3) Third quartile (Q_3): 75th percentile
 - When the number of observations is even, include all the observations when you locate the quartiles

Practice Question 2-3

- Question: Consider the following data sets, each arranged in ascending order.

- **Odd number of observations**

3, 5, 7, 9, 11, 13, 15

- Find the first quartile Q_1 .
- Find the second quartile Q_2 (the median).
- Find the third quartile Q_3

- **Even number of observations**

4, 6, 8, 10, 12, 14

- Find the first quartile Q_1 .
- Find the second quartile Q_2 (the median).
- Find the third quartile Q_3

The Five-Number Summary and Boxplots

- The Five-Number Summary
 - The five-number summary of a distribution consists of the smallest observation, the first quartile (Q_1), the median (M), the third quartile (Q_3), and the largest observation, written in order from smallest to largest.
 - Minimum, Q_1 , M , Q_3 , Maximum

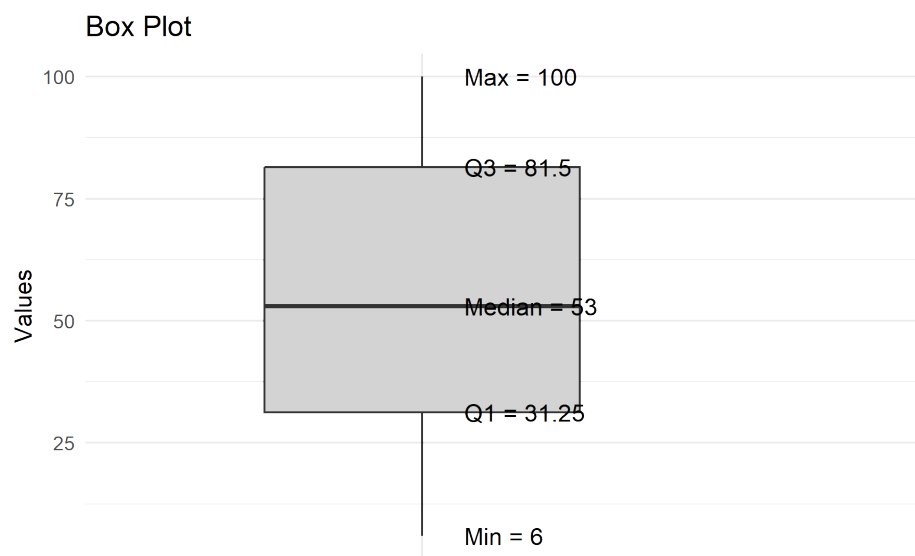
Example: Summary of sample in R

Figure 10: Five-number summary

```
> # Summary statistics
>
> set.seed(123)
> sample_1 <- sample(0:100, size = 30, replace = TRUE)
> sample_1
[1] 30 78 50 13 66 41 49 42 100 13 24 89 90 68 90 56 91 8 92 98 71 25 6 41 8 82 35 77 80 42
>
> summary(sample_1)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   6.00  31.25   53.00   55.17   81.50   100.00
```

- Boxplot
 - A boxplot is a graph of the five-number summary
 - A central box spans the quartiles Q_1 and Q_3
 - A line in the box marks the median M
 - Lines extend from the box out to the smallest and largest observations

Figure 11: Boxplot



Spotting Suspected Outliers and Modified Boxplots

- In data, the smallest and largest observations can be extremes that don't describe the variability of the majority of the data.
- The distance between the quartiles (the range of the center half of the data) is a more resistant measure of variability.
- **The Interquartile Range (IQR)**
 - The interquartile range (IQR) is the distance between the first and third quartiles,

$$IQR = Q_3 - Q_1$$

- The $1.5 \times IQR$ Rule for Outliers
 - Call an observation a suspected outlier if it falls more than $1.5 \times IQR$ below the first quartile or above the third quartile
- In a modified boxplot, the suspected outliers are identified in the boxplot with a special plotting symbol such as a dot (•)

Example

- In the boxplot figures below, which sample is left-skewed and which sample is right-skewed?

Figure 12: Boxplots

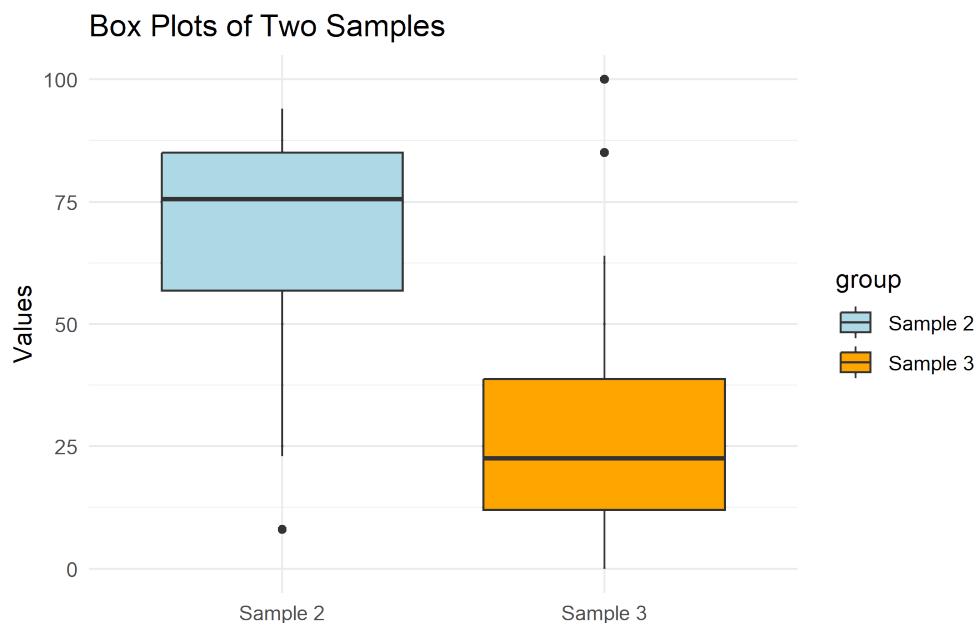


Figure 13: Histogram of Sample 2

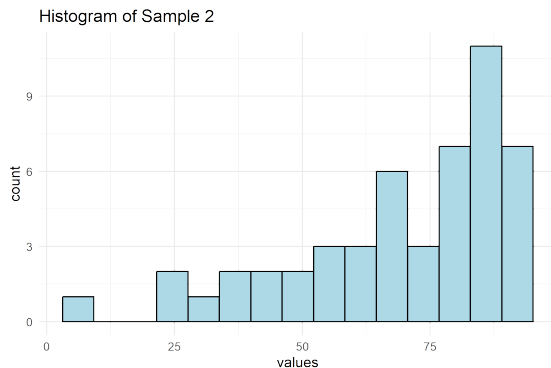
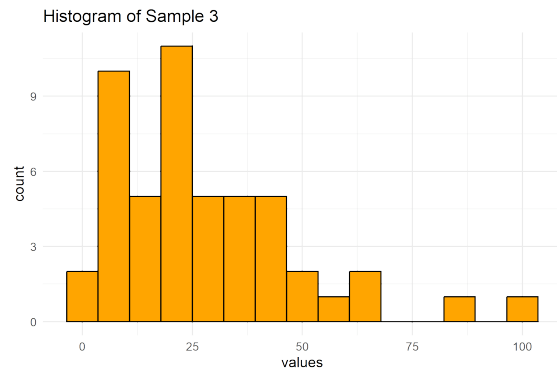


Figure 14: Histogram of Sample 3



Measuring Variability: The Standard Deviation

- The *standard deviation* and its close relative the *variance* measure variability by looking at how far the observations are from their mean
- The Variance s^2 and the Standard Deviation s
 - (1) **Variance:** The *variance* s^2 of a set of observations is an average of the squares of the deviations of the observations from their mean. In symbols, the variance of n observations $x_1, x_2, x_3, \dots, x_n$ is

$$s^2 = \frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \dots + (x_n - \bar{x})^2}{n - 1}$$

or, more compactly,

$$s^2 = \frac{1}{n - 1} \sum (x_i - \bar{x})^2$$

- (2) **Standard Deviation:** is the square root of the variance s^2

$$s = \sqrt{\frac{1}{n - 1} \sum (x_i - \bar{x})^2}$$

- Notice that the “average” in the variance s^2 divides the sum by one fewer than the number of observations $n - 1$ rather than n
 - The reason is that the deviations $x_i - \bar{x}$ always sum to exactly 0 so that knowing $n - 1$ of them determines the last one
 - Only $n - 1$ of the squared deviations can vary freely, and we average by dividing the total by $n - 1$
 - The number $n - 1$ is called the ***degrees of freedom*** of the variance or standard deviation

Example.

Consider the following sample of $n = 10$ observations:

$$x = \{2, 4, 6, 8, 10, 12, 14, 16, 18, 20\}$$

The sample mean is

$$\bar{x} = \frac{2 + 4 + 6 + 8 + 10 + 12 + 14 + 16 + 18 + 20}{10} = \frac{110}{10} = 11$$

Deviations from the Mean

The deviations of each observation from the mean are:

x_i	$x_i - \bar{x}$
2	-9
4	-7
6	-5
8	-3
10	-1
12	1
14	3
16	5
18	7
20	9

Observe that the sum of the deviations is

$$\sum_{i=1}^{10} (x_i - \bar{x}) = 0$$

Explanation of Degrees of Freedom

Once the mean $\bar{x}$ is fixed, only $n - 1 = 9$ of the deviations $x_i - \bar{x}$ can vary freely. The final deviation is completely determined by the requirement that the deviations sum to zero.

Therefore, the variance divides by $n - 1$ rather than n :

$$s^2 = \frac{1}{n - 1} \sum_{i=1}^{10} (x_i - \bar{x})^2$$

The number $n - 1 = 9$ is called the **degrees of freedom** of the sample variance.

- Properties of Standard Deviation s

- (1) s measures variability about the mean ($\bar{x}$) and should be used only when the mean is chosen as the measure of center
- (2) s is always zero or greater than zero ($s \geq 0$).
 - $s = 0$: only when there is no variability. This happens only when all observations have the same value
 - $s > 0$: As the observations become more variable about their mean, s gets larger
- (3) s has the same units of measurement as the original observations
 - For example, if you measure weight in pounds, both the mean ($\bar{x}$) and standard deviation (s) are in pounds.
- (4) Like the mean ($\bar{x}$), s is not resistant. A few outliers can make s very large.

Figure 15: Histogram of small standard deviation

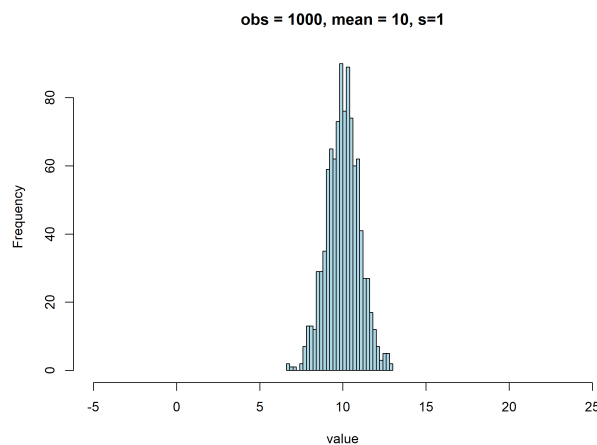
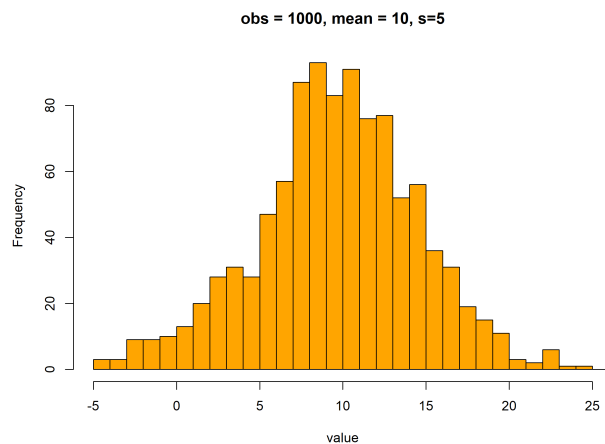


Figure 16: Histogram of large standard deviation



Practice Question 2-4

The following two samples each contain $n = 5$ observations.

1. $\{9, 10, 10, 10, 11\}$

2. $\{2, 6, 10, 14, 18\}$

- (a) Compute the sample mean $\bar{x}$ for each sample.
- (b) Compute the sample standard deviation s for each sample.
- (c) Compare the two standard deviations. Which sample shows greater variability?

Choosing Measures of Center and Variability

- We now have a choice between two descriptions of the center and variability of a distribution: (1) the five-number summary or (2) $\bar{x}$ and s
 - The five-number summary is usually better than the mean and standard deviation for describing a skewed distribution or a distribution with strong outliers
 - Use $\bar{x}$ and s only for reasonably symmetric distributions that are free of outliers
 - * This is because outliers can greatly affect the values of the mean $\bar{x}$ and standard deviation s

CH 3. The Normal Distributions

Density Curves

- Density Curve
 - Is always on or above the horizontal axis
 - **Has area exactly 1 underneath it**
 - A density curve describes the overall pattern of distribution. The area under the curve and above any range of values is the **proportion** of all observations that fall in that range
 - The areas under the density curve give quite **good approximations** to the actual distribution of the data

Example

Figure 17-Figure 19 are the histograms and density curves of the scores of 1,000 students in a math exam (0–20). The smooth curve (*density curve*) drawn closely approximates the tops of the histogram bars and provides a good description of the overall pattern of the data

- The proportion of scores less than or equal to 9.5 in the actual data is 0.418
- The proportion of scores less than or equal to 9.5 from the density curve is 0.427

Figure 17

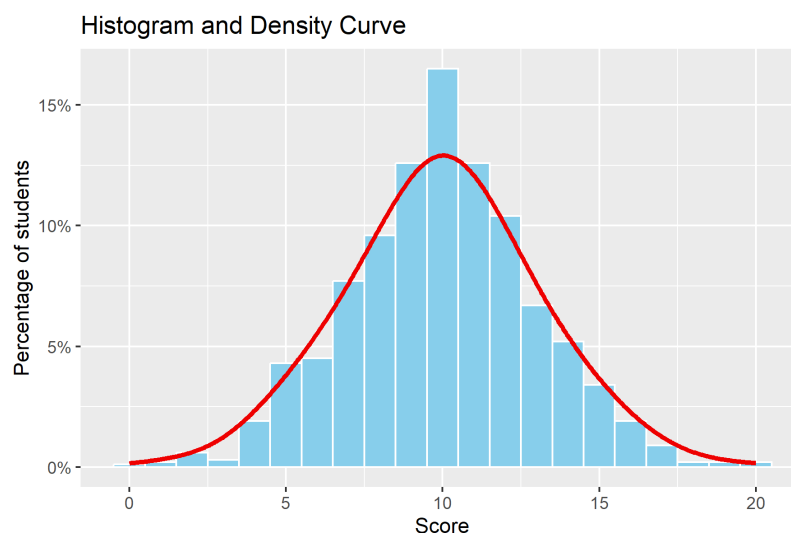


Figure 18

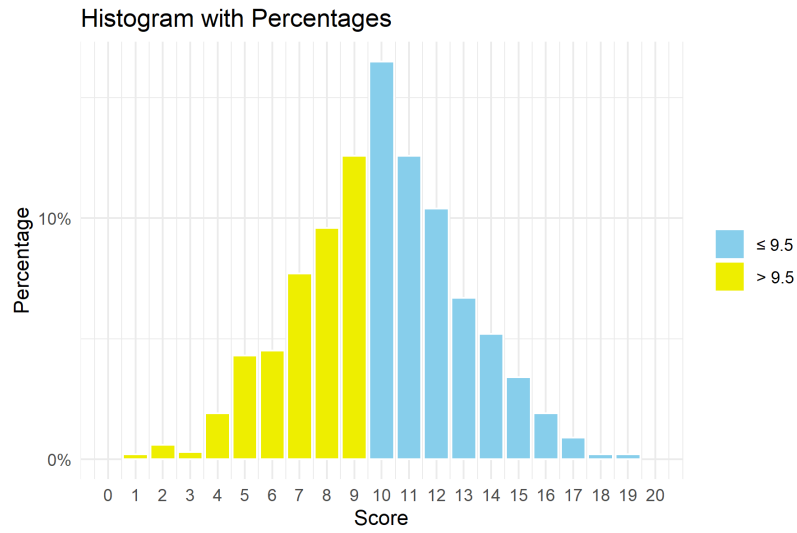
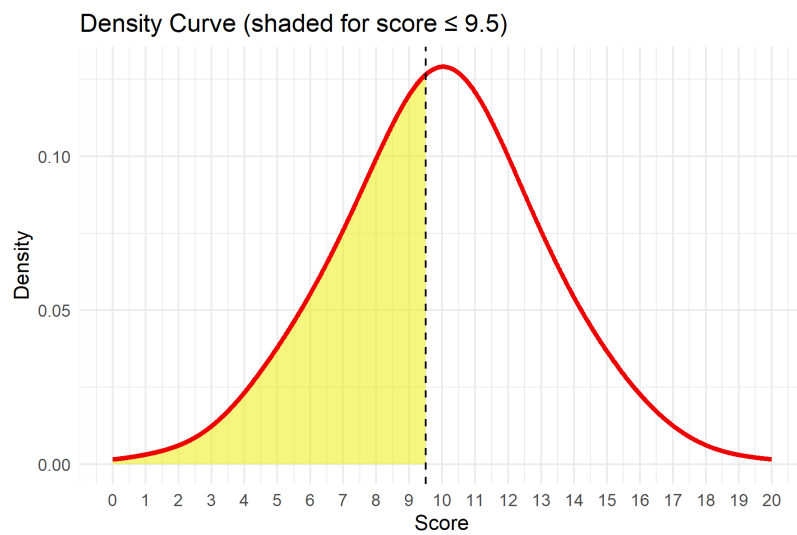


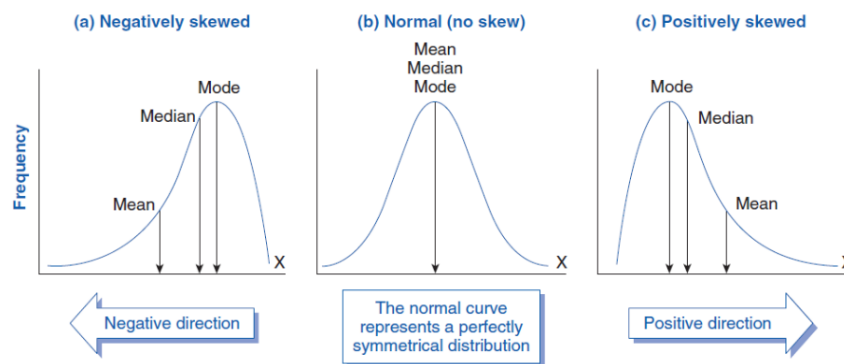
Figure 19



Describing Density Curves

- The measures of center and variability apply to density curves as well as to actual sets of observations
 - (1) Median: the point with half the observations on either side. So the median of a density curve is the *equal-areas point*
 - (2) Quartiles: divide the area under the curve into quarters
 - For example, one-fourth of the area under the density curve is to the left of the Q_1
 - (3) Mean (μ): the balance point at which the curve would balance if made of solid material. The usual notation for the mean of a density curve is μ
 - (4) Standard Deviation (σ): we use the standard deviation of a density curve as a σ

Figure 20: Density Curves



Normal Distributions: $N(\mu, \sigma)$

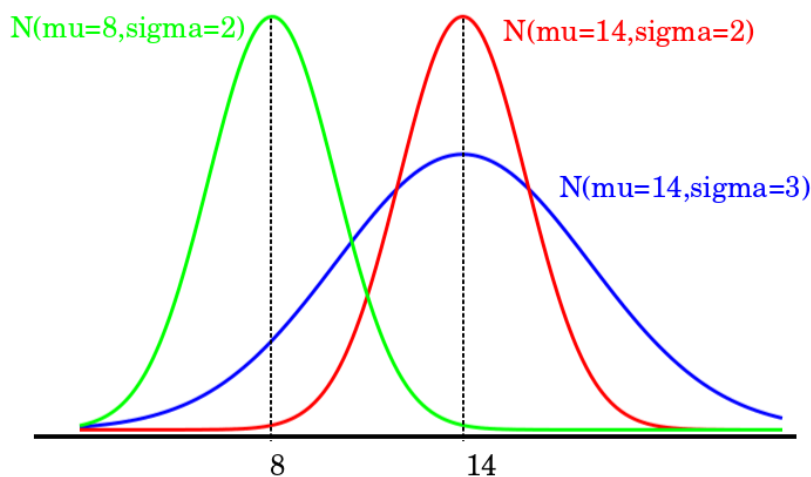
- Normal Curves

- All normal curves have the same overall shape: **symmetric, single-peaked, bell-shaped**
- Any specific Normal curve is completely described by giving its mean μ and its standard deviation σ
- The mean is located at the center of the symmetric curve and is the same as the median
- The standard deviation σ controls the variability of a Normal curve
 - * If σ is large, the area under the normal curve is less concentrated about the mean (dispersed)
 - * If σ is small, the area under the normal curve is more concentrated about the mean (concentrated)

- Normal Distributions

- A normal distribution is described by a Normal density curve
- Any particular normal distribution is completely specified by two numbers: (1) mean μ and (2) standard deviation σ . We abbreviate the Normal distribution with mean μ and standard deviation σ as $N(\mu, \sigma)$
- The mean of a normal distribution is at the **center of the symmetric normal curve**.
- The standard deviation is the **distance from the center** to the change-of-curvature points on either side

Figure 21: Normal Curves

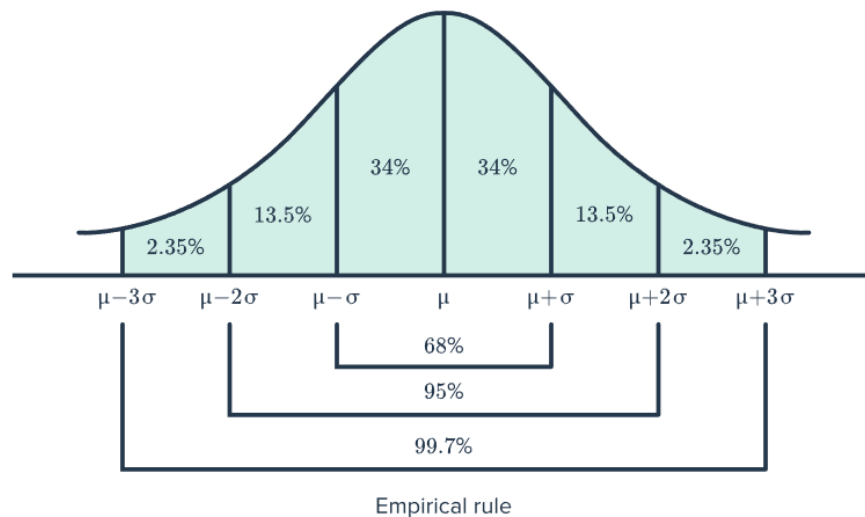


- Why are the Normal distributions important in statistics?
 - (1) Normal distributions are good descriptions for some distributions of *real data*
 - (2) Normal distributions are good approximations to the results of many kinds of *chance outcomes* e.g., the proportion of heads in many tosses of a coin
 - (3) Many statistical inference procedures are based on the normal distribution and work well for other roughly symmetric distribution

The 68-95-99.7 Rule

- The 68-95-99.7 Rule (empirical rule)
 - In the Normal Distribution with mean μ and standard deviation σ
 - * Approximately 68% of the observations fall within 1σ of the mean μ
 - * Approximately 95% of the observations fall within 2σ of μ
 - * Approximately 99.7% of the observations fall within 3σ of the μ

Figure 22



Empirical Rule
(68 – 95 – 99.7%)

A statistical rule that provides an estimate for the distribution of approximately normal data.

The Standard Normal Distribution

- Standardizing and z -Scores
 - If x is an observation from a distribution that has mean μ and standard deviation σ , the **standardized value** of x is
$$z = \frac{x - \mu}{\sigma}$$
 - We standardize observations from symmetric distributions to express them in a **common scale**
- A standardized value is often called a **z -score**
 - A z -score tells us how many standard deviations the original observation falls away from the mean and in which direction
 - Observations larger than the mean are positive when standardized, and observations smaller than the mean are negative

Practice Question 3-1

Researchers collected the heights of adult men in a city. Heights are approximately normally distributed with $\mu = 175$ cm, $\sigma = 7$ cm.

- (1) What is the standardized (z -score) value for a man who is 182 cm tall? Show your steps.
- (2) Interpret your result in context. What does this z -score mean?
- (3) Another man is 168 cm tall. Compute his z -score. Who is more “unusual” relative to the population: the 182 cm man or the 168 cm man?

- **Standard Normal Distribution:** $N(0, 1)$
 - The Standard Normal Distribution is the Normal distribution $N(0, 1)$ with mean 0 and standard deviation 1.
 - If a variable x has any Normal distribution $N(\mu, \sigma)$ with mean μ and standard deviation σ , then the standardized variable

$$z = \frac{x - \mu}{\sigma}$$

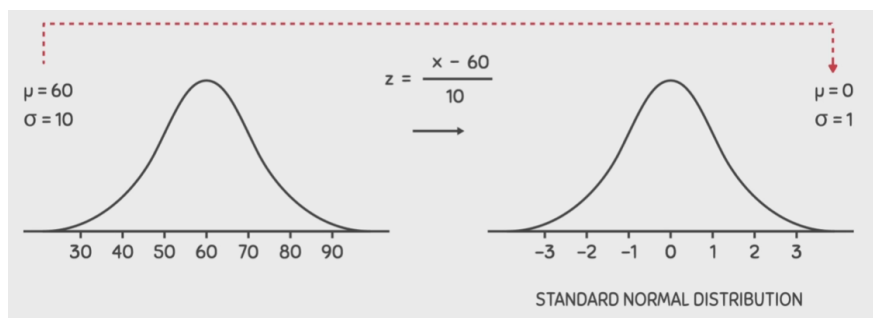
has the standard Normal distribution

Figure 23: Standardization

**Example**

Suppose that I gathered data from last year's final macroeconomics exam and found that it followed a normal distribution with a mean of 60 and a standard deviation of 10

Figure 24

**Practice Question 3-2**

In 2018, when she was a high school senior, Linda scored 680 on the mathematics part of the SAT. The distribution of SAT math scores in 2018 was normal with mean 528 and standard deviation 117. Jack took the ACT and scored 26 on the mathematics portion. ACT math scores for 2018 were normally distributed with mean 20.5 and standard deviation 5.5.

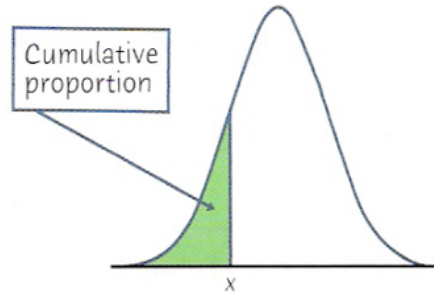
- (1) Find the standardized scores for both students
- (2) Assuming that both tests measure the same kind of ability, who had the higher score?

Finding Normal Proportions

- Cumulative Proportions

- The cumulative proportion for a value x in a distribution is the proportion of observations in the distribution that are less than or equal to x

Figure 25: $P(X \leq x)$



- There is *no* probability concentrated at any single point on a continuous density curve.
 - * For a normal distribution (and all continuous distributions), the probability of any exact value is

$$P(X = x) = 0.$$

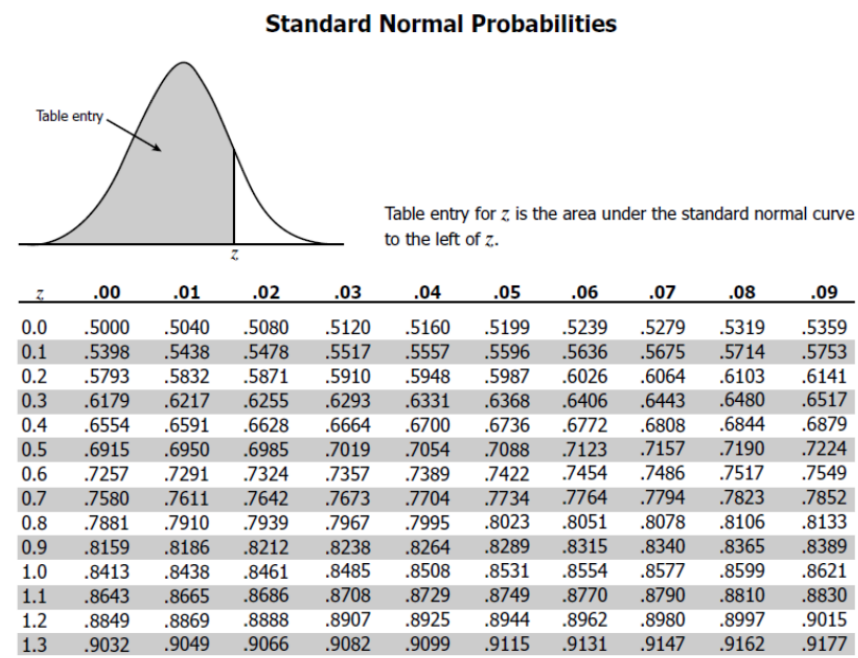
This is because probability corresponds to the *area* under the density curve, and a single point has zero width, so its area—and therefore its probability—is zero.

Using the Standard Normal Table (z-Table)

- Steps for Using the z-Table to Find Normal Proportions

- (1) State the problem in terms of the observed variable x . Draw a picture that shows the proportion you want in terms of cumulative proportions
- (2) Standardize x to restate the problem in terms of a standard Normal variable z
- (3) Use z-Table to find areas to the left of z . The fact that the total area under the curve is 1 may also be necessary to find the required data

Figure 26: Standard Normal Table (z-Table)

**Example**

Scores of high school seniors on the SAT follow the Normal distribution, with $\mu = 1059$ and $\sigma = 210$. What proportion of seniors have scores less than or equal to 1232?

$$\begin{aligned}
 X &\sim N(1059, 210), \quad P(X \leq 1232) \\
 z &= \frac{1232 - 1059}{210} \approx 0.824 \approx 0.82 \\
 P(Z \leq 0.82) &\approx 0.794
 \end{aligned}$$

Practice Question 3-3

The lifetimes of LED light bulbs follow a Normal distribution with mean $\mu = 18,000$ hours and standard deviation $\sigma = 2,400$ hours. What proportion of bulbs last greater than or equal to 20,500 hours?

Practice Question 3-4

The lengths of manufactured metal rods follow a Normal distribution with mean $\mu = 60$ cm and standard deviation $\sigma = 8$ cm. Find the rod length x that marks the **third quartile** (the 75th percentile).

$$X \sim N(60, 8)$$

$$P(X \leq x) = 0.75$$

From the z-table: $z_{0.75} \approx 0.67$ (rounded to two decimals)

$$\text{Use } z = \frac{x - \mu}{\sigma} :$$

$$0.67 = \frac{x - 60}{8}$$

$$x = 60 + (0.67)(8)$$

$$x \approx 65.36$$

CH 4. Scatterplots and Correlation

- In Chapters 1 through 3, we focused on exploring features of a *single variable*.
- In this chapter, we continue our study of exploratory data analysis but for the purpose of examining *relationship between variables*.

Explanatory and Response Variables

- Response Variable, Explanatory Variable
 - **Response Variable (Dependent Variable)**: measure an outcome of a study
 - **Explanatory Variable (Independent Variable)**: explain or influence changes in a response variable
- Examples
 - Beer and Blood Alcohol
 - Stress level, debt, and income
- In many studies, the goal is to show that changes in one or more explanatory variables actually *cause* changes in a response variable.
 - However, not every explanatory–response relationship implies direct causation.
 - For example, data show that nations with more television sets per person tend to have higher life expectancies. Is this relationship truly causal?

Displaying Relationships: Scatterplots

- Scatterplot
 - A scatterplot is a plot that displays the relationship between two quantitative variables measured on the same individuals.
 - The values of one variable appear on the horizontal axis, and the value of the other appear on the vertical axis
 - Each individual in the data appears as the point in the plot fixed by the values of both variables for that individual

Example

Figure 27: Average Income and Illiteracy

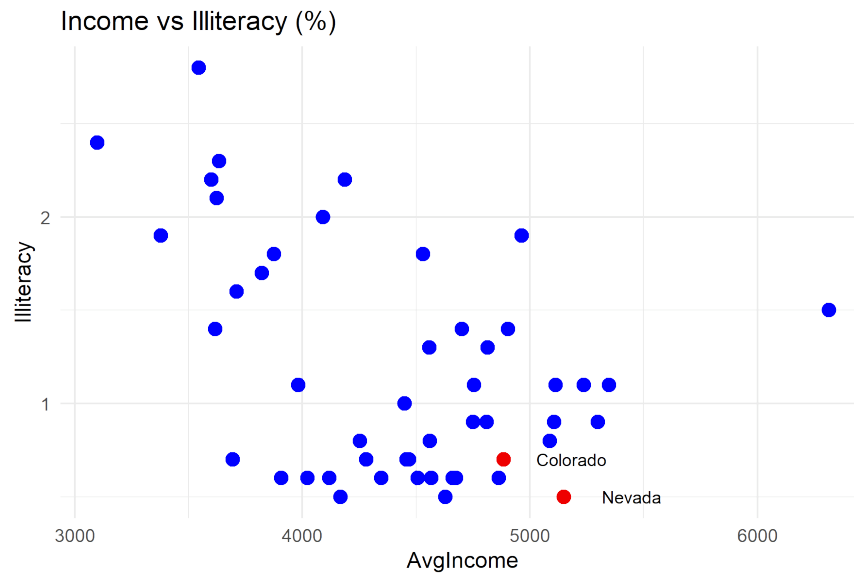
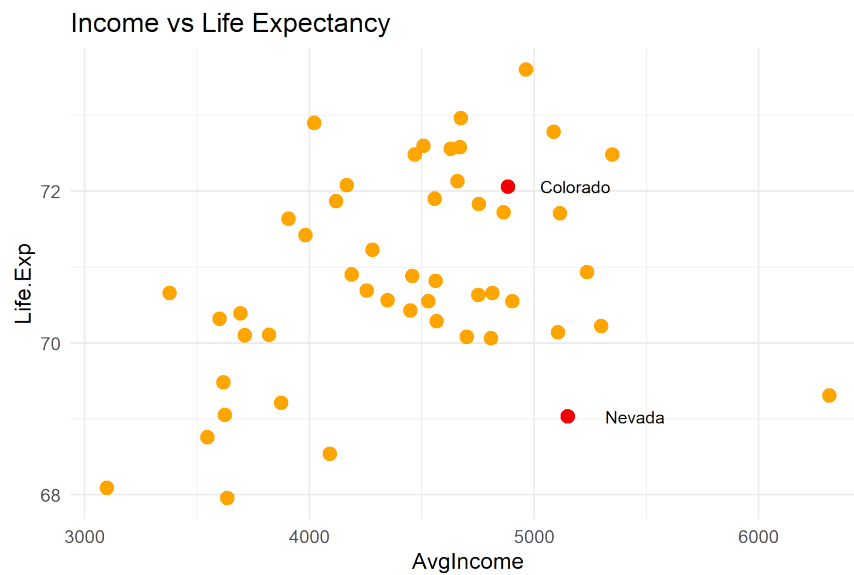


Figure 28: Average Income and Life Expectancy



Interpreting Scatterplots

- Examining a Scatterplot
 - Describe the overall pattern of the plot by the direction, form, and strength.
 - (1) *Direction*: indicates whether the overall pattern moves from lower left to upper right, from upper left to lower right, or neither
 - (2) *Form*: refer to the approximate functional form e.g., linear, nonlinear
 - (3) *Strength*: refers to how closely the points in the plot follow the form. If they fall almost perfectly on a straight line, we say there is a strong straight-line relationship
 - (4) *Outlier*: an important kind of deviation that an individual value falls outside the overall pattern of the relationship
- Positive Association, Negative Association
 - (1) *Positive Association*: two variables are positively associated when above-average values of one tend to accompany above-average values of other, and below-average values also tend to occur together
 - (2) *Negative Association*: two variables are negatively associated when above-average values of one tend to accompany below-average values of the other and vice versa

Figure 29: Linear Form (Positive)

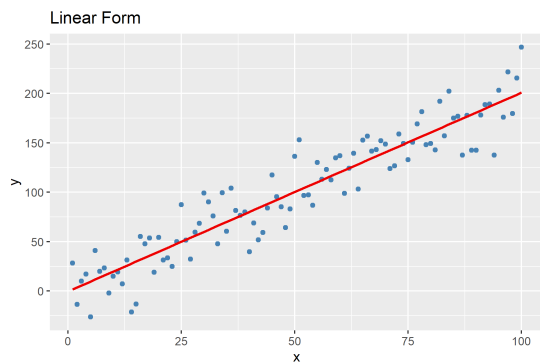


Figure 30: Nonlinear Form (Quadratic)

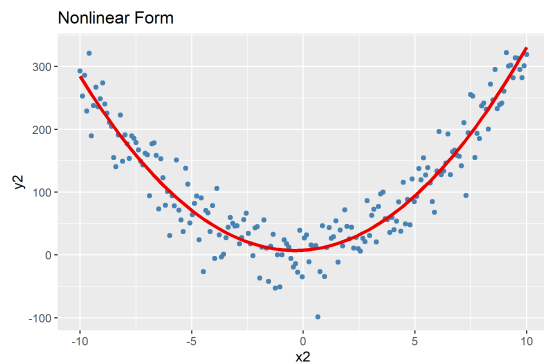


Figure 31: Linear Form (Weak Strength)

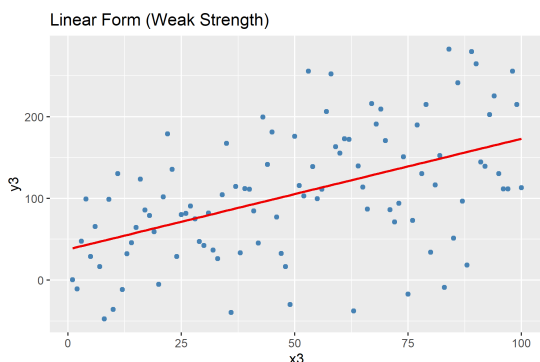
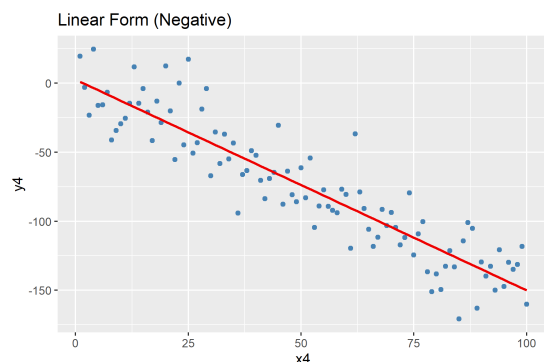


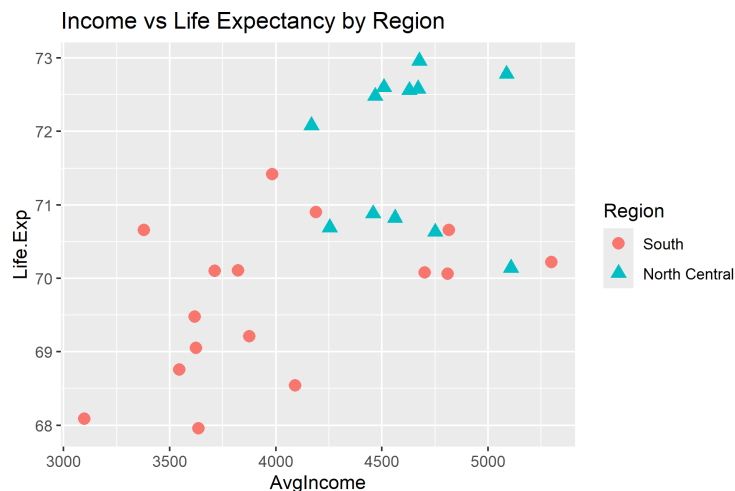
Figure 32: Linear Form (Negative)



Adding Categorical Variables to Scatterplots

- Categorical Variables in Scatterplots
 - To add a categorical variable to a scatterplot, use a different plot color or symbol for each category
 - For example, dividing the states into regions (categorical variable) introduces a third variable into the scatterplot (Figure 33)

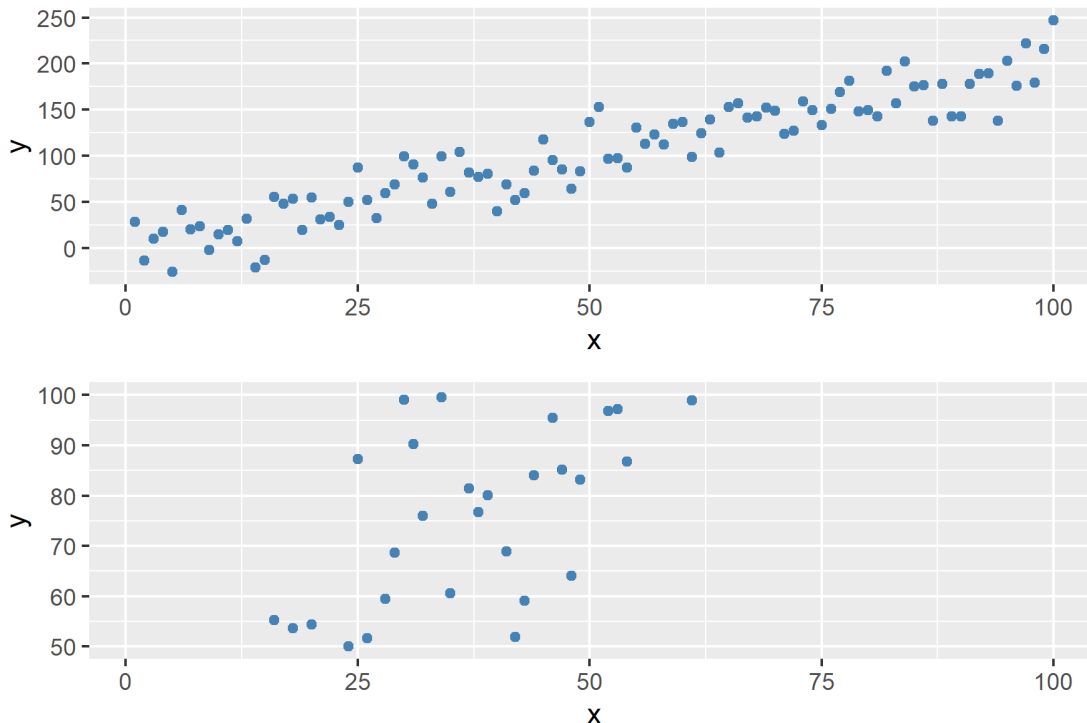
Figure 33: Income and Life Expectancy by Region



Measuring Linear Association: Correlation

- Linear (straight-line) relations are particularly important because a straight line is a simple pattern that is quite common
- A linear relationship between two variables is *strong* if the points lie close to a straight line and *weak* if they are widely scattered about a line
- However, our eyes can be fooled by changing the plotting scales or the amount of space around the cloud of points in a scatterplot (Figure 34)
- Therefore, we need to follow our strategy for data analysis by using a numerical measure to supplement the graph. **Correlation** is the measure we use.

Figure 34: Two Scatterplots of the same data



• Correlation

- The **correlation** measures the *direction* and *strength* of the linear relationship between two quantitative variables
- Correlation is usually written as r
- The correlation r between two variables x and y is

$$r = \frac{1}{n-1} \left\{ \left(\frac{x_1 - \bar{x}}{s_x} \right) \left(\frac{y_1 - \bar{y}}{s_y} \right) + \left(\frac{x_2 - \bar{x}}{s_x} \right) \left(\frac{y_2 - \bar{y}}{s_y} \right) + \cdots + \left(\frac{x_n - \bar{x}}{s_x} \right) \left(\frac{y_n - \bar{y}}{s_y} \right) \right\}$$

or, more compactly,

$$r = \frac{1}{n-1} \sum \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

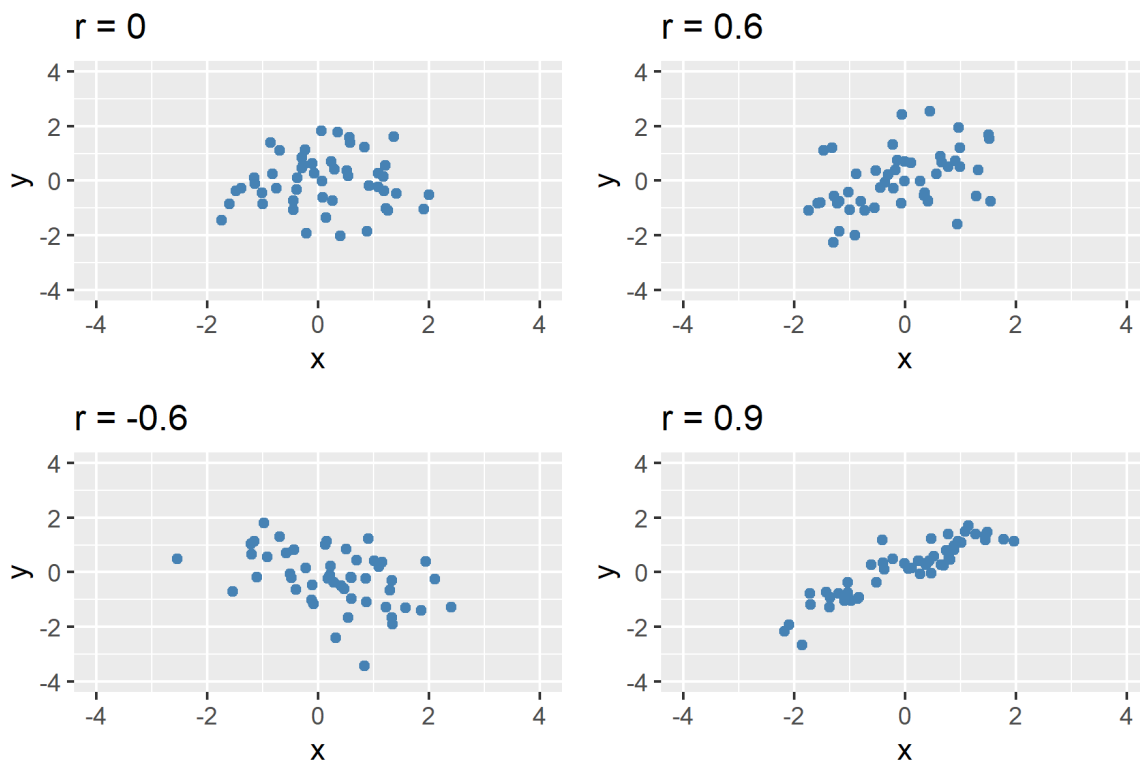
- * n : number of observations
 - * x_i, y_i : two quantitative variables
 - * $\bar{x}, \bar{y}$: mean
 - * s_x, s_y : standard deviation
 - * $\frac{x_i - \bar{x}}{s_x}, \frac{y_i - \bar{y}}{s_y}$: standardized value of x and y for individual i
- The correlation r is an average of the products of the standardized x and the standardized y for all the individual i .

Facts about Correlation

- Properties of Correlation (r)

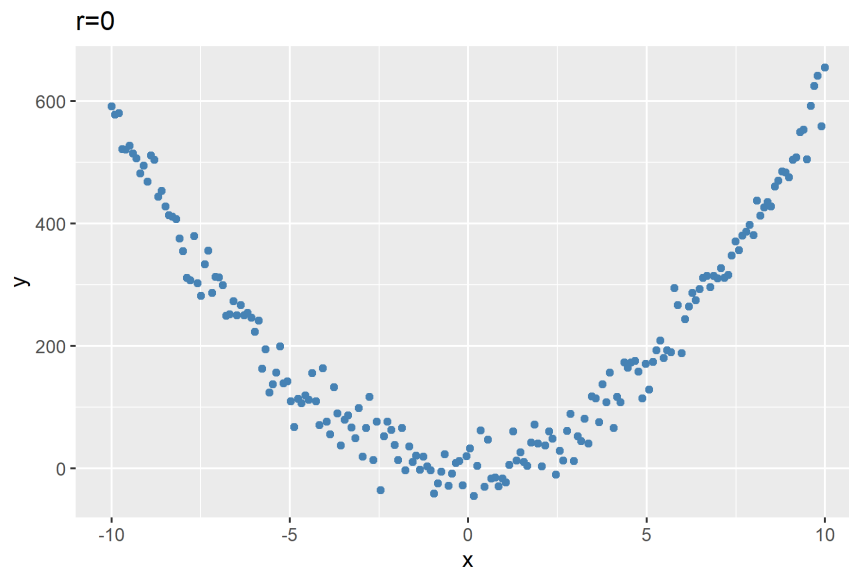
- (1) *Correlation makes no distinction between and response variables*
 - It makes no difference which variable you call x and which you call y in calculating the correlation
- (2) *Because r uses the standardized values of the observations, r does not change when we change the units of measurement of x , y , or both*
 - For example, measuring height in inches rather than centimeters and weight in pounds rather than kilograms does not change the correlation between height and weight
 - The correlation r itself has no unit of measurement: it is just a number
- (3) *Positive r indicates a positive association between the variables, and negative r indicates a negative association*
- (4) *The correlation r is always a number between -1 and 1 ($-1 \leq r \leq 1$)*
 - **r is near 0**: indicate a very weak linear relationship between the two variables
 - **r close to -1 or 1**: indicate that the points in a scatterplot lie close to a straight line
 - **$r = 1$, or $r = -1$** : occur only in the case of a perfect linear relationship between the two variables, when the points lie exactly along a straight line

Figure 35: Scatterplots with Different Levels of Correlation



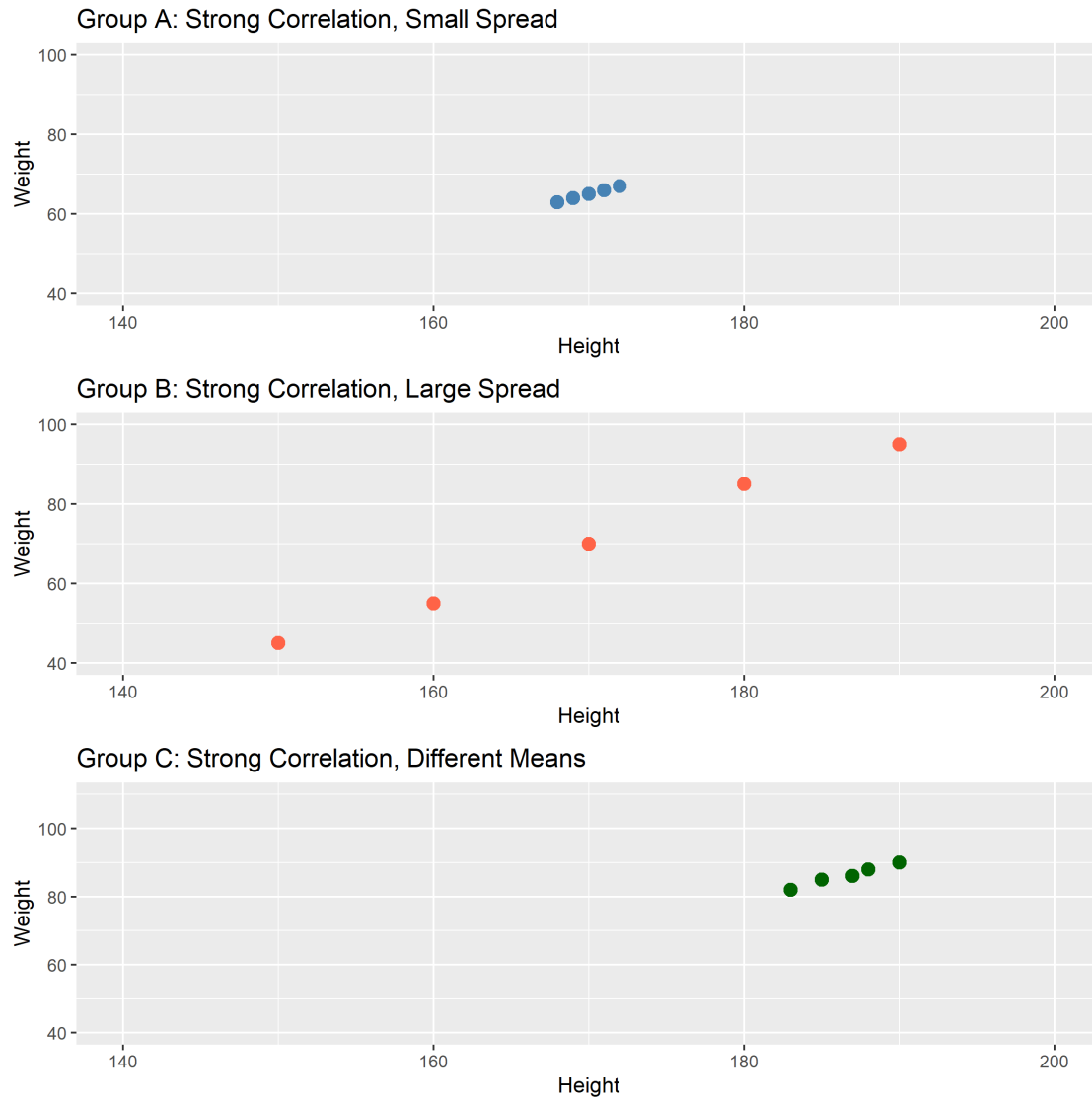
- Cautious to keep in mind when you use correlation r
 - (1) Correlation requires that both variables be quantitative, so it makes sense to do the arithmetic indicated by the formula for r
 - (2) Correlation measures the strength of only the linear relationship between two variables. Correlation does not describe curved relationship between variables, no matter how strong they are
 - For example, below is a scatter plot of $r=0$

Figure 36



- A correlation of 0 means there is no linear relationship between two variables, indicating they don't consistently increase or decrease together in a straight line, making one a poor linear predictor for the other.
 - However, it doesn't necessarily mean they are unrelated; a strong non-linear pattern (like a U-shape) can also result in a correlation near zero
- (3) Like the mean and standard deviation, the correlation is not resistant: r is affected by a few outliers
 - (4) Correlation is not a complete summary of two-variable data
 - You should consider the means and standard deviations along with the correlation (Figure 37)

Figure 37



CH 5. Regression

- When there is a clear explanatory variable and a strong straight-line relationship, it is tempting to assume that large values of the correlation imply that there is a cause-and-effect relationship between explanatory variable and the response variable
 - This need not be true
 - **Correlation does not imply causation**
 - Correlation measures the direction and strength of these linear relationship
- When a scatterplot shows a linear relationship, we would like to summarize the overall pattern by drawing a line of best fit (*regression line*) on the scatterplot

Regression

- **Regression**
 - Regression is a statistical method used to describe and quantify how one variable (response variable) changes in relation to one or more other variables (explanatory variable)
- **Regression Line**
 - A regression line summarizes the relationship between two variables—but only in a specific setting: when one of the variables helps explain or predict the other
 - A regression line is a straight line that describes how a response variable y changes as an explanatory variable x changes
 - We often use a regression line to *predict* the value of y for a given value of x when we believe the relationship between y and x is linear
- **Review of Straight Lines**
 - Suppose that y is a response variable and x is an explanatory variable. A straight line relating y to x has an equation of the form

$$y = a + bx$$

In this equation, b is the **slope**, the amount by which y changes when x increases by one unit. The a is the **intercept**, the value of y when $x = 0$

Example

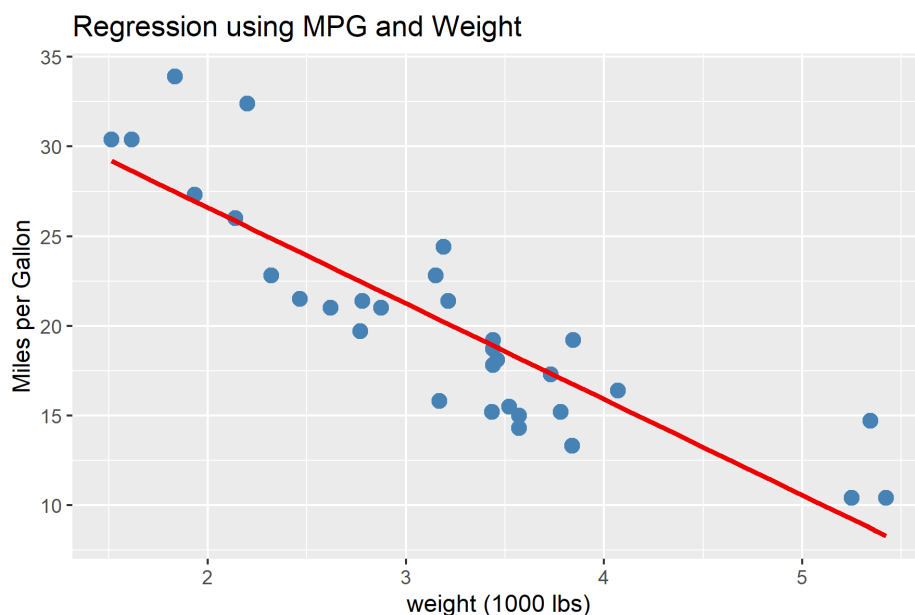
Below are the data and the corresponding scatterplot with the fitted regression line. Based on the data, the regression line is $\widehat{mpg} = 37.28 - 5.34 \times wt$

- **Slope (Coefficient of wt = -5.34):** For each increase of 1 unit in weight (that is, for each additional 1000 pounds), the predicted fuel efficiency decreases by approximately 5.34 miles per gallon. This indicates a negative relationship between car weight and fuel economy.
- **Intercept (37.28):** The intercept represents the predicted fuel efficiency when $wt = 0$. Although a car cannot realistically have zero weight, this value is necessary for the mathematical model and helps determine the position of the regression line.

Table 2: MPG and Weight

Model	mpg	wt	Model	mpg	wt	Model	mpg	wt
Mazda RX4	21.0	2.620	Merc 450SE	16.4	4.070	AMC Javelin	15.2	3.435
Mazda RX4 Wag	21.0	2.875	Merc 450SL	17.3	3.730	Camaro Z28	13.3	3.840
Datsun 710	22.8	2.320	Merc 450SLC	15.2	3.780	Pontiac Firebird	19.2	3.845
Hornet 4 Drive	21.4	3.215	Cadillac Fleetwood	10.4	5.250	Fiat X1-9	27.3	1.935
Hornet Sportabout	18.7	3.440	Lincoln Continental	10.4	5.424	Porsche 914-2	26.0	2.140
Valiant	18.1	3.460	Chrysler Imperial	14.7	5.345	Lotus Europa	30.4	1.513
Duster 360	14.3	3.570	Fiat 128	32.4	2.200	Ford Pantera L	15.8	3.170
Merc 240D	24.4	3.190	Honda Civic	30.4	1.615	Ferrari Dino	19.7	2.770
Merc 230	22.8	3.150	Toyota Corolla	33.9	1.835	Maserati Bora	15.0	3.570
Merc 280	19.2	3.440	Toyota Corona	21.5	2.465	Volvo 142E	21.4	2.780
Merc 280C	17.8	3.440	Dodge Challenger	15.5	3.520			

Figure 38: Scatterplot with Regression Line



Practice Question 5-1

An online article suggested that for each additional person who took up regular running for exercise, the number of cigarettes smoked daily would decrease by 0.178. If we assume that 48 million cigarettes would be smoked per day if nobody ran, what is the equation of the regression line for predicting the number of cigarettes smoked per day from the number of people who regularly run for exercise?

Practice Question 5-2

We expect a car's highway gas mileage to be related to its city gas mileage (in mpg). Data for all 1259 vehicles in the government's *2019 Fuel Economy Guide* give the regression line

$$\text{highway mpg} = 8.720 + 0.914 \times \text{city mpg}$$

for predicting highway mileage from city mileage

- (a) What is the slope of this line? Interpret the numerical value
- (b) What is the intercept? Explain why the value of the intercept is not statistically meaningful
- (c) Find the predicted highway mileage for a car that gets 16 mpg in the city. Do the same for a car with city mileage of 28 mpg
- (d) Draw a graph of the regression line for city mileages between 10 and 50 mpg

The Least-Squares Regression Line

- We need a way to draw a regression line that doesn't depend on our guess of where the line should go
 - Because we use the regression line to predict y from x , the prediction errors we make are errors in y , the vertical direction in the scatterplot
- **A good regression line makes the vertical distances of the points from the line as small as possible**
 - There are many ways to make the collection of vertical distances “as small as possible”. The most common is the *least-squares* method
- **Least-Squares Regression Line**
 - The **least-squares regression line** of y on x is on the line that makes the sum of the squares of the vertical distances of the data points from the line as small as possible
- **Equation of the Least-Squares Regression Line**
 - From the data, calculate the means $\bar{x}$ and $\bar{y}$ and the standard deviations s_x and s_y , of the two variable and their correlation r . The least-squares regression line is the line

$$\hat{y} = a + bx$$

with slope

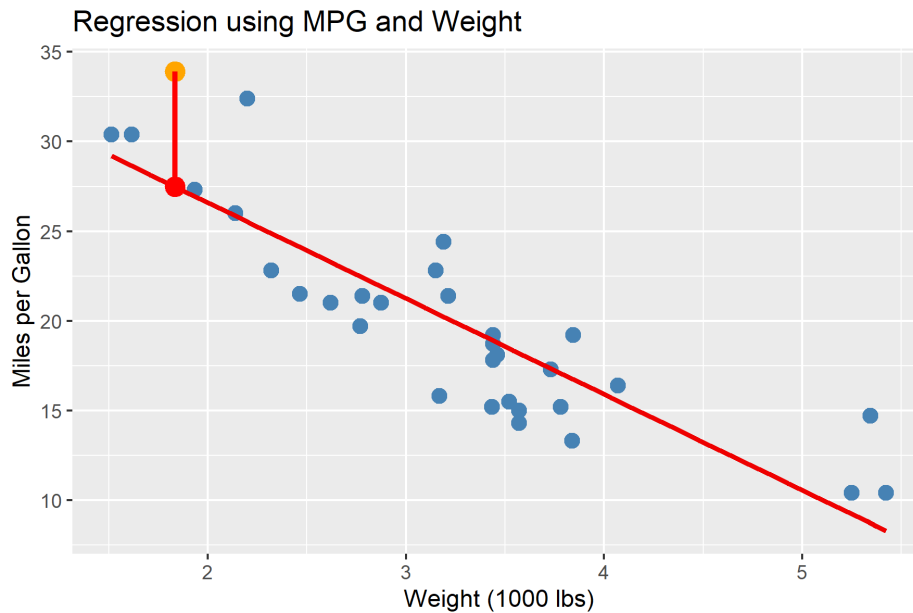
$$b = r \cdot \frac{s_y}{s_x}$$

and intercept

$$a = \bar{y} - b\bar{x}$$

- y : a observed response (outcome)
- $\hat{y}$: a predicted response (outcome) for any x

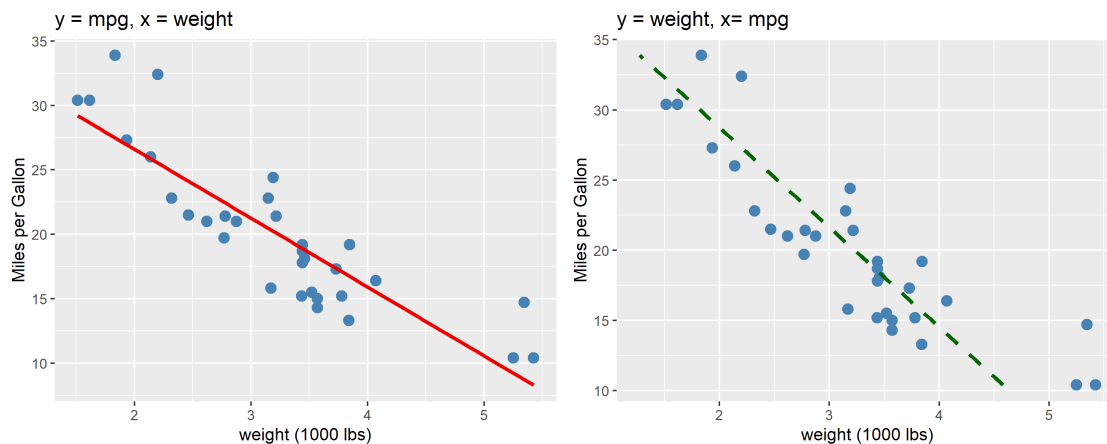
Figure 39



Facts About Least-Squares Regression

- Special Properties of least-squares regression
 - (1) The distinction between explanatory and response variables is essential in regression
 - Least-squares regression makes the distances of the data points from the line small only in the y direction
 - If we reverse the roles of the two variables, we get a different least-squares regression line (Figure 40)

Figure 40



(2) There is a close connection between correlation and the slope of the least-squares line

- The slope and the correlation always have the same sign
- Along the regression line, a change of one standard deviation in x (s_x) corresponds to a change of r standard deviations in y (s_y)
 - * For example, if $r = 0.5$, $s_y = 6$ and $s_x = 2$, then

$$b = r \cdot \frac{s_y}{s_x} = 0.5 \cdot \frac{6}{2} = 1.5$$

In terms of standard deviations, being one standard deviation above the mean in X (i.e., 2 units) corresponds to being $r = 0.5$ standard deviations above the mean in Y , which is $0.5 \times 6 = 3$ units.

- (3) The least-squares regression line always passes through the point $(\bar{x}, \bar{y})$ on the graph of y against x
- (4) The correlation r describes the strength of a straight-line relationship

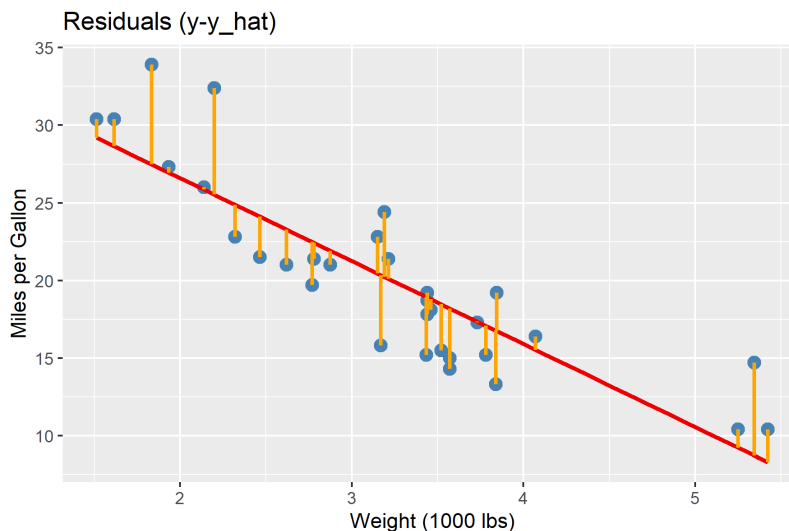
- Residuals

- A residual is the difference between an observed value of the response variable and the value predicted by the regression line.
- That is, a residual is the prediction error that remains after we have chosen the regression line:

$$\begin{aligned} \text{residual} &= \text{observed } y - \text{predicted } y \\ &= y - \hat{y} \end{aligned}$$

- Because they represent “leftover” variation in the response after fitting the regression line, these distances are called residuals

Figure 41: Residuals



- Residual Plots

- A residual plot is a scatterplot of the regression residuals against the explanatory variable. Residual plots help us assess how well a regression line fits the data
- It magnifies the deviations of the points from the line and makes it easier to see unusual observations and patterns

Figure 42: Scatter Plot

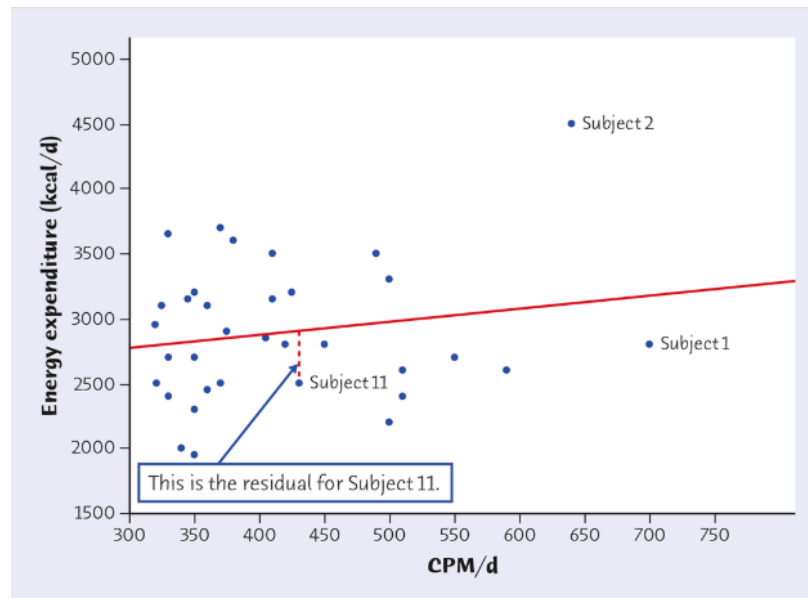
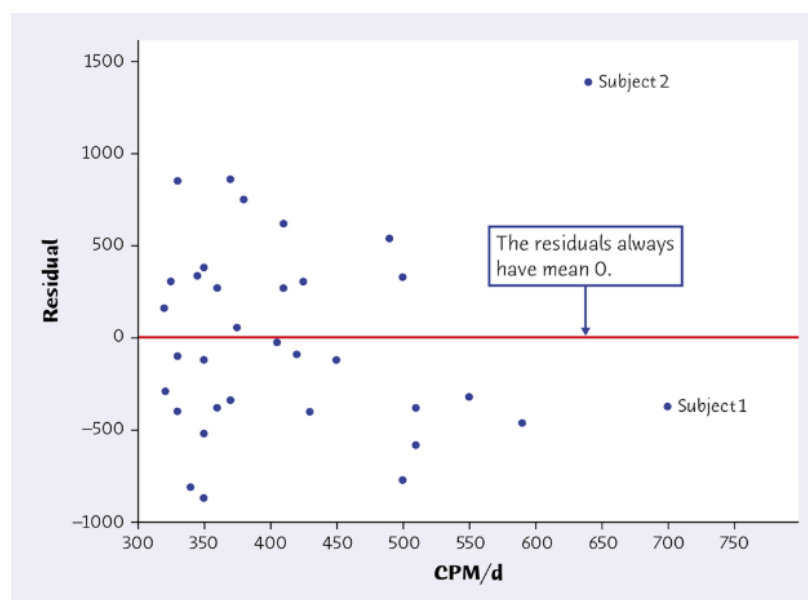
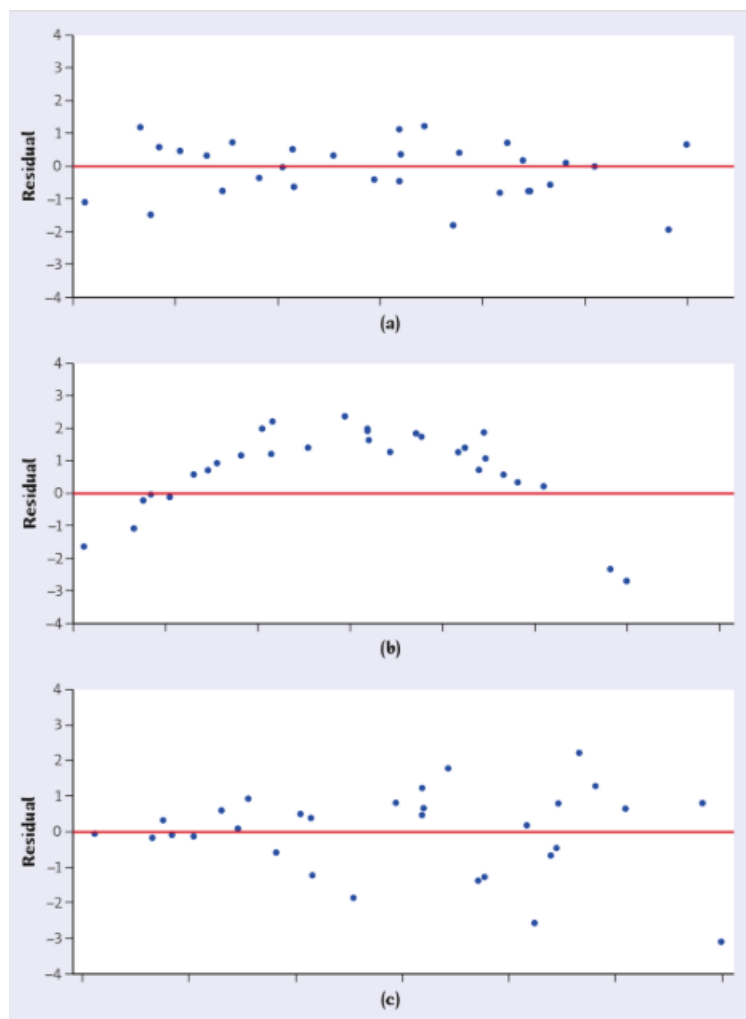


Figure 43: Residual Plot



- Patterns of some typical residual plots
 1. (a): An unstructured horizontal band centered at 0 and symmetric about 0
 2. (b): A curved pattern indicates that the relationship between the response and explanatory variables is curved rather than linear
 - A straight line is not a good description of such a relationship
 3. (c): A fan-shaped pattern shows that the variation of the response about the least-squares line increases as the explanatory variable increases
 - Predictions of the response will be more precise for smaller values of the explanatory variable, where the response shows less variability about the line

Figure 44: Residuals



Influential Observations

- Influential Observations

- An observation is influential for a statistical calculation if removing it would markedly change the result of the calculation
- The result of a statistical calculation may be of little practical use if it depends strongly on a few influential observations
- Points that are **outliers** in a scatter plot can greatly influence both the correlation and the least-squares regression line. In particular, points that lie far in the x -direction (high leverage) or far from the regression pattern in the y -direction can pull the fitted line toward themselves and substantially change the correlation.

Example

Figure 45 illustrates the effect of an outlier on the regression line. In this dataset, there are 14 regular observations and 1 outlier. When the outlier is included, the fitted regression line is $\hat{y} = -0.12 + 1.11x$. However, when the outlier is removed, the fitted regression line becomes $\hat{y} = 2.49 + 0.46x$. This comparison shows how a single outlier can substantially change both the slope and intercept of the regression line.

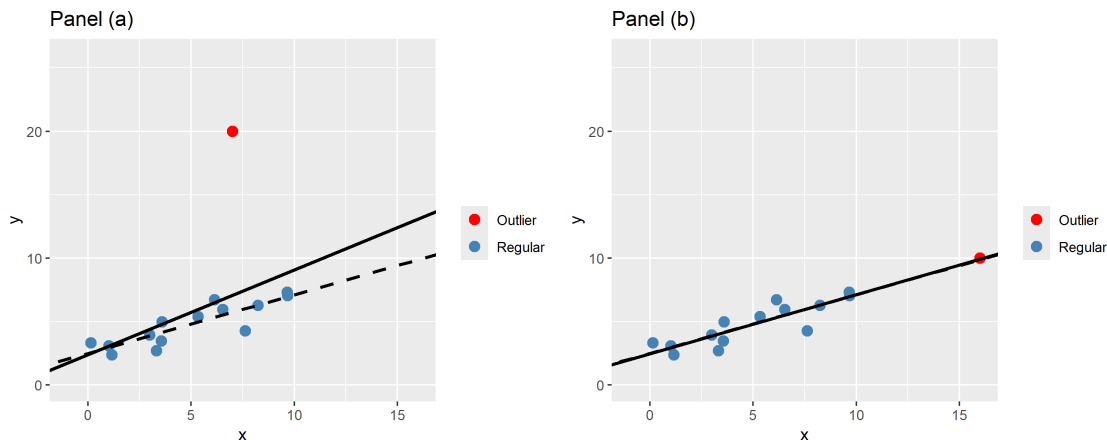
Figure 45: Influential Observation



Example

The regression line with the outlier is $\hat{y} = 2.40 + 0.66x$ for Panel (a), and $\hat{y} = 2.47 + 0.46x$ for Panel (b) respectively. This Figure 46 illustrates that not all outliers are influential observations. Although both panels contain an outlier, the effect on the fitted regression line differs, showing that an outlier does not necessarily have a strong impact on the correlation or the least-squares regression line.

Figure 46



Cautious About Correlation and Regression

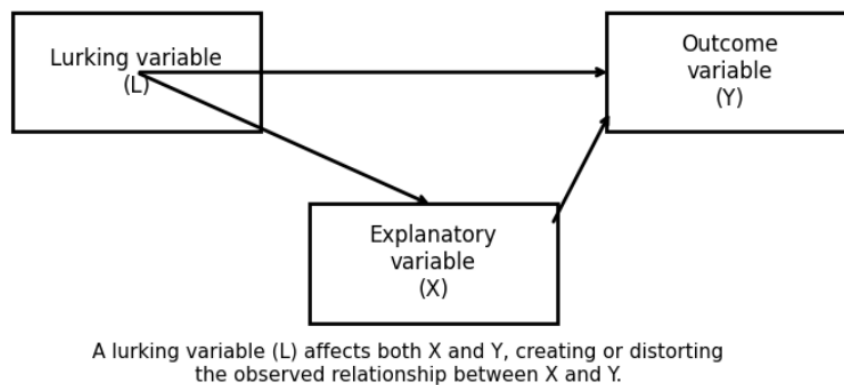
- When you use correlation and regression, you must be aware of their limitations
 - (1) Correlation and regression lines describe **only linear relationships**
 - (2) Correlation and least-squares regression lines are not resistant
 - Always plot your data and look for observations that may be influential
- **Extrapolation**
 - Extrapolation is the use of a regression line for prediction far outside the range of values of the explanatory variable x that you used to obtain the line
 - Such predictions are often not accurate
 - For example, you have data on a child's growth between 3 and 8 years age. If you fit a regression line to these data and use it to predict height at age 25 years, you may well predict inaccurately.

- **Lurking Variable**

- A lurking variable is a variable that is not among the explanatory or response variables in a study and yet may influence the interpretation of relationships among those variables (Figure 47)
- For example, suppose we study the relationship between the minimum wage and employment.
 - * Explanatory variable (X): minimum wage level in a city or state
 - * Outcome variable (Y): employment in low-wage industries
 - * Lurking variable (L): local economic conditions (e.g., economic growth or demand)

If the local economy is strong, policymakers may be more likely to raise the minimum wage ($L \rightarrow X$), and employment may also rise because firms are hiring more ($L \rightarrow Y$). As a result, we may observe a positive association between the minimum wage and employment even if the direct causal effect of the minimum wage on employment is weak or negative.

Figure 47: Lurking Variables



Association Does Not Imply Causation

- A strong association between two variables is not enough to draw conclusions about cause and effect
- A least-squares regression line that fits the data well and gives accurate predictions is also not enough to draw conclusions about cause and effect

CH 6. Two-Way Tables

- **Two-way Table**

- In this chapter, we use two-way tables to describe relationships between two categorical variables
- A table of counts used to organize data about two categorical variables
- Entries in the table are the counts of how often each combination of the corresponding row and column variables occurs
- Two-way tables are often used to summarize large amounts of information by grouping observations into categories

Example

Table 3: Two-way table of Titanic passengers

Sex	Total				Sum
	1st	2nd	3rd	Crew	
Male	180	179	510	862	1731
Female	145	106	196	23	470
Total	325	285	706	885	2201

Marginal Distributions

- **Marginal Distributions**

- The **marginal distribution** of one of the categorical variables in a two-way table of counts is the distribution of values of that variable among all individuals described by the table
- Each marginal distribution from a two-way table is a distribution for a single categorical variable
- Percentages are often more informative than counts.
- However, marginal distributions tell us nothing about the relationships between two variables

Example

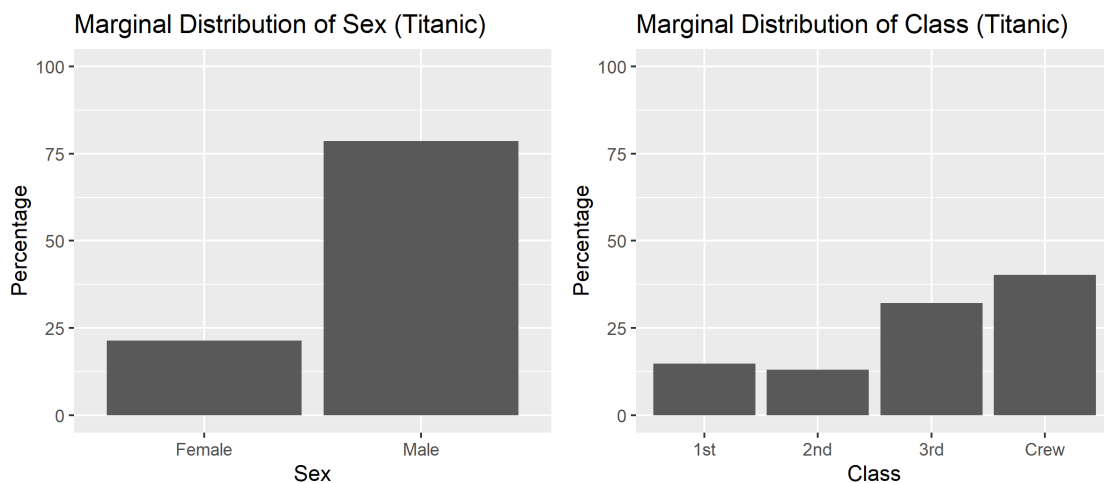
Table 4: Marginal distribution of Sex (Titanic passengers) in percentages.

Sex	Count	Percent
Male	1731	78.65%
Female	470	21.35%
Total	2201	100.00%

Table 5: Marginal distribution of Class (Titanic passengers) in percentages.

Class	Count	Percent
1st	325	14.77%
2nd	285	12.95%
3rd	706	32.08%
Crew	885	40.21%
Total	2201	100.00%

Figure 48



Conditional Distribution

- Conditional Distributions
 - A conditional distribution of a variable is the distribution of values of that variable among only individuals who have a given value of the other variable
 - There is a separate conditional distribution of each value of the other variable
 - To describe a relationship between two categorical variables, we must calculate some well-chosen percentages from the counts given in the body of the table

Example

The conditional distribution of passenger class for the male and female groups.

Table 6: Conditional distribution of sex given class (Titanic passengers)

Sex	1st	2nd	3rd	Crew
Male	55.38%	62.81%	72.24%	97.40%
Female	44.62%	37.19%	27.76%	2.60%
Total	100.00%	100.00%	100.00%	100.00%

Table 7: Conditional distribution of class given sex (Titanic passengers)

Sex	1st	2nd	3rd	Crew	Total
Male	10.40%	10.34%	29.46%	49.80%	100.00%
Female	30.85%	22.55%	41.70%	4.89%	100.00%

Figure 49: Panel (a)

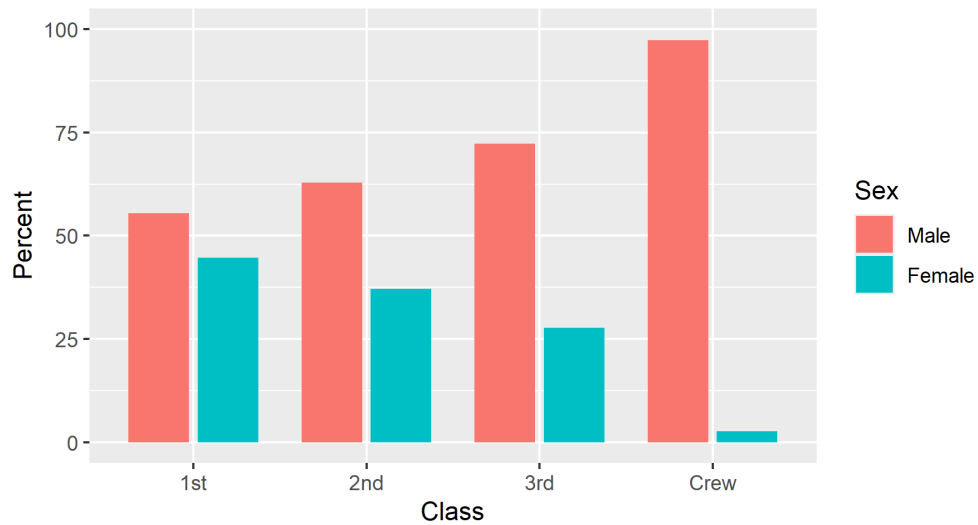
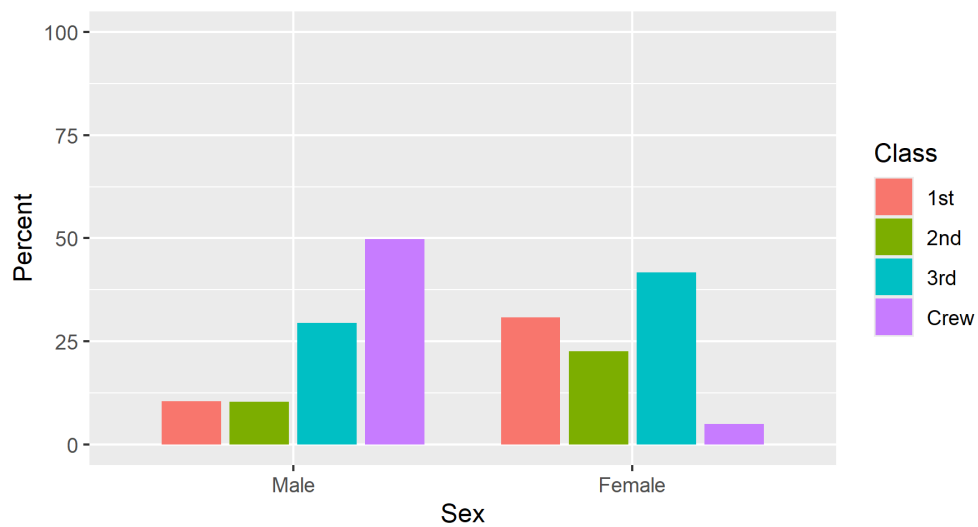


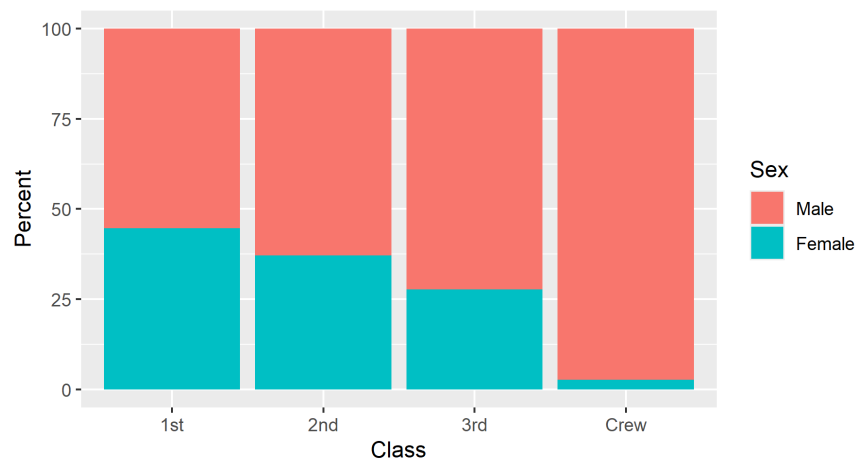
Figure 50: Panel (b)



- Segmented Bar Graph

- A segmented bar graph is a bar graph for presenting data about two categorical variables in which each bar is divided into parts (Figure 51)
- Each bar represents the observations that make a particular value of one variable, and the length of each part of the bar represents the proportion of those observations that take a specific value of the second variable

Figure 51: Segmented Bar Graph



Simpson's Paradox

- As in the case with quantitative variables, the effects of lurking variables can change or even reverse relationships between two categorical variables.
- **Simpson's Paradox**
 - An association or comparison that holds for all or several groups can reverse direction when the data are combined to form a single group.
 - This reversal is called **Simpson's paradox**

Example: Do Medical Helicopters Save Lives?

- Suppose we compare outcomes for patients transported by **helicopter** versus **road**. A natural question is whether helicopter transport improves survival.
- However, the comparison is complicated by a **lurking variable: accident seriousness** (Serious vs. Minor). Serious accidents have lower survival rates overall, and helicopters are used more often for serious cases.

Table 8: Overall outcomes by transport mode

Transport	Survived	Died	Total
Helicopter	32	28	60
Road	78	27	105

Table 9: Outcomes by transport mode and accident seriousness (Simpson's paradox example)

Seriousness	Transport	Survived	Died	Total
Serious	Helicopter	30	20	50
Serious	Road	6	9	15
Minor	Helicopter	2	8	10
Minor	Road	72	18	90

- **Overall (marginal) results.** If we ignore seriousness and combine all cases, the overall survival rates are:

$$\text{Helicopter: } \frac{32}{60} \approx 53.3\%, \quad \text{Road: } \frac{78}{105} \approx 74.3\%.$$

So, in the aggregated data, road appears to have a higher survival rate.

- **Stratified (conditional) results.** When we compare *within* each seriousness group, we get:
 - **Serious accidents:** Helicopter survival rate = $30/50 = 60\%$, Road survival rate = $6/15 = 40\%$.
 - **Minor accidents:** Helicopter survival rate = $2/10 = 20\%$, Road survival rate = $72/90 = 80\%$.

These conditional comparisons show that the survival rates depend strongly on seriousness.

- **Why this is Simpson's paradox.**
 - The overall comparison is misleading because helicopters are used disproportionately for **serious** accidents (which have worse outcomes regardless of transport). Road transport is used mostly for **minor** accidents (which have better outcomes regardless of transport). Because the two transport modes have different mixes of serious and minor cases, the marginal (overall) rates can suggest a different conclusion than the conditional (within-group) rates.
- **Key idea:**
 - Always check for a lurking variable and compare conditional distributions before drawing conclusions from overall averages.

Part II

Producing Data

- Exploratory Data Analysis
 - Purpose is unrestricted exploration of the data, searching for interesting patterns
 - Conclusions apply only to the individual and circumstances for which we have data in hand
 - Conclusions are informal, based on what we see in the data
- **Statistical Inference**
 - Purpose is to answer specific questions, posed before the data were produced
 - Conclusions apply to a larger group of individuals or a broader class of circumstances
 - Conclusions are formal, backed by a statement of our confidence in them
- Ch 8 - Ch 10
 - describes statistical designs for producing data by samples and experiments and ethical issues involved
 - The quality of the inferences we make from data depends heavily on how the data are produced

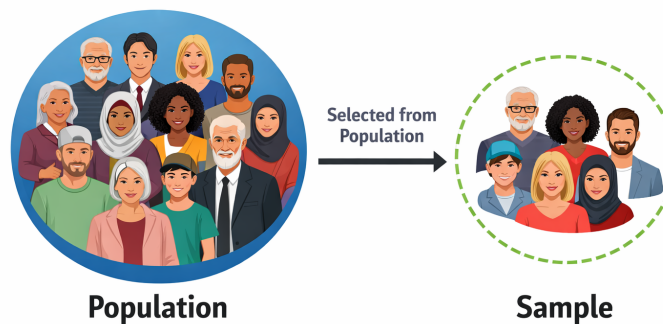
CH 8. Producing Data: Sampling

- A sound statistical design is necessary if we are to trust data from a *sample* for drawing sound about the *population*
- When sampling from large human populations, even with a sound design, there are still many practical difficulties that arise
- The impact of technology (particularly cell phones and the Internet) is making it harder to produce trustworthy national data by sampling

Population vs. Sample

- Population, Sample, and Sampling Design
 - (1) **Population:** is the entire group of individuals about which we want information
 - (2) **Sample:** a part of the population from which we actually collect information. We use a sample to draw conclusions about the entire population
 - (3) **Sampling Design:** describes exactly how to choose a sample from the population

Figure 52: Population vs Sample

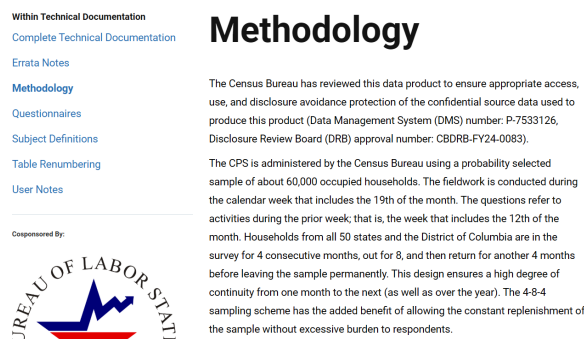


- Sample Survey
 - A survey conducted on a sample from the population of all individuals about which we desire information
 - * What population we want to describe?
 - * What we want to measure?
 - We base conclusions about the population on data from the sample

Example: The Current Population Survey (CPS)

- The CPS contacts about 60,000 households each month
- It produces the monthly unemployment rate and much other economic and social information
- Define its population as all U.S. residents 16 years of age and over who are civilians and are not in an institution such as a prison
- Variables—e.g., employed, unemployed, in the labor force, not in the labor force, etc.

Figure 53: CPS



How to Sample Badly

- **Bias**
 - The design of a statistical study is **biased** if it systematically favors certain outcomes
 - A systematic error caused by a bad sampling design

Example

What is a potential source of bias in each of the following sampling designs?

- (1) A phone survey about employment conducted during weekday work hours.
- (2) Campus survey on voting preferences
- (3) Question wording: “Do you support banning dangerous chemicals?” vs “Do you support banning chemicals used in manufacturing?”
- (4) Estimating average income by surveying people at an upscale mall
- (5) A news website posts a poll: “Should the city ban e-scooters? Click to vote.”

- Convenience Sample
 - A sample composed of members of the population who are selected by the researcher because they are easy to reach
- Voluntary Response Sample
 - A voluntary response sample consists of people who choose themselves by responding to a broad appeal
 - Voluntary response samples are biased because people with strong opinions are most likely to respond

Simple Random Samples

- **The first question to ask about any sample is whether it was chosen at random**
- Simple Random Sampling (SRS)
 - A **simple random sample** of size n consists of n individuals from the population chosen in such a way that every set of n individuals has an equal chance to be the sample actually selected
 - a basic statistical method where a subset (sample) is chosen from a larger group (population) ensuring every individual has an equal chance of selection, making it unbiased and representative

Example: Simple Random Sample (SRS)

A university has $N = 12,000$ students. To estimate the average hours of sleep per night:

- (1) Obtain a complete list (roster) of all 12,000 students and label them $1, 2, \dots, 12,000$.
- (2) Use a random number generator to select $n = 300$ *distinct* integers from $\{1, 2, \dots, 12,000\}$.
- (3) Survey the 300 students whose labels were selected.

This is a simple random sample because every student has an equal chance of being chosen.

Trustworthiness of Inference from Samples

- **Inference**

- Process of drawing conclusions about a population on the basis of sample data
- We infer information about the population from what we now about the sample

- Why we use Random Sampling?

- (1) The first reason to rely on random sampling is to eliminate bias in selecting samples from the list of available individuals (unbiased sample)
- (2) The second reason to use random sampling is that the laws of probability allow trustworthy inference about the population
 - The results of random sampling don't change haphazardly from sample to sample
 - The results obey the laws of probability that govern chance behavior. And these laws allow us to say how likely it is that sample results are close to the truth

Other Sampling Designs

- Stratified Random Sample

- To select a stratified random sample, first classify the population into groups of similar individuals, called **strata**.
- Then, choose a separate SRS in each stratum and combine these SRSs to form the full sample

Example: Stratified Random Sample

- Suppose a school has 1,000 students:

Freshmen	600
Sophomores	250
Juniors	100
Seniors	50

- We want a stratified random sample of size $n = 100$ using proportional allocation (10% from each grade).

Stratum (Grade)	Population Size (N_h)	Sample Size (n_h)
Freshmen	600	60
Sophomores	250	25
Juniors	100	10
Seniors	50	5

- Then, select n_h students *at random* within each grade.
- Practical Reasons why we use stratified sampling
 - (1) Every grade is represented
 - (2) To make the sample match the school's real proportions
 - (3) To get a more accurate overall estimate
 - (4) Compare grades between groups

Cautions About Sample Surveys

- **Undercoverage and Nonresponse**
 - **Undercoverage:** occurs when some groups in the population are left out of the process of choosing the sample
 - **Nonresponse:** occurs when an individual chosen for a sample can't be contacted or refuses to participate
- **Response Bias**
 - Occurs when survey respondents answer falsely, either deliberately or by mistake
 - **Wording effects**
 - * the influence of question wording on survey responses, in which different phrasings of the same question can evoke different responses

Example: Undercoverage

- You want to estimate “average internet speed in the state,” but you sample using an email list. Households without reliable internet (often rural/low-income) are less likely to be on that list → they're undercovered, and your estimate will be biased upward.

Example: Nonresponse

- You randomly select 1,000 households to ask about income, but high-income and very low-income households are more likely to refuse. Even though the sample was random, the final responding group isn't representative → your income estimate gets biased.

Example: Response Bias

- Social desirability: Asking students, “Do you cheat on exams?” Many will say “no” even if they have.
- Leading question: “Don't you agree the new policy is improving the school?” pushes people toward “yes.”
- Recall error: “How many hours did you study last month?” People guess poorly.

Practice Question 8-1

A researcher wants to estimate the average number of hours of sleep per night for students at a high school. They randomly select 200 students from the school roster and email them a survey. Only 60 students respond. Many respondents report sleeping 8–9 hours, but the researcher suspects some students exaggerated because they think that is the “healthy” answer. Which option best matches each problem described?

- (a) **Undercoverage** = emailing the survey; **Nonresponse** = only 60 respond; **Response bias** = exaggerating sleep hours
- (b) **Undercoverage** = only 60 respond; **Nonresponse** = exaggerating sleep hours; **Response bias** = emailing the survey
- (c) **Undercoverage** = exaggerating sleep hours; **Nonresponse** = emailing the survey; **Response bias** = only 60 respond
- (d) **Undercoverage** = using a school roster; **Nonresponse** = exaggerating sleep hours; **Response bias** = only 60 respond

CH 9. Producing Data: Experiments

- Experiments are used when the situation calls for a conclusion about whether a treatment causes a change in a response

Observation vs. Experiment

- Observation vs. Experiment
 - **Observational Study:** observes individuals and measures variables of interest but does not attempt to influence the responses
 - * The purpose of an observational study is to describe some group or situation
 - **Experiment:** deliberately imposes some **treatment** on individuals to observe their responses.
 - * The purpose of an experiment is to study whether the treatment causes a change in the response
 - * When our goal is to understand cause and effect, experiments are the preferred source for fully convincing data
- **Confounding**
 - Two variables (explanatory variables or lurking variables) are confounded when their effects on a response variable cannot be distinguished from each other
 - Observational studies of the effect of one variable on another often fail because the explanatory variable is confounded with lurking variables

Example: Coffee and Heart Disease

(a) Observational Study Setting

- Sample: 10,000 adults from a health system
- Variables: cups of coffee/day, smoking status, age, diet, exercise, heart disease records
- Compare heart disease rates across coffee-intake groups,

(b) Experiment Setting

- Sample: 2,000 adults who currently drink 0-2 cups/day
- Randomly assign
 - Group A: drink 2 cups of caffeinated coffee daily
 - Group B: drink 0 cups (or decaf placebo)
- Measure heart disease biomarkers and heart disease events of two groups and compare

Practice Question 9-1: More Education Improves Driving?

A recent study shows that among those over 25, as education level increased from less than high school, to high school grad, to some college, to college grad, the rate of motor vehicle crash deaths decreased

- (b) What are the explanatory and response variables?
- (b) Those with less education tend to drive cars that are older, have poorer crash test ratings, and have fewer safety features such as side airbags. Are the variables age of car, crash test rating, and presence of safety features explanatory variables, response variables, or lurking variables?

Subjects, Factors, and Treatments

- Basic vocabulary of experiments
 - Subjects: the individuals studied in an experiment
 - Factors: the explanatory variables in an experiment
 - **Treatment**: any specific experimental condition applied to the subjects.

Example: Cash transfer experiment

- Cash transfer experiments show up in economics because governments and NGOs have been giving people money to reduce poverty for decades. Economists want to know what the money actually causes
- Does giving cash cause improvements in outcomes like consumption, assets, food security, schooling, health, or psychological well-being? Random assignment is used to make “cash vs no cash” comparisons fair
 - **Subjects**: the individuals/households (or villages) enrolled in the study.
 - **Factor**: whether the subject is assigned to receive a cash transfer.
 - * Levels could be: *no transfer* vs. *transfer*.
 - * (Sometimes more levels: \$0, \$250, \$500; or *cash now* vs. *cash later*.)
 - **Treatment**: a specific condition applied to subjects, e.g.
 - * **Treatment group**: receive a \$500 cash transfer.
 - * **Control group**: receive \$0 (no cash transfer).

Randomized Comparative Experiments

- Randomized Comparative Experiment
 - An experiment that uses both comparison of two or more treatments of subjects to treatments is a randomized comparative experiment

Example: New Teaching Method vs. Standard (Education)

- **Population:** All students in a school district.
- **Sample:** $n = 200$ students.
- **Random assignment:** Assign each student at random to one of two groups:
 - **Group A (Treatment):** New teaching method.
 - **Group B (Control):** Standard teaching method.
- **Outcome measured:** Final exam score after 8 weeks.
- **Comparison:** Compare the average exam scores:

$$\bar{x}_A \quad \text{vs.} \quad \bar{x}_B$$

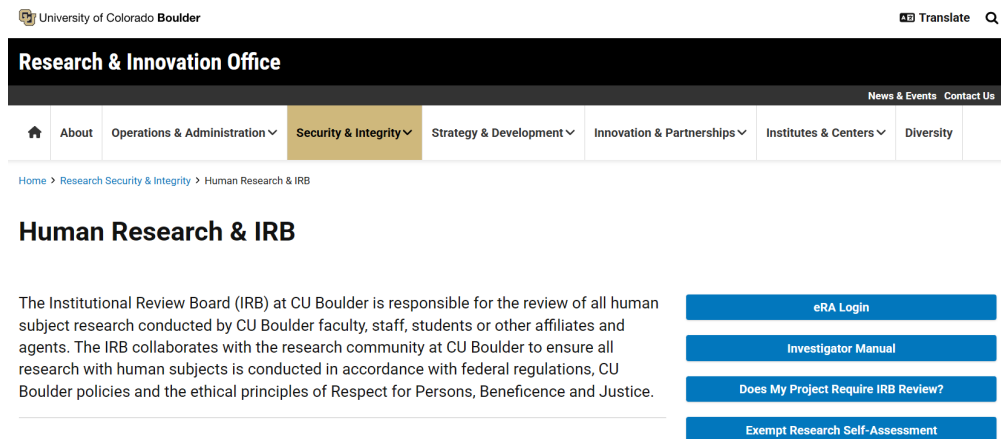
The Logic of Randomized Comparative Experiments

- Randomized comparative experiments are designed to give good evidence that differences in the treatments actually **cause** the differences we see in the response
 - (1) Random assignment of subjects forms groups that should be similar in all respects before the treatments are applied
 - (2) A comparative experiment with randomization ensures that influences other than the experimental treatments operate equally on all groups
 - (3) Therefore, differences in average response must be due either to the treatments or to the play of chance in the random assignment of subjects to the treatments
- Statistical Significance
 - An observed effect so large that it would rarely occur by chance is called **statistically significant**
 - If we observe statistically significant differences among the groups in a randomized comparative experiment, we have good evidence that the treatments actually caused these differences
 - **A statistically significant association in data from a well-designed experiment does imply causation**

CH 10. Data Ethics

- Basic Data Ethics for Human Subjects
 - All planned studies must be reviewed in advance by an *institutional review board (IRB)* charged with protecting the safety and well-being of the subjects
 - All individuals who are subjects in a study must give their *informed consent* before data are collected
 - All individual data must be kept *confidential*. Only statistical summaries for groups of subjects may be made public

Figure 54: IRB



Part III

From Data Production to Inference

- Generalizing the results of sampling or experiments to a larger group of individuals or a broader class of circumstances is one goal of statistical inference
- The conclusions of inference use the language of *probability*, the mathematics of chance

CH 12. Introducing Probability

- Why is probability, the mathematics of chance behavior, needed to understand statistics, and data science?
 - This is because deciding what is true about this population by observing only a portion of it involves some degree of uncertainty
- **Random samples eliminate bias from the act of choosing a sample, but they can still fail to perfectly agree with the true population proportion because of the variability that results when we choose at random**
 - If the variation when we take repeated samples from the same population is too great, we can't trust the results of any one sample

Example

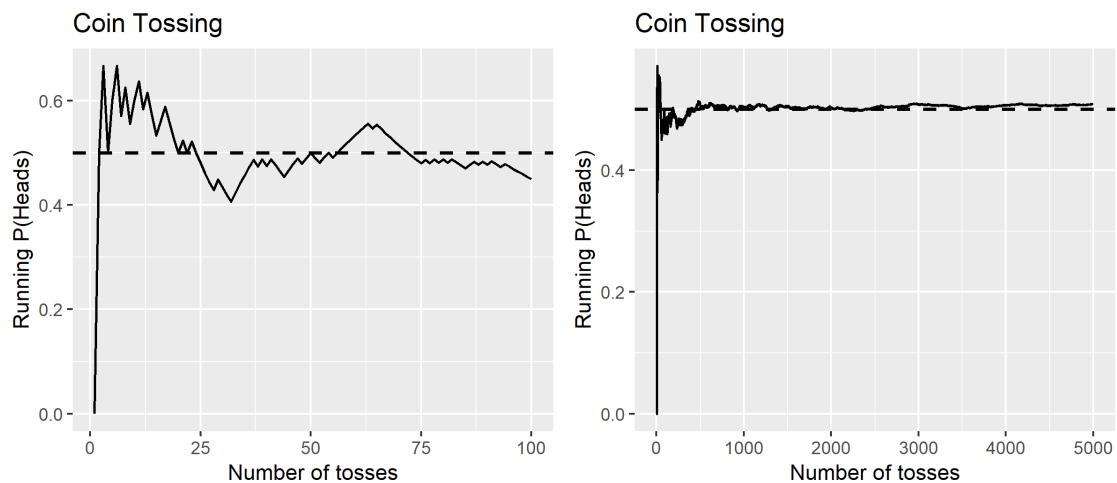
Suppose we have results from a Washington Post News poll. The poll took a random sample of 1000 adults. The poll found that 460 of the people in the sample said that someone in their household owned a gun. It means the sample proportion who said that someone in their household owned a gun is 0.46 (46%). If the sample were a simple random sample of all adults, based on the previous chapters

- It's a fact that 46% as an estimate of the unknown proportion in the population
- We don't know what percentage of all adults would say that someone in their household owned a gun, but we *estimate* that about 46% did at the time of the poll
- What if the second random sample found that the proportion of people who owned a gun is 60%?

The Idea of Probability

- **Chance behavior is unpredictable in the short run but has a regular and predictable pattern in the long run**
- For example, the result of coin tossing can't be predicted in advance because the result will vary when you toss the coin. But there is a regular pattern in the results, a pattern that emerges clearly only after many repetitions (Figure 55)
- The idea of probability is **empirical**
 - It is based on observation rather than theory
 - Probability describes what happens in very many trials, and we must actually observe many trials to pin down a probability

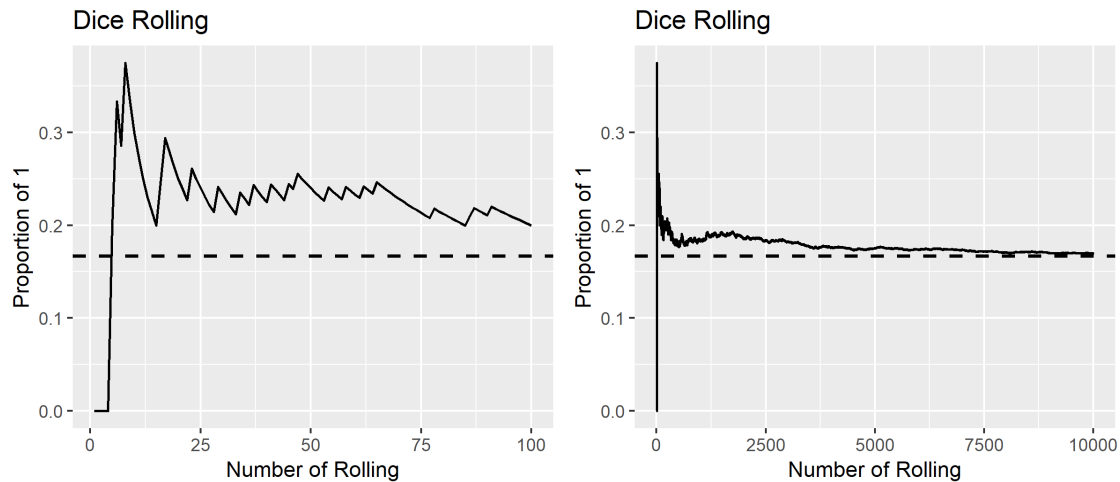
Figure 55: Coin Tossing



- Randomness and Probability
 - (1) Random: if individual outcomes are uncertain but there is nonetheless a regular pattern or distribution of outcomes in a large number of repetitions
 - (2) Probability: the probability of any outcome of a random phenomenon is the proportion of times the outcome would occur in a very long series of repetitions
 - For example, the proportion in a small or moderate number of tosses can be far from the probability (Figure 55)
 - Probability describes only what happens in the long run

Example: Dice Rolling

Figure 56: Dice Rolling



Probability Models

- Probability Models

- (3) The **sample space** S of a random phenomenon is the set of all possible outcomes
- (3) An **event** is an outcome or a set of outcomes of a random phenomenon. That is, an event is a subset of the sample space
- (3) A **probability model** is a mathematical description of a random phenomenon consisting of two parts: a sample space S and a way of assigning probabilities to events

- If all outcomes in a sample space are **equally likely**, we find the probability of any event by

$$P(\text{event}) = \frac{\text{number of ways the event could occur}}{\text{total number of outcomes in the sample space}}$$

Example

(a) One coin toss

- Sample space: $S = \{H, T\}$
- Total Outcomes: $|S| = 2$
- $P(H) = \frac{1}{2} = 0.5$

(b) One die roll

- Sample space: $S = \{1, 2, 3, 4, 5, 6\}$
- Total Outcomes: $|S| = 6$
- $P(\text{roll } 1) = \frac{1}{6} \approx 0.1667$

(c) Two dice rolls (ordered pair)

- Sample space:

$$S = \{(i, j) \mid i \in \{1, 2, 3, 4, 5, 6\}, j \in \{1, 2, 3, 4, 5, 6\}\}$$

- Total outcomes: $|S| = 6 \cdot 6 = 36$
- $P(\text{sum of two outcomes equals } 7)$:

$$P(i + j = 7) = \frac{6}{36} = \frac{1}{6} \approx 0.1667$$

Practice Question 12-1: One coin and one die

(a) What is the sample space of one coin and one die?

The sample space for the coin is $S_C = \{H, T\}$ and for the die is $S_D = \{1, 2, 3, 4, 5, 6\}$.
The combined sample space is:

$$S = \{(H, 1), (H, 2), (H, 3), (H, 4), (H, 5), (H, 6), (T, 1), (T, 2), (T, 3), (T, 4), (T, 5), (T, 6)\}.$$

Therefore, $|S| = 12$.

(b) What is the probability of getting Heads on the coin and an even number on the die?

Let A be the event “Heads and an even number”:

$$E = \{(H, 2), (H, 4), (H, 6)\}.$$

Then $|A| = 3$ and $|S| = 12$, so

$$P(A) = \frac{|A|}{|S|} = \frac{3}{12} = \frac{1}{4} = 0.25.$$

Probability Rules

- Probability Rules

1. The probability $P(A)$ of any event A satisfies $0 \leq P(A) \leq 1$
2. If S is the sample space in a probability model, then $P(S) = 1$
3. Two event A and B are **disjoint (mutually exclusive)** if they have no outcomes in common and so can never occur together. If A and B are disjoint,

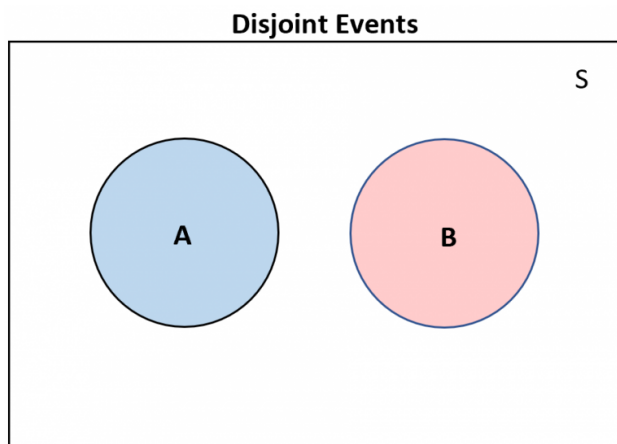
$$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B)$$

This is the **addition rule for disjoint events**(Figure 57)

4. For any event A ,

$$P(A \text{ does not occur}) = 1 - P(A)$$

Figure 57: Disjoint Events



Example

Draw one card from a standard deck of 52 cards.

Let

$$A = \{\text{draw an Ace}\}, \quad B = \{\text{draw a King}\}.$$

These events are disjoint because a card cannot be both an Ace and a King. Since there are 4 Aces and 4 Kings,

$$P(A) = \frac{4}{52}, \quad P(B) = \frac{4}{52}.$$

By the addition rule for disjoint events, the probability that the card you draw is an Ace or a King

$$P(A) + P(B) = \frac{4}{52} + \frac{4}{52} = \frac{8}{52} = \frac{2}{13}.$$

Practice Question 12-2

Choose an American adult aged 20 years or over at random. Define two events:

A = the person chosen is obese

B = the person chosen is overweight but not obese

According to the National Center for Health Statistics, $P(A) = 0.40$ and $P(B) = 0.32$

- (a) Explain why events A and B are disjoint
- (b) Say in plain language what the event “ A or B ” is
- (c) If C is the event that the person chosen has normal weight or less, what is $P(C)$?

Finite Probability Models

- Finite Probability Model
 - A probability model with a finite sample space is called **finite**
 - To assign probabilities in a finite model, list the probabilities of all the individual outcomes (countable)
 - These probabilities must be numbers between 0 and 1 that add to exactly 1
 - The probability of any event is the sum of the probabilities of the outcomes making up the event

Practice Question 12-3

Choose an adult age 18 or over in the United States at random and ask, “How many cups of coffee do you drink on average per day.” Call the response χ for short. Based on a large sample survey, here is a probability model for the answer you will get:

Number	0	1	2	3	4+
Probability	0.36	0.26	0.19	0.08	0.11

- (a) Verify that this is a valid finite probability model
- (b) Describe the event $\chi < 4$ in words. What is $P(\chi < 4)$?
- (c) Express the event “have at least one cup of coffee on an average day” in terms of χ . What is the probability of this event?

Continuous Probability Models

- **Continuous probability model**

- A **continuous** probability model assigns probabilities using areas under a density curve
 - * **Remember that any density curve has area exactly 1 underneath it**
- The probability that X falls in a range is the area under the curve over that interval:

$$P(a \leq X \leq b) = \int_a^b f(x) dx.$$

- In a continuous model, the probability of any *single exact value* is 0:

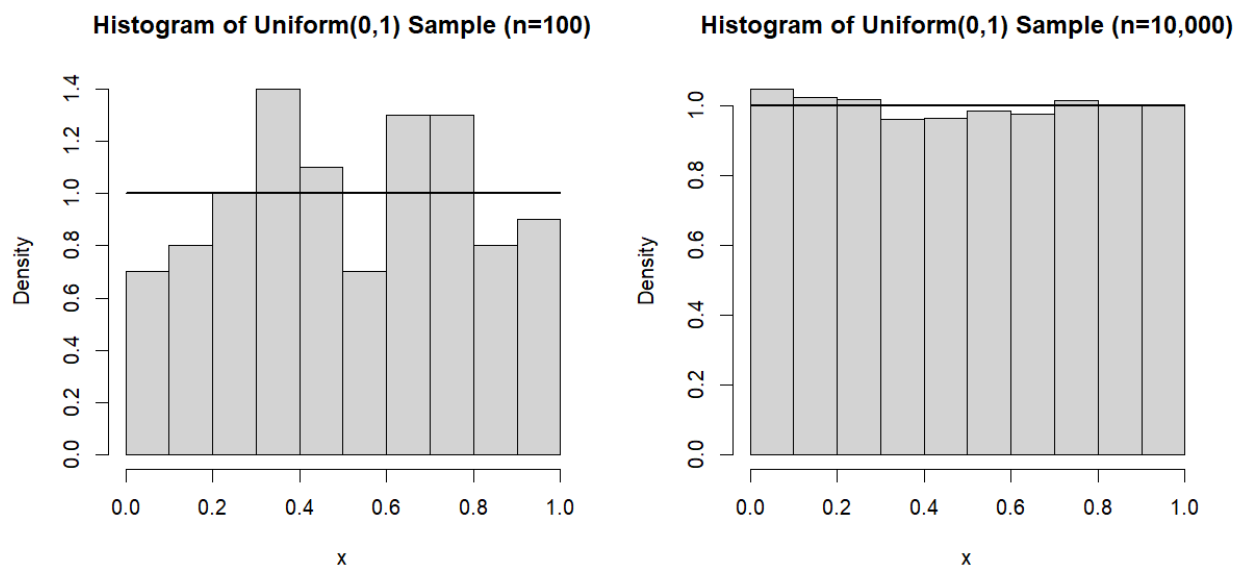
$$P(X = x) = 0 \quad \text{for any specific } x.$$

- This does *not* mean x is impossible; it means probability is assigned to **intervals**, not individual points (uncountable)

- **Uniform Distribution: $Y \sim \text{Uniform}$**

- A uniform distribution is a continuous probability distribution that assigns equal probability to every number (or to every interval of a given length) in the range of values over which it is defined

Figure 58: Uniform Density Curve

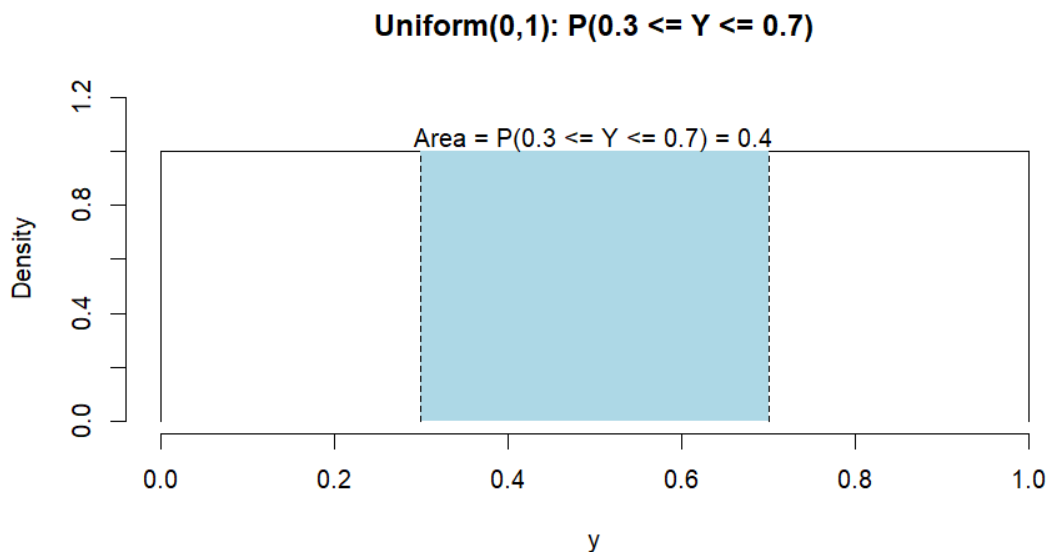


Example

Suppose that we want to choose a number at random between 0 and 1, allowing any number between 0 and 1 as the outcome (Y), and all possible outcomes to be equally likely (so $Y \sim \text{Uniform}(0, 1)$).

- S = all numbers between 0 and 1
- Probability of a single value: In a continuous model, $P(Y = 0.3) = 0, P(Y = 0.7) = 0, \dots$
- Probability of an interval: For a $\text{Uniform}(0, 1)$ distribution Y , $P(0.3 \leq Y \leq 0.7) = 0.7 - 0.3 = 0.4$

Figure 59: $Y \sim \text{Uniform}(0, 1)$

**Practice Question 12-4**

Let T be the bus lateness (in minutes) on a given morning. Assume the lateness is uniformly distributed between 0 and 10 minutes: $T \sim \text{Uniform}(0, 10)$. You will be late if the bus is more than 6 minutes late.

- What is $P(T = 6)$?
- What is the probability you will be late, i.e. $P(T > 6)$?

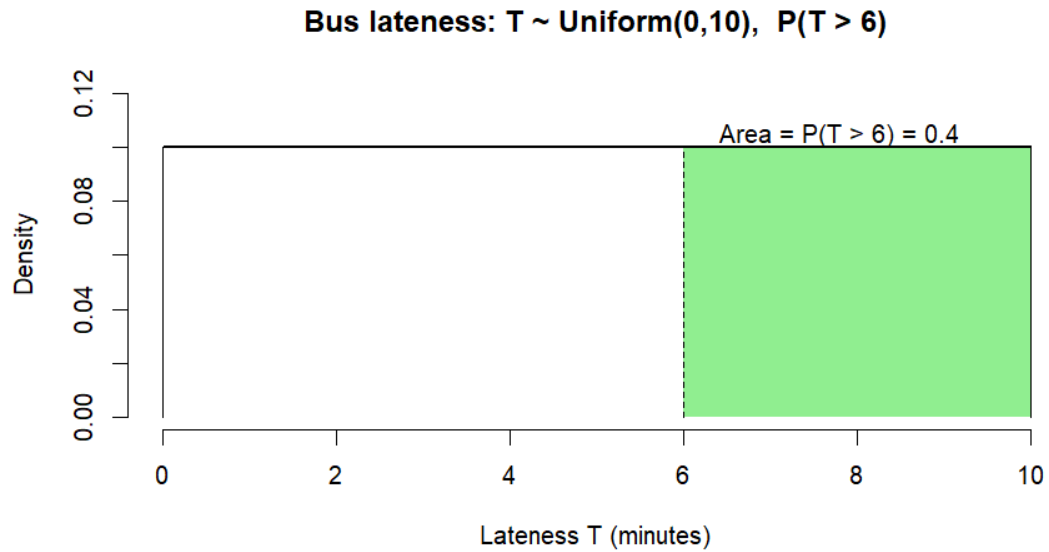
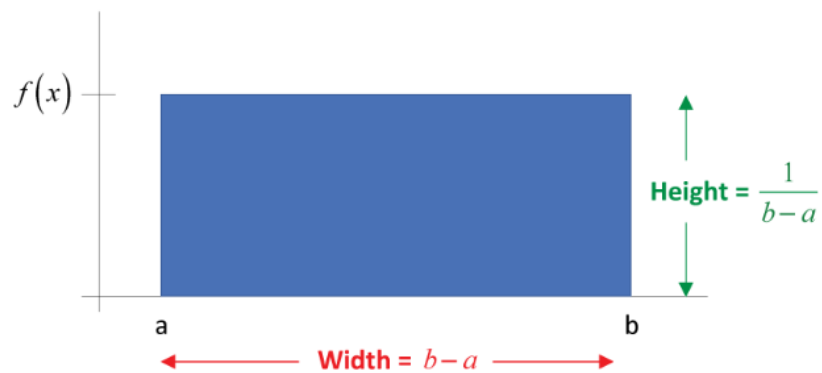
Figure 60: $T \sim \text{Uniform}(0, 10)$ 

Figure 61: Continuous Uniform Distribution Graph



$$\text{Area} = \text{width} \times \text{height}$$

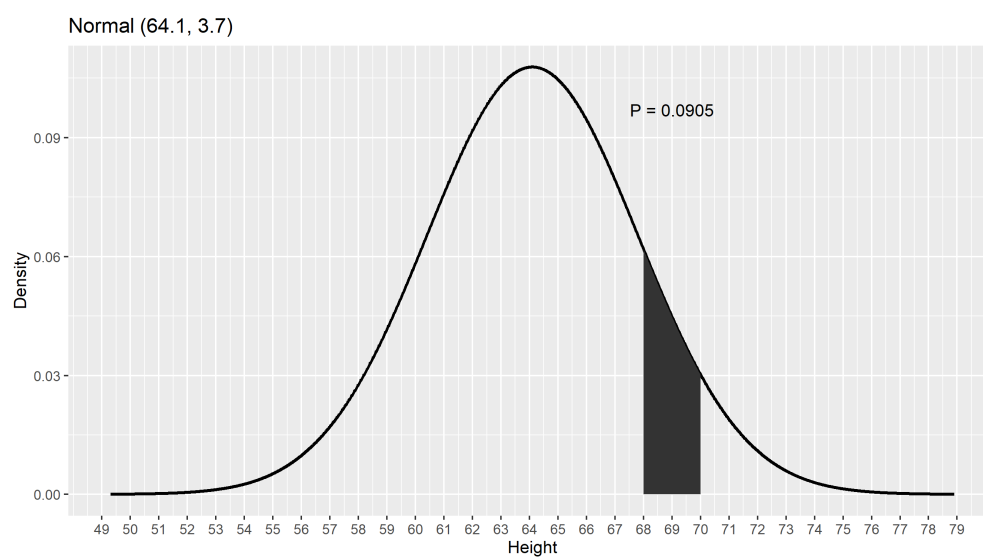
$$\text{Area} = (b - a) \left(\frac{1}{b - a} \right) = 1$$

Practice Question 12-5

What is the probability that a randomly chosen women aged 20-29 has a height between 68 and 70 inches? The height χ of the woman has the $N(64.1, 3.7)$.

Figure 62: z-table

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767

Figure 63: $\chi \sim N(64.1, 3.7)$, $P(68 \leq \chi \leq 70)$ 

Random Variable

- Random Variable
 - A random variable is a variable whose value is a numerical outcome of a random phenomenon
 - The probability distribution of a random variable χ tells us that values χ can take and how to assign probabilities to those values
 - We use denote random variables by using capital letters near the end of the alphabet, such as χ or Y
- A random variable χ and its distribution can be **discrete** or **continuous**
 - (1) The distribution of a **discrete random variable with finitely many possible values** gives the probability of each value
 - (2) A **continuous random variable** takes all values in some interval or numbers. A density curve describes the probability distribution of a continuous random variable

Additional Part: Basic Probability Theory

- Outcomes and Events
 1. Outcome: a specific result. For example, in a coin flip, an outcome is either H or T. A roll of a six-sided die has the six outcomes 1,2,3,4,5,6
 2. Sample Space (S): is the set of all possible outcomes. In a coin flip, the sample space is $S = \{H, T\}$. If two coins are flipped, the sample space is $S = \{HH, HT, TH, TT\}$
 3. Event (A): is a subset of outcomes in S . An example event from the roll of a die is $A = \{1, 2\}$
- Definitions. For events A and B :
 1. A is a **subset** of B , written $A \subset B$, if every element of A is an element of B
 2. The event with no outcomes $\emptyset = \{\}$ is called the **null** or **empty set**
 3. The **union** $A \cup B$ is the collection of all outcomes that are in either A or B (or both)
 4. The **intersection** $A \cap B$ is the collection of elements that are in both A and B
 5. The **complement** A^c of A are all outcomes in S which are not in A
 6. The events A and B are **disjoint** if they have no outcomes in common: $A \cap B = \emptyset$
 7. The events $A_1, A_2, A_3, \dots$ are a **partition** of S if they are mutually disjoint and their union is S

- Partitioning Theorem

- If $\{B_1, B_2, B_3, \dots\}$ is a partition of S , then for any event A ,

$$A = \bigcup_{i=1}^{\infty} (A \cap B_i)$$

The sets $(A \cap B_i)$ are mutually disjoint.

- Probability Function

- A function P which assigns a numerical value to events is called a **probability function** if it satisfies the following axioms of probability:

1. $P(A) \geq 0$
2. $P(S) = 1$
3. If $A_1, A_2, A_3, \dots$ are disjoint, then $P[\bigcup_{j=1}^{\infty} A_j] = \sum_{j=1}^{\infty} P(A_j)$

- Properties of the Probability Function

- For events A and B , the following properties hold:

1. $P(A^c) = 1 - P(A)$
2. $P(\emptyset) = 0$
3. $P(A) \leq 1$
4. Monotone Probability Inequality: If $A \subset B$, then $P(A) \leq P(B)$
5. Inclusion-Exclusion Principle: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
6. Boole's Inequality: $P(A \cup B) \leq P(A) + P(B)$
7. Bonferroni's Inequality: $P(A \cap B) \geq P(A) + P(B) - 1$

CH 13. General Rules of Probability

The General Addition Rule

- Addition Rule for Any Two Events

- For any two events A and B,

$$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B) - P(A \text{ and } B) = P(A) + P(B) - P(A \cap B)$$

Figure 64: Venn diagram of disjoint events A and B

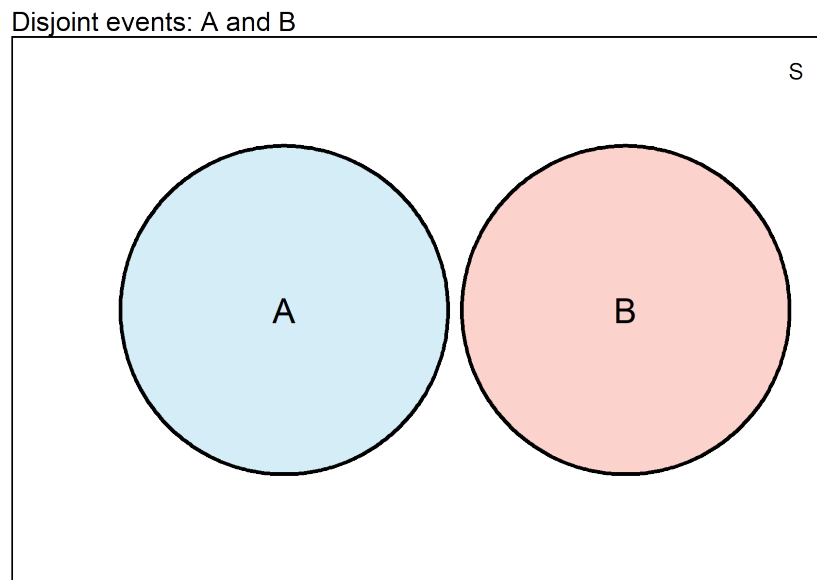
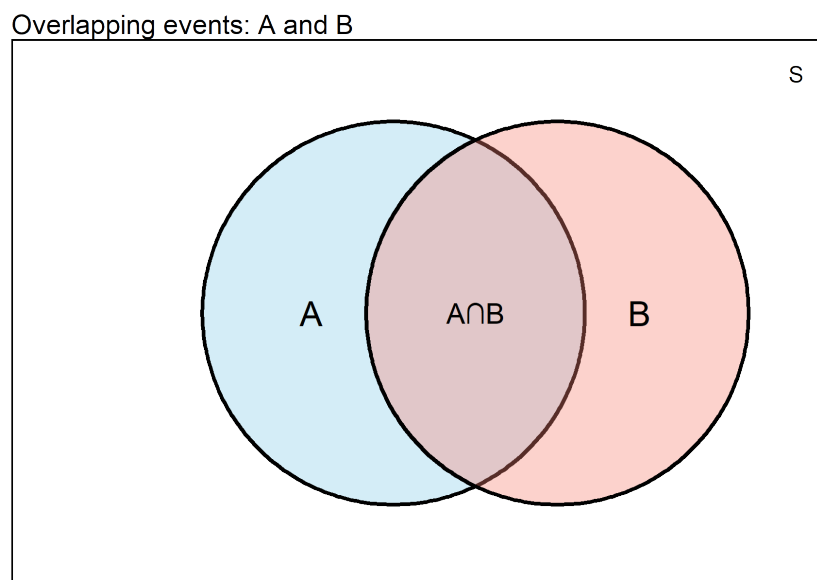


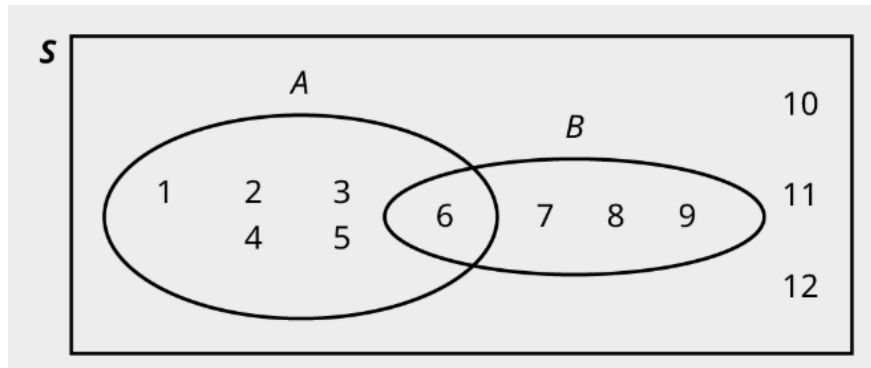
Figure 65: Venn diagram of joint events A and B



Example

Suppose an experiment has the outcomes $1, 2, 3, 4, \dots, 12$ where each outcome has an equal chance of occurring. Let event $A = \{1, 2, 3, 4, 5, 6\}$, and event $B = \{6, 7, 8, 9\}$. Then Venn diagram is as follows:

Figure 66

**Practice Question 13-1**

During the first week in 2019, the Pew Research Center tracked the types of English language YouTube videos posted. Let A be the event that the video posted involved games of some sort, including sports. Let B be the event that the video posted involved hobbies and skills, including sports. Pew Research Center finds that $P(A) = 0.30$, $P(B) = 0.31$, and $P(A \cup B) = 0.34$.

- Make a Venn diagram showing the events $\{A \text{ and } B\}$, $\{A \text{ and not } B\}$, $\{\text{not } A \text{ and } B\}$ and $\{\text{neither } A \text{ nor } B\}$
- Describe each of these events in words
- Find the probabilities of all four events

Independence and the Multiplication Rule

- **Multiplication Rule for Independent Events**

- Two events A and B are **independent** if knowing that one occurs does not change the probability that the other occurs.
- If A and B are independent,

$$P(A \text{ and } B) = P(A \cap B) = P(A) \cdot P(B)$$

- Independent Events A and B

- Example: Flip a fair coin and roll a fair die.
 - * Let A = “the coin lands Heads”.
 - * Let B = “the die shows a 6”.
- Example: Two separate flips of the same fair coin.
 - * Let A = “the first flip is Heads”.
 - * Let B = “the second flip is Heads”.
 - * Since the second flip is not affected by the first flip, A and B are independent.

- Dependent Events A and B

- Example: Draw two cards from a standard deck *without replacement*.
 - * Let A = “the first card drawn is an Ace”.
 - * Let B = “the second card drawn is an Ace”.
- Example: Student attendance and quiz score.
 - * Let A = “the student attended class last week”.
 - * Let B = “the student scores 90% or higher on the quiz”.

Example

Flip a fair coin and roll a fair six-sided die.

$$A = \{\text{coin lands Heads}\}, \quad B = \{\text{die shows an even number } (2, 4, 6)\}.$$

$$P(A) = \frac{1}{2}, \quad P(B) = \frac{3}{6} = \frac{1}{2}.$$

$$P(A \cap B) = P(\text{Heads and even}) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

Check independence:

$$P(A \cap B) = P(A)P(B) \implies \frac{1}{4} = \left(\frac{1}{2}\right) \left(\frac{1}{2}\right).$$

Therefore, A and B are independent.

1) Sample space Ω

Every outcome is a pair (coin, die):

$$\Omega = \{(H, 1), (H, 2), (H, 3), (H, 4), (H, 5), (H, 6), (T, 1), (T, 2), (T, 3), (T, 4), (T, 5), (T, 6)\}.$$

There are $2 \cdot 6 = 12$ equally likely outcomes, so each has probability $1/12$.

2) Define events A and B

$$A = \{\text{coin lands Heads}\} = \{(H, 1), (H, 2), (H, 3), (H, 4), (H, 5), (H, 6)\}$$

$$B = \{\text{die shows an even number (2, 4, 6)}\} = \{(H, 2), (H, 4), (H, 6), (T, 2), (T, 4), (T, 6)\}.$$

3) Compute $P(A)$

$$P(A) = \frac{|A|}{|\Omega|} = \frac{6}{12} = \frac{1}{2}.$$

4) Compute $P(B)$

$$P(B) = \frac{|B|}{|\Omega|} = \frac{6}{12} = \frac{1}{2}.$$

5) Find the intersection $A \cap B$

This means Heads AND even:

$$A \cap B = \{(H, 2), (H, 4), (H, 6)\}.$$

6) Compute $P(A \cap B)$

$$P(A \cap B) = \frac{|A \cap B|}{|\Omega|} = \frac{3}{12} = \frac{1}{4}.$$

7) Check independence

Events A and B are independent if

$$P(A \cap B) = P(A)P(B).$$

Compute:

$$P(A)P(B) = \left(\frac{1}{2}\right) \left(\frac{1}{2}\right) = \frac{1}{4}.$$

Since

$$P(A \cap B) = \frac{1}{4} = P(A)P(B),$$

we conclude that A and B are independent.

Practice Question 13-2

An instant lottery game gives you probability 0.02 of winning on any one play. Plays are independent of each other. If you play three times, the probability that you win on none of your play is about

- (a) 0.98
- (b) 0.94
- (c) 0.000008
- (d) 0.06

Practice Question 13-3

The probability that you win on one or more of your three plays of the game in the previous question is about

- (a) 0.02
- (b) 0.06
- (c) 0.94
- (d) 0.999992

Conditional Probability

- **Conditional Probability**

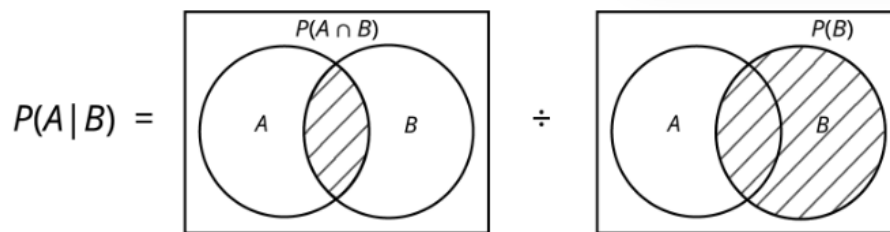
- The probability of one event A , given that another event B occurs
- The idea of a conditional probability $P(A|B)$ of one event A , given that another event B occurs, is the proportion of all occurrences of B for which A also occurs

- **Conditional Probability Formula**

- When $P(B) > 0$, the conditional probability of A , given B is

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Figure 67: $P(A|B)$



Example

A 2019 survey found 65% of adults had read a print book (A) in the preceding 12 months, 25% had read a book in digital format (B), and 18% had read both a print book and a book in digital format ($A \cap B$). If we choose an adult at random, the conditional probability that he or she had read a print book, given that he or she had read a digital book, is

$$P(\text{print}|\text{digital}) = P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.18}{0.25} = 0.72$$

- Is the event A and B are independent in this case?

Practice Question 13-4

Flip a fair coin and roll a fair six-sided die. Given that the die shows an even number (2, 4, or 6), what is the probability that the coin lands heads?

Let $A = \{\text{coin lands heads}\}$, $B = \{\text{die shows an even number}\}$.

$$P(A | B) = \frac{P(A \cap B)}{P(B)}.$$

$$P(A) = \frac{1}{2}, \quad P(B) = \frac{3}{6} = \frac{1}{2}.$$

$$P(A \cap B) = P(A) P(B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} \quad (\text{coin and die are independent}).$$

$$P(A | B) = \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{1}{2}.$$

Equivalently, learning that one event happened does not change the probability of the other:

$$P(A | B) = P(A) \quad \text{and} \quad P(B | A) = P(B),$$

The General Multiplication Rule

- **Multiplication Rule for any Two Events**

- The probability that both of two events A and B happen together can be found by

$$P(A \text{ and } B) = P(A \cap B) = P(A) \cdot P(B|A)$$

- Here, $P(B|A)$ is the conditional probability that B occurs, given the information that A occurs
- In words, this rule says that for both of two events to occur, first one must occur and then, given that the first event has occurred, the second must occur

- **Independent Events**

- Two events A and B that both have positive probability are **independent** if

$$P(B|A) = P(B) \text{ or } P(A|B) = P(A)$$

- This gives us a way to check whether two events are independent

Tree Diagrams

- Probability models often have several stages, with probabilities at each stage conditional on the outcomes of earlier stages
- **Tree Diagram**
 - A tree diagram is a graphical way to organize a probability model that has several stages in which leftmost branch segments represent probabilities of outcomes, and branches to the right represent conditional probabilities, given those outcomes

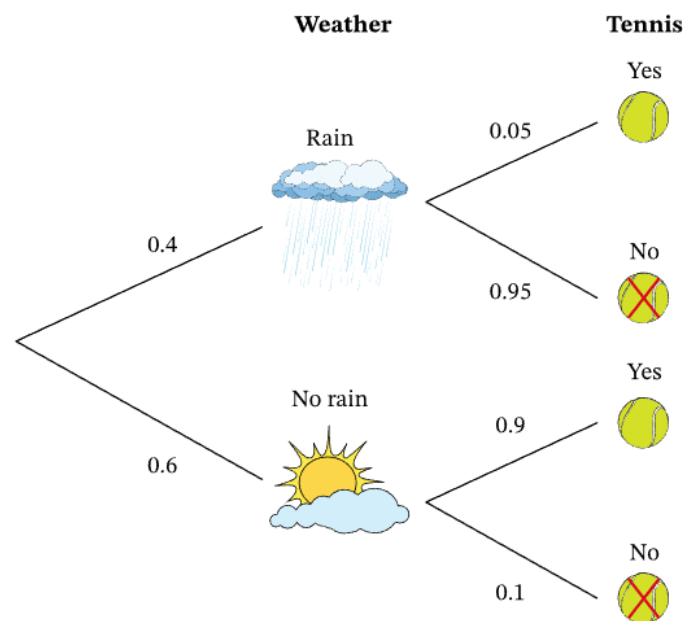
Example

Ramy has entered a tennis tournament and wants to practice as often as possible to prepare for it. Whether his coach Bassem will play tennis with him on any given day, however, depends on the weather.

Based on past data, the probability that it will rain on any given day before the tournament is 0.4. If it rains, the probability that Bassem will play tennis with Ramy is 0.05. If it does not rain, however, the probability that Bassem will play tennis with Ramy is 0.9.

- What is the probability that it rains and they play tennis?
- What is the probability that they play tennis?

Figure 68: Tree Diagram



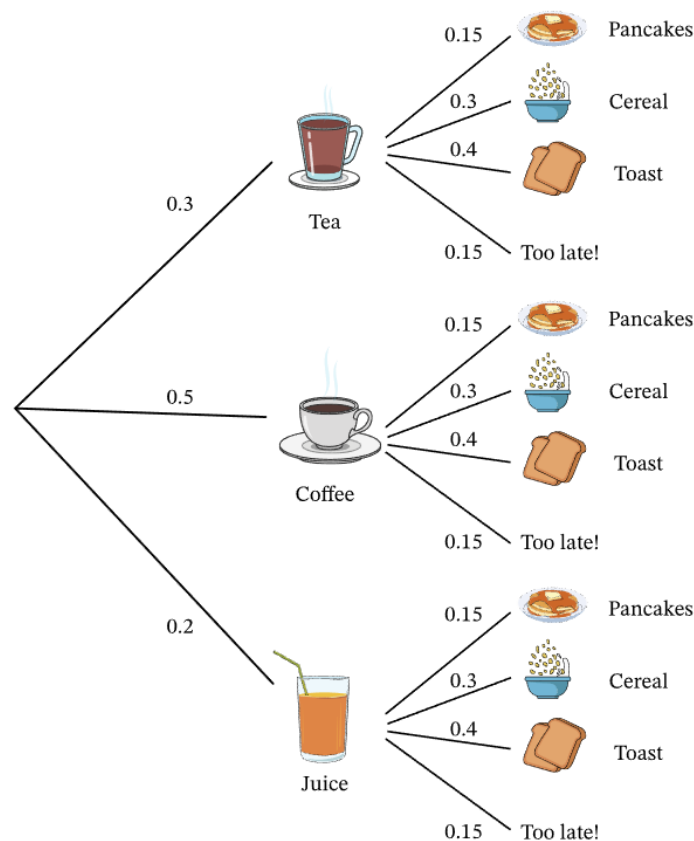
Practice Question 13-5

When Fady gets up, on any given morning, there is a 30% chance he will drink tea, a 50% chance he will choose coffee, and a 20% chance he will opt for juice.

For breakfast, Fady has toast 40% of the time, pancakes 15% of the time, cereal 30% of the time, and nothing if he gets up too late and has to rush (15%).

- What is the difference compared to the previous tree diagram?
- What is the probability that Fady has tea and cereal in the morning?
- What is the probability that Fady has toast in the morning?

Figure 69: Tree Diagram



Bayes' Rule

- Law of Total Probability

- An important relationship can be derived from the partitioning theorem which states that if $B_1, B_2, B_3, \dots$ is a partition of the sample space S , and $P(B_i) > 0$ for all i , then

$$A = \bigcup_{i=1}^{\infty} (A \cap B_i)$$

- Since the events $(A \cap B_i)$ are disjoint,

$$P(A) = \sum_{i=1}^{\infty} P(A \cap B_i) = \sum_{i=1}^{\infty} P(A|B_i)P(B_i)$$

- Bayes' Rule

- If $P(A) > 0$ and $P(B) > 0$, then

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|A^c)P(A^c)}$$

- * $P(A | B)$: **posterior** probability of A after observing B
- * $P(A)$: **prior** probability of A (belief before seeing B)
- * $P(A^c) = 1 - P(A)$: probability that A is **not true**
- * $P(B | A)$: **likelihood** of observing B if A is true
- * $P(B | A^c)$: likelihood of observing B if A is false
- * $P(B) = P(B | A)P(A) + P(B | A^c)P(A^c)$: **evidence** / normalizing constant
- Bayes' rule tells you how to update the probability of a claim when you get new evidence.
 - * You start with a guess about how likely A is. Then you see some information B . You should believe A more if B would be common when A is true, and less if B would also happen a lot when A is false.

Example.

A university counseling center uses a short screening questionnaire to flag students who may have *high stress*.

Let

$$A = \{\text{student has high stress}\}, \quad B = \{\text{student screens positive}\}.$$

Step 1: Prior (before seeing the screen). From past semesters, about 10% of students have high stress:

$$P(A) = 0.10.$$

Step 2: Screening accuracy (how evidence behaves).

$$P(B | A) = 0.80 \quad \text{and} \quad P(B | A^c) = 0.15.$$

(High-stress students screen positive 80% of the time; non-high-stress students still screen positive 15% of the time.)

Step 3: Posterior (after observing a positive screen).

$$P(A | B) = \frac{P(B | A)P(A)}{P(B | A)P(A) + P(B | A^c)P(A^c)}.$$

Compute $P(A^c) = 1 - P(A) = 1 - 0.10 = 0.90$. Then

$$P(A | B) = \frac{(0.80)(0.10)}{(0.80)(0.10) + (0.15)(0.90)} = \frac{0.08}{0.08 + 0.135} = \frac{0.08}{0.215} \approx 0.372.$$

Interpretation. Before seeing the questionnaire result, we think a randomly chosen student has high stress with probability $P(A) = 0.10$ (the *base rate*).

After we observe a *positive* screen (B), we update to

$$P(A | B) \approx 0.37.$$

This means: among students who screen positive, about 37% actually have high stress (and about 63% do not). In other words, a positive screen *raises* our belief from 10% to 37%, but it is not a confirmation.

Example (blind date: updating belief about being from Texas)

Story. You meet someone on a blind date. Before you notice anything, you have a baseline belief about whether they are from Texas. Then you see they are wearing a cowboy hat, and you update your belief.

Let

$$A = \{\text{she is from Texas}\}, \quad B = \{\text{she is wearing a cowboy hat}\}.$$

Step 1: Prior (base rate). Suppose (before seeing the hat) you think there is a 5% chance she is from Texas:

$$P(A) = 0.05 \quad \Rightarrow \quad P(A^c) = 0.95.$$

Step 2: How informative is the evidence? Assume:

$$P(B \mid A) = 0.20 \quad \text{and} \quad P(B \mid A^c) = 0.02.$$

Meaning:

- If she is from Texas, there is a 20% chance she wears a cowboy hat.
- If she is not from Texas, there is a 2% chance she wears a cowboy hat.

Step 3: Posterior (updated belief after seeing the hat).

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B \mid A)P(A) + P(B \mid A^c)P(A^c)} = \frac{(0.20)(0.05)}{(0.20)(0.05) + (0.02)(0.95)}.$$

$$P(A \mid B) = \frac{0.010}{0.010 + 0.019} = \frac{0.010}{0.029} \approx 0.345.$$

Interpretation (what is happening?). Before seeing the cowboy hat, your belief that she is from Texas was 5%. After seeing the hat, your belief increases to about 34.5%. This increase happens because the hat is *more likely* if she is from Texas than if she is not.

Is the cowboy hat a “good” piece of evidence? The evidence is good when it is *much more common* under A than under A^c . A quick way to see this is the likelihood ratio:

$$\frac{P(B \mid A)}{P(B \mid A^c)} = \frac{0.20}{0.02} = 10.$$

So wearing a cowboy hat is about 10× more likely if she is from Texas than if she is not, which makes it fairly strong evidence. However, it is still not proof: even after updating, the probability is not 100%.

Practice Question 13-6

A store manager wants to know whether a customer is a *loyalty program member* based on what they buy.

Let

$A = \{\text{customer is a loyalty program member}\}$, $B = \{\text{customer buys a premium product}\}$.

Suppose the following probabilities are known:

$$P(A) = 0.25, \quad P(B | A) = 0.50, \quad P(B | A^c) = 0.20.$$

Questions:

1. Compute the updated probability $P(A | B)$.
2. Compare $P(A | B)$ to the prior $P(A)$. Did observing B increase or decrease your belief that the customer is a loyalty program member?
3. Compute the likelihood ratio $\frac{P(B | A)}{P(B | A^c)}$ and explain (in one sentence) what it says about how informative B is.

CH 14. Binomial Distributions

The Binomial Setting and Binomial Distributions

- **The Binomial Setting**

1. There are a fixed number of observations, n
2. The n observations are all **independent**
 - That is, knowing the result of one observation does not change the probabilities we assign to other observations
3. Each observation fall into **one of just two outcomes “success” and “failure”**
4. The probability of a success, call it p , is the same for each observation
 - That is, the probability of a failure is $1 - p$

- **Binomial Distribution:** $\chi \sim \text{Bin}(n, p)$

- The count χ of successes in the binomial setting has the binomial distribution with parameters n and p
- The parameter n is the number of observations, and p is the probability of a success on any one observation
- The possible values of χ are the whole numbers from 0 to n

Example

- A basketball player shoots 5 free throws. Let X be the number she makes. If her probability of making a free throw is p , then

$$X \sim \text{Bin}(5, p).$$

- You plant 10 dogwood trees. Let Y be the number that live through the winter. If each tree survives with probability p , then

$$Y \sim \text{Bin}(10, p).$$

- You toss a coin 50 times. Let Z be the number of heads. If the probability of heads is p , then

$$Z \sim \text{Bin}(50, p).$$

Binomial Probabilities

• Binomial Coefficient

- The number of ways of arranging k successes among n observations is given by the **binomial coefficient**

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

for $k = 0, 1, 2, \dots, n$

- Factorial Notation $n!$

$$n! = n \times (n-1) \times (n-2) \times (n-3) \times \dots \times 3 \times 2 \times 1$$

• Binomial Probability

- If χ has the binomial distribution with n observations and probability p of success on each observation, the possible values of χ are $0, 1, 2, 3, \dots, n$. If k is any one of these values,

$$P(\chi = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Example

Flip a fair coin $n = 4$ times. Let X be the number of heads. We want to find $P(X = 2)$.

- In this example, we want the probability of getting exactly two heads in four coin tosses.
- The possible arrangements (outcomes) with exactly two heads are:

$$\{ \text{HHTT}, \text{HTHT}, \text{HTTH}, \text{THHT}, \text{THTH}, \text{TTHH} \}.$$

There are 6 of them.

- We can calculate this number (the binomial coefficient) using the formula:

$$\binom{4}{2} = \frac{4!}{2!(4-2)!} = \frac{4!}{2! \cdot 2!} = \frac{4 \times 3 \times 2 \times 1}{(2 \times 1)(2 \times 1)} = 6.$$

- Each arrangement has probability

$$(0.5)^2 \cdot (0.5)^2 = (0.5)^4 = \frac{1}{16},$$

since there are 4 tosses and each toss has probability 0.5.

- As a result,

$$P(X = 2) = 6 \times \frac{1}{16} = \frac{6}{16} = \frac{3}{8} = 0.375.$$

- We can also compute this directly using the binomial probability formula:

$$\begin{aligned} P(X = 2) &= \binom{4}{2} (0.5)^2 (1 - 0.5)^{4-2} \\ &= 6(0.5)^2 (0.5)^2 \\ &= 6(0.5)^4 \\ &= \frac{6}{16} = 0.375. \end{aligned}$$

Practice Question 14-1

Flip a fair coin $n = 4$ times. Let X be the number of heads. We want to find $P(X \leq 2)$

Practice Question 14-2

The blood types of successive children born to the same parents are independent and have fixed probabilities that depend on the genetic makeup of the parents. Each child born to a particular set of parents have probability 0.25 of having blood type O. If these parents have five children, what is the probability that exactly two of them have type O blood? The count of children with type O blood is a binomial random variable Y with $n = 5$ tries and probability $p = 0.25$ of a success on each try. We want $P(Y = 2)$

Binomial Mean and Standard Deviation

- Binomial Mean and Standard Deviation
 - If a random variable χ has the binomial distribution with number of observation n and probability of success p , the mean and standard deviation of χ are

$$\mu = n \cdot p$$

$$\sigma = \sqrt{n \cdot p \cdot (1 - p)}$$

Example

Suppose CU Boulder kicker Alejandro Mata has a success rate of 80% on kicks, and he attempts $n = 10$ kicks. Let X be the number of successful kicks. Then

$$X \sim \text{Bin}(10, 0.8).$$

- **Mean (expected number of successes):**

$$\mu = np = 10(0.8) = 8.$$

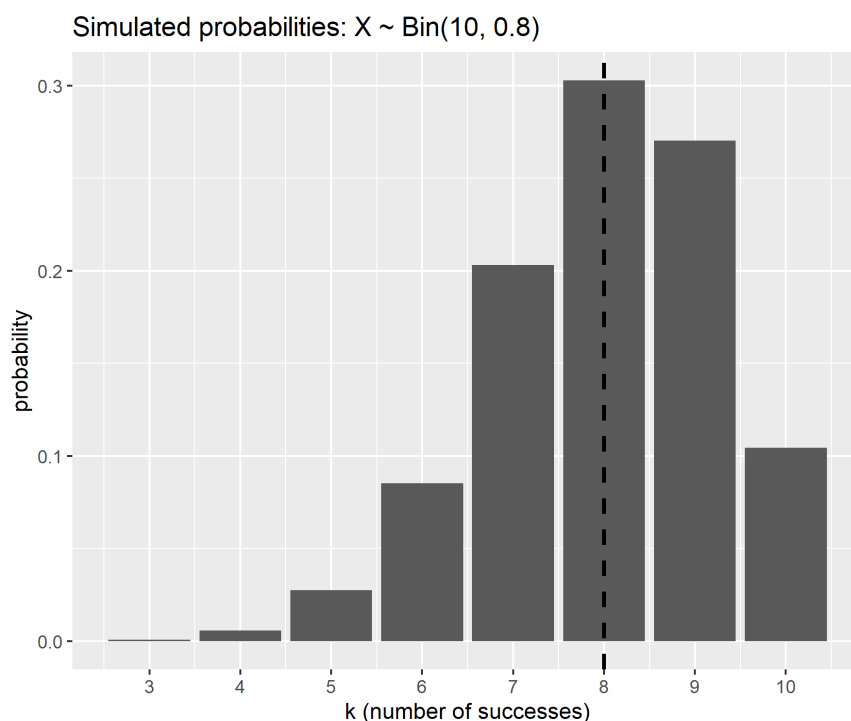
This means that, on average, he is expected to make about 8 kicks out of 10.

- **Standard deviation (typical variation around the mean):**

$$\sigma = \sqrt{np(1 - p)} = \sqrt{10(0.8)(0.2)} = \sqrt{1.6} \approx 1.265.$$

This measures how much the number of made kicks typically varies from the mean.

Figure 70: Probability histogram for $X \sim \text{Bin}(10, 0.8)$



Practice Question 14-3

When an opinion poll uses random digit dialing to select respondents for polls, the response rate is approximately 10% for people contacted by cell phone. You watch a pollster dial 20 cell numbers many times using random digit dialing

- What is the mean number of calls that yield a response?
- What is the standard deviation σ of the count of calls that yield a response?

The Normal Approximation to Binomial Distributions

- Normal Approximation for Binomial Distributions
 - Suppose that a count χ has the binomial distribution with n observations and success probability p
 - When n is large, the distribution of χ is approximately Normal

$$\chi \sim N(np, \sqrt{np(1-p)})$$

- As a rule of thumb, the Normal approximation can be used when n is so large that $np \geq 10$ and $n(1-p) \geq 10$

Figure 71: $X \sim \text{Bin}(20, 0.8)$

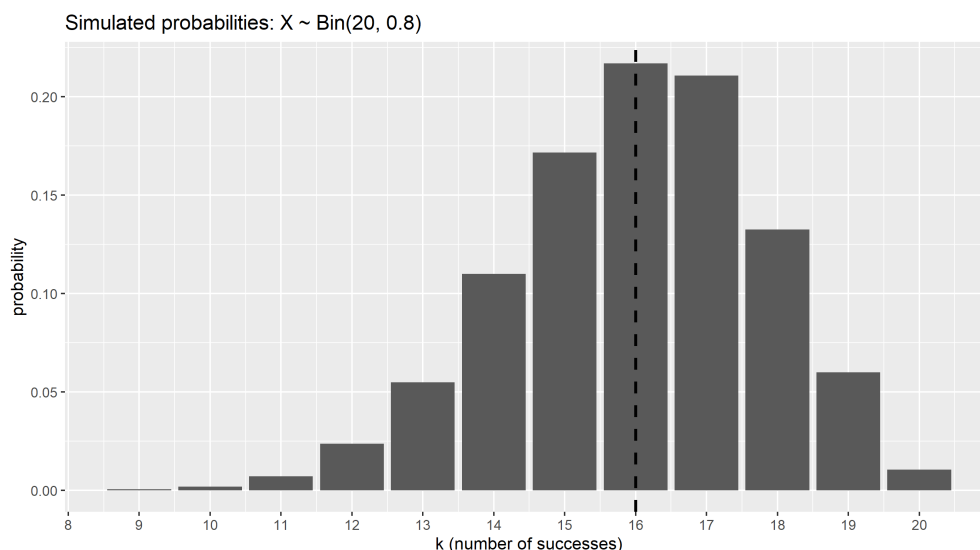
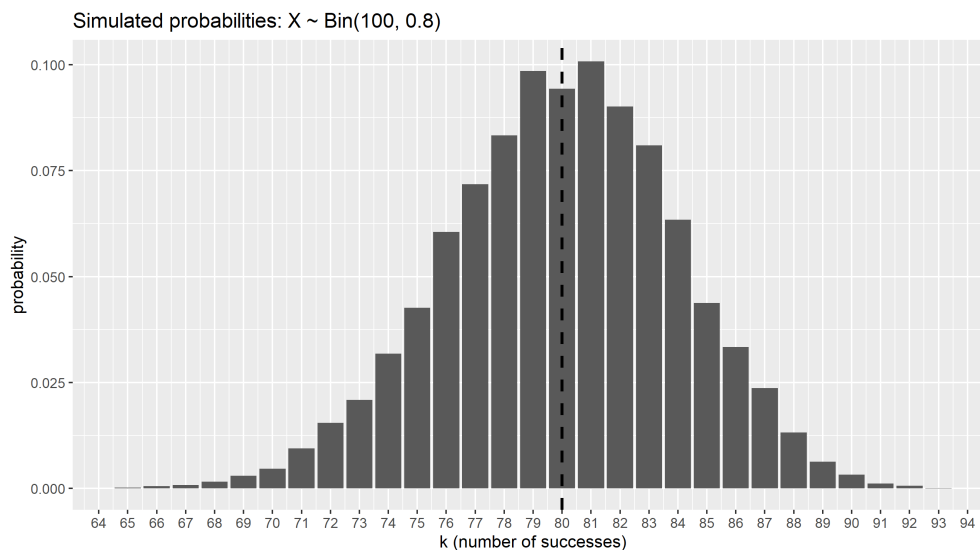


Figure 72: $X \sim \text{Bin}(100, 0.8)$ 

Example

Young adults aged 18 to 34 are more likely to be living with a romantic partner in their own home. Although approximately 32% of young adults now live with a parent. In a nationwide random sample of 1200 young adults, what is the probability that 400 or more live with a parent?

- Let X be the number of young adults (out of 1200) who live with a parent. Since each person either lives with a parent or not, and $p = 0.32$, we model

$$X \sim \text{Bin}(1200, 0.32).$$

We want to find $P(X \geq 400)$.

- Since $np = 1200(0.32) = 384 \geq 10$ and $n(1 - p) = 1200(0.68) = 816 \geq 10$, the normal approximation is appropriate. The mean and standard deviation are

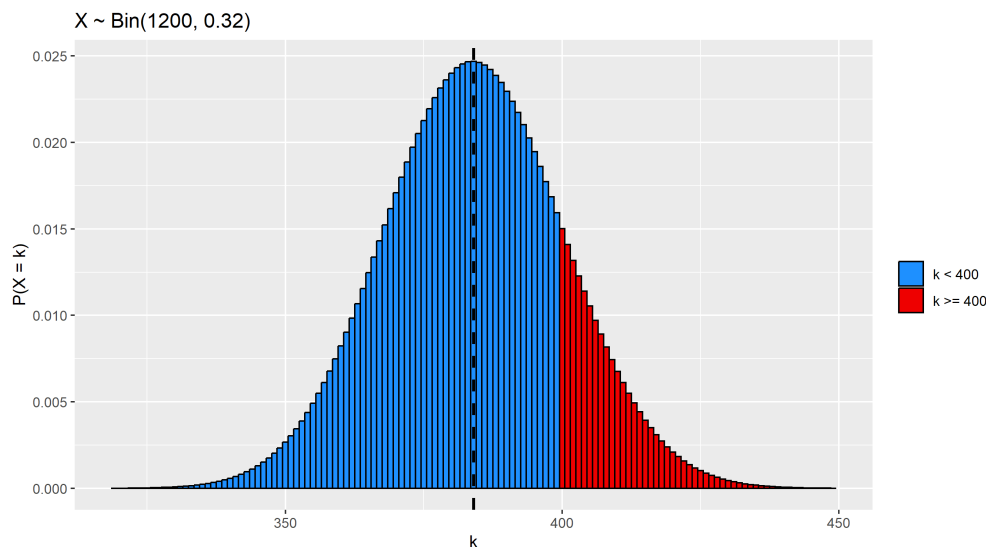
$$\mu = np = 384, \quad \sigma = \sqrt{np(1 - p)} = \sqrt{1200(0.32)(0.68)} \approx 16.159.$$

Therefore,

$$X \approx N(\mu = 384, \sigma = 16.159).$$

- Standardizing X gives a standard Normal variable Z . The probability we want is

$$P(X \geq 400) = P\left(Z \geq \frac{400 - 384}{16.159}\right) = P(Z \geq 0.99) = 1 - 0.8389 = 0.1611$$

Figure 73: $X \sim \text{Bin}(1200, 0.32)$ **Practice Question 14-4**

A basketball player makes a free throw about 50% of the time. She shoots $n = 100$ free throws. What is the probability that she makes 60 or more?

- **Write this question in statistical expressions.** Let X be the number of made free throws. Then

$$X \sim \text{Bin}(100, 0.50), \quad \text{and we want } P(X \geq 60).$$

- **Is it appropriate to approximate using a Normal distribution? Express it.** Check:

$$np = 100(0.50) = 50 \geq 10, \quad n(1 - p) = 100(0.50) = 50 \geq 10,$$

so the Normal approximation is appropriate. The mean and standard deviation are

$$\mu = np = 50, \quad \sigma = \sqrt{np(1 - p)} = \sqrt{100(0.50)(0.50)} = \sqrt{25} = 5.$$

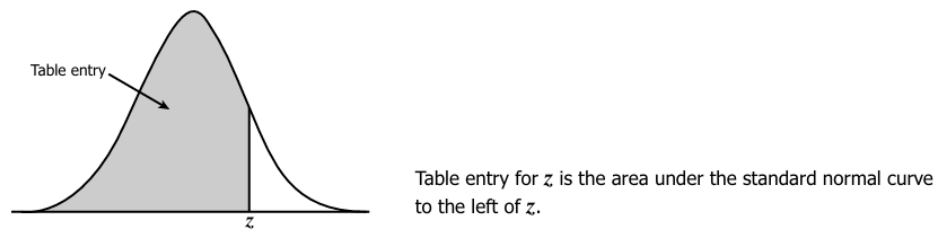
Therefore,

$$X \approx N(\mu = 50, \sigma = 5).$$

- **Find $P(X \geq 60)$ using a z-score.**

$$P(X \geq 60) \approx P(N \geq 60) = P\left(Z \geq \frac{60 - 50}{5}\right) = P(Z \geq 2) = 1 - 0.9772 = 0.0228 (2.28\%).$$

Figure 74



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857

CH 15. Sampling Distributions

- Probability is a tool we can use to generalize from data produced by random samples and randomized comparative experiments → some wider population
 - For example, how the mean of a sample can provide information about the mean of the population?
- In this chapter, we begin to formalize this process
- The reasoning of statistical inference rests on determining the answer to the question “How often would this method give an accurate answer if we used it very many times?”
- If our data come from random sampling or randomized comparative experiments, the laws of probability answer the question

Parameters and Statistics

- Parameter, Statistic
 1. **Parameter:** a number that describes the **population**. In practice, the value of a parameter is not known because we can rarely examine the entire population
 2. **Statistic:** a number that can be computed from the **sample data** without making use of any unknown parameters. In practice, we often use a statistic to estimate an unknown parameter
- Parameter, Statistics: Notation
 - Parameters
 - * μ : population mean
 - * σ : population standard deviation
 - Statistics
 - * $\bar{x}$: sample mean
 - * s : sample standard deviation

Example

The mean income of the sample of 128,759 household included in the 2018 CPS was $\bar{x}$ =\$90,021. This number is **statistic** because it describes this one CPS sample. The parameter μ , the mean income of all U.S. households is unknown.

Statistical Estimation and the Law of Large Numbers

- Statistical inference uses sample data to draw conclusions about the entire population
- We can describe the behavior of a sample statistic by using probability model that answer the question “What would happen if we did this many times?”
- **Law of Large Numbers (LLN)**
 - If we repeatedly draw observations at random from a population with mean μ , then as the sample size n increases, the sample mean $\bar{x}$ tends to get closer to μ .
 - If $X_1, X_2, \dots$ are independent draws from the same population with finite mean μ , then as $n \rightarrow \infty$, the sample mean

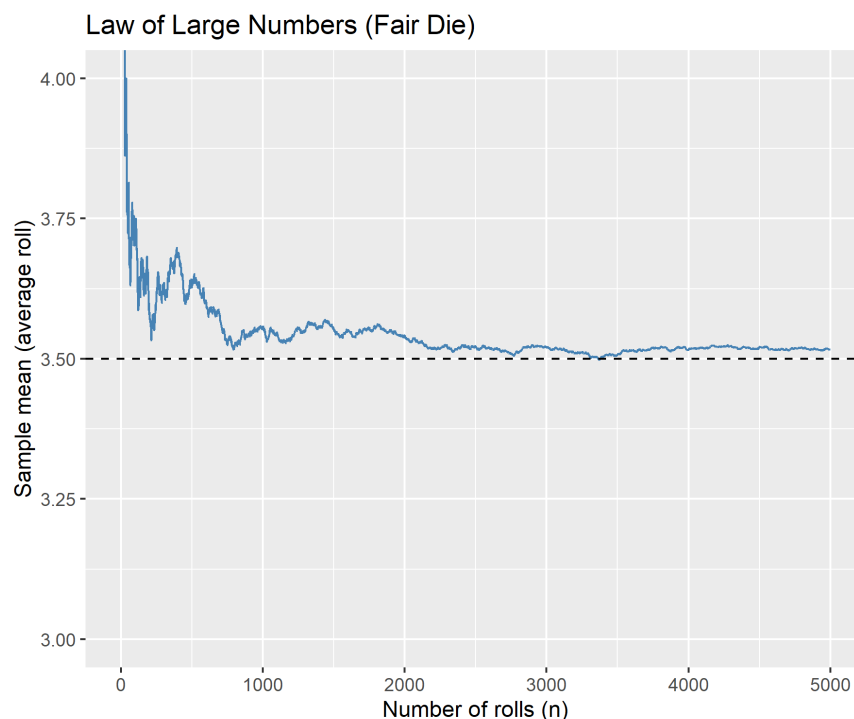
$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

becomes closer to μ (with high probability).

Example

For a fair die, the population mean is $\mu = 3.5$. As shown in the graph, the running sample mean $\bar{X}_n$ fluctuates for small n , but as the number of rolls increases, it becomes more stable and approaches 3.5, illustrating the Law of Large Numbers.

Figure 75: Example of LLN



Sampling Distributions

- **Population Distribution, Sampling Distribution**

- **Population Distribution:** is the distribution of values of the variable among all the individuals in the population
- **Sampling Distribution:** is the distribution of values taken by the statistic in all possible samples of the same size from the same population

The Sampling Distribution of $\bar{X}$

- Mean and Standard Deviation of a Sample Mean

- Suppose that $\bar{X}$ is the mean of an SRS of size n drawn from a population with mean μ and standard deviation σ .
- Then the sampling distribution of $\bar{X}$ has

mean: μ

standard deviation: $\frac{\sigma}{\sqrt{n}}$

- These facts about the mean and the standard deviation of the sampling distribution of $\bar{X}$ are true for **any** population, not just for the population has Normal Distribution

- Implications for statistical inference

1. The mean of the statistic $\bar{X}$ is always equal to the mean μ of the population
 - That is, the sampling distribution of $\bar{X}$ is centered at μ
 - In repeated sampling, $\bar{X}$ will sometimes fall above the true value of the parameter μ and sometimes below, but there is no systematic tendency to overestimate or underestimate the parameter
 - We say that the statistic $\bar{X}$ is an **unbiased estimator** of the parameter μ
2. Averages are less variable than individual observations
 - If individual observations have standard deviation σ , then sample means $\bar{X}$ from samples of size n have standard deviation $\frac{\sigma}{\sqrt{n}}$.
3. The results of large samples are less variable than the results of small samples
 - For example, if $n_1 = 10$, then $SD(\bar{X}) = \sigma/\sqrt{10}$. If the sample size increases to $n_2 = 100$, then $SD(\bar{X}) = \sigma/\sqrt{100}$, which is smaller

- Unbiased Estimator
 - An unbiased estimator is a statistic used to estimate a parameter where the mean of the statistic's sampling distribution equals the true value of the population parameter being estimated
- Sampling Distribution of a Sample Mean for Normally Distributed Populations
 - If population observations have the $N(\mu, \sigma)$ distribution, then the sample mean $\bar{X}$ of an SRS of size n has the $N(\mu, \sigma/\sqrt{n})$
 - If the population distribution is Normal $\rightarrow$ the sampling distribution of the sample mean is Normal, regardless of the sample size n

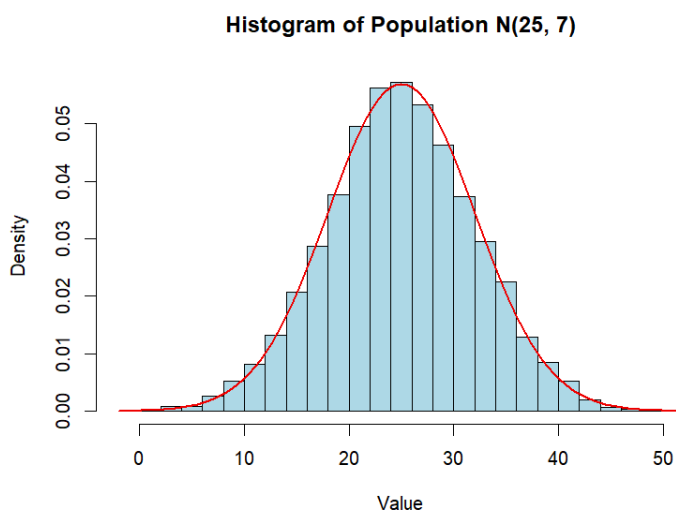
Example

Suppose a population of size $N = 10,000$ is generated from a normal distribution $X \sim N(25, 7)$.¹ Figure 76 shows the population distribution.

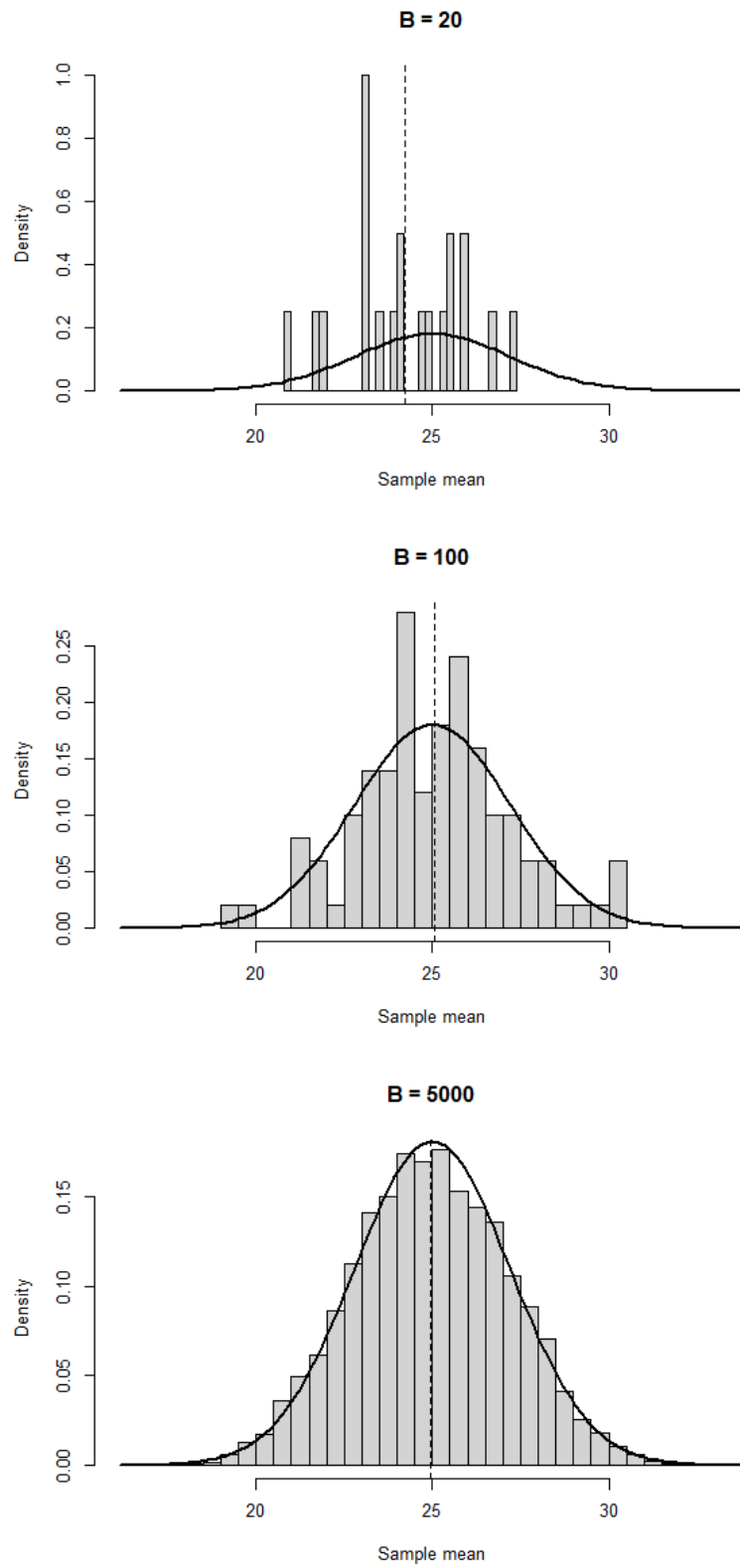
Next, we repeatedly draw samples of size $n = 10$ (with replacement) from this population. For each sample, we compute the sample mean $\bar{X}$. Repeating this procedure B times produces a simulated sampling distribution of $\bar{X}$. Figure 77 shows the sampling distributions for $B = 20$, $B = 200$, and $B = 5000$.

As B increases, the histogram of $\bar{X}$ becomes smoother and more closely matches the theoretical sampling distribution, which is approximately normal with mean 25 and standard deviation $7/\sqrt{10}$.

Figure 76: Population Distribution



¹Here $N(25, 7)$ means mean 25 and standard deviation 7.

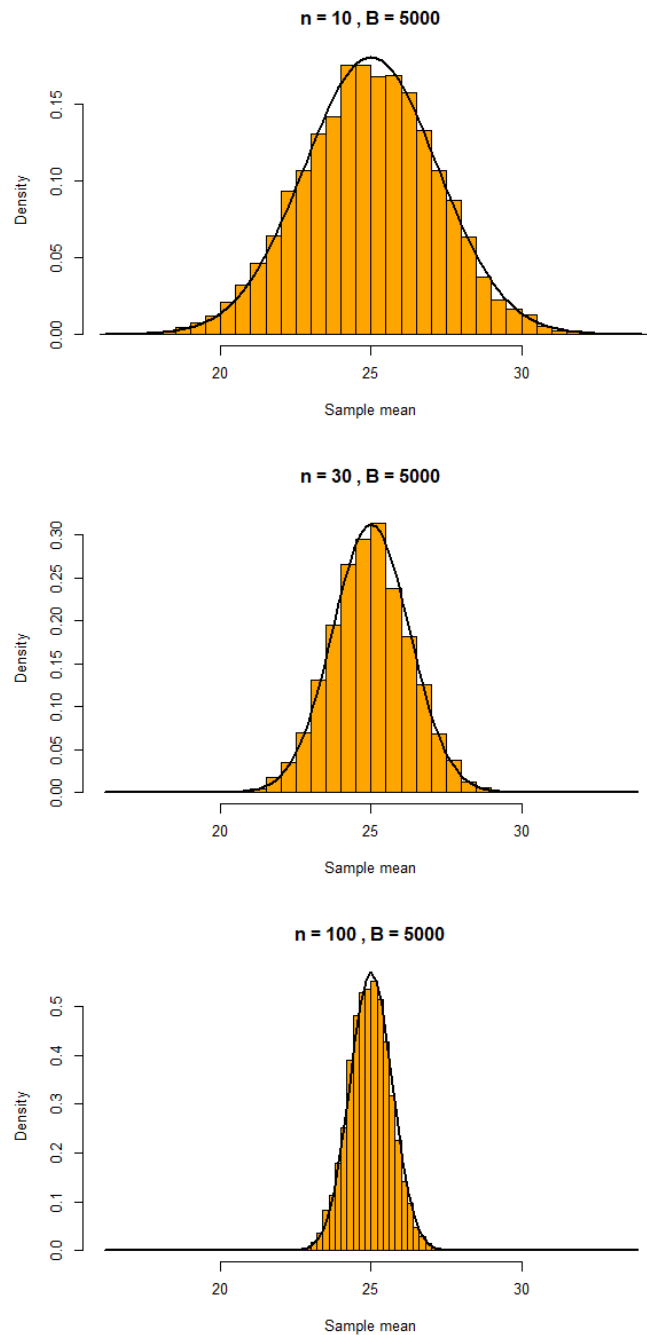
Figure 77: Sampling distributions of $\bar{X}$ for different number of repeated samples

Example (continued)

Next, the figure below shows how the variability (standard deviation) of the sampling distribution of the sample mean changes as the sample size increases ($n = 10$, $n = 30$, and $n = 100$), while the number of repeated samples is held fixed at $B = 5000$.

As n increases, the sampling distribution becomes narrower because $SD(\bar{X}) = \sigma/\sqrt{n}$.

Figure 78: Sampling distributions of $\bar{X}$ for different sample sizes n (with B fixed).



The Central Limit Theorem (CLT)

- But what is the shape of the sampling distribution when the population distribution is not Normal?
- **It is a remarkable fact that as the sample size (n) increases, the distribution of $\bar{X}$ changes shape: it looks less like that of the population and more like a Normal Distribution**
- This is true no matter what shape the population distribution has, as long as the population has a finite standard deviation σ

- **Central Limit Theorem (CLT)**

- Draw an SRS of size n from any population with mean μ and finite standard deviation σ .
- The central limit theorem says that when n is large, the sampling distribution of the sample mean $\bar{X}$ is approximately Normal:

$$\bar{X} \approx N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

- The central limit theorem allows us to use Normal probability calculations to answer questions about sample means from many observations even when the population distribution is **not** normal
- How large a sample size n is needed for $\bar{X}$ to be close to Normal depends on the population distribution
 - * More observations are required if the shape of the population distribution is far from Normal

Figure 79: Central Limit Theorem (right-skewed)

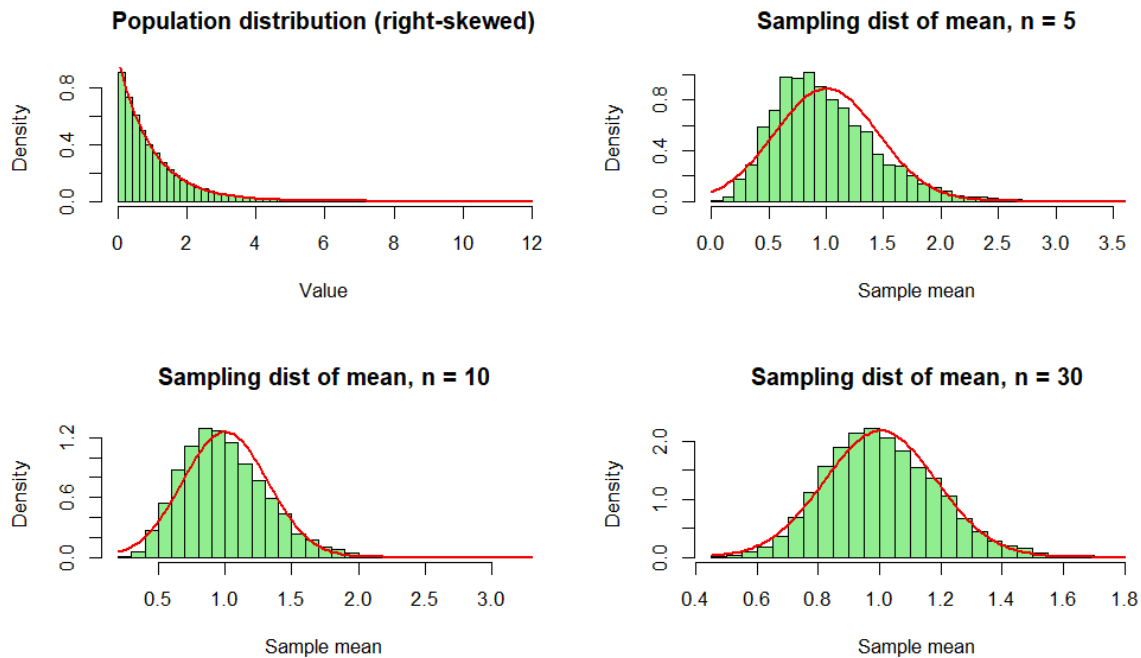
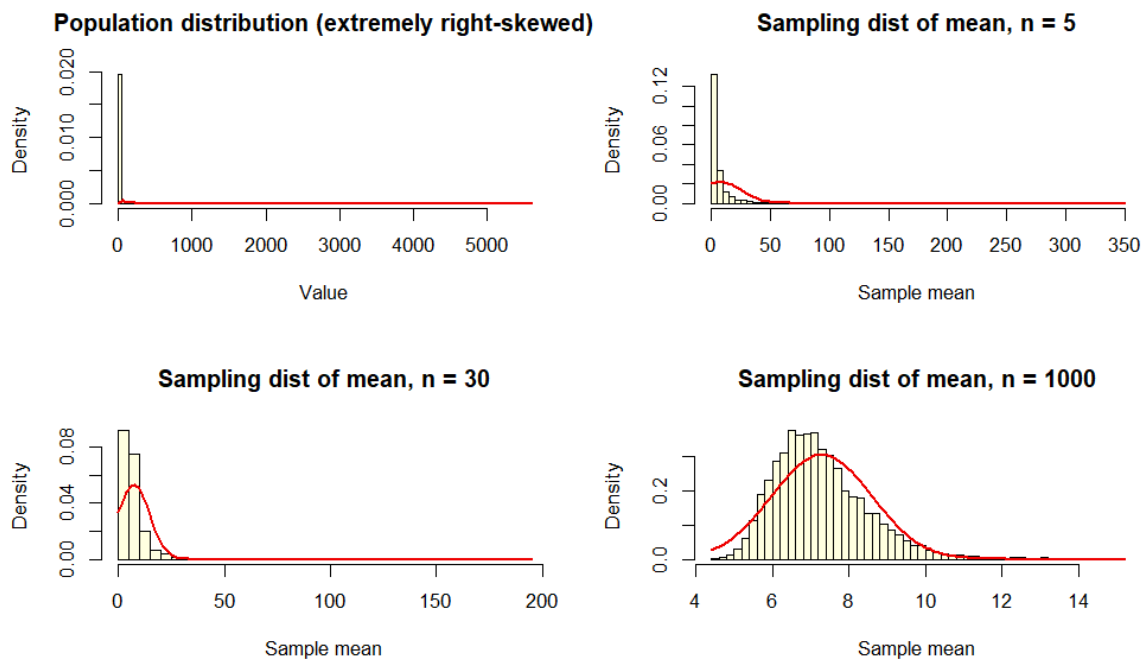


Figure 80: Central Limit Theorem (extremely right-skewed)



- Thinking about Sample Means ($\bar{X}_1, \bar{X}_2, \bar{X}_3, \dots$)
 1. Means of random samples are **less variable** than individual observations
 2. Means of random samples are **more Normal** than individual observations

Example

An economist is studying how long it takes freelancers to get paid after submitting an invoice. For a certain platform, the delay X (in weeks) is well modeled by an exponential distribution with $\mu = 1$ week and $\sigma = 1$ week. In a monthly audit, the economist randomly samples $n = 70$ invoices and computes the average payment delay $\bar{X}$. The platform triggers an internal review if the average delay exceeds 1.1 weeks. What is the probability that an internal review is triggered (i.e., $P(\bar{X} > 1.1)$)?

Solution. By the Central Limit Theorem,

$$\bar{X} \approx N\left(\mu, \frac{\sigma}{\sqrt{n}}\right) = N\left(1, \frac{1}{\sqrt{70}}\right).$$

Standardizing,

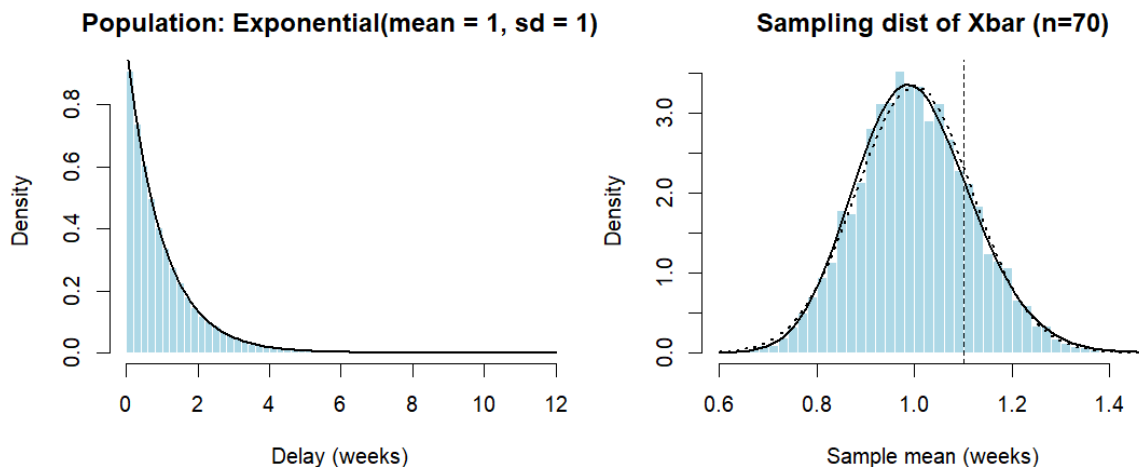
$$P(\bar{X} > 1.1) \approx 1 - \Phi\left(\frac{1.1 - 1}{1/\sqrt{70}}\right) = 1 - \Phi(0.10\sqrt{70}) = 1 - \Phi(0.8367).$$

Thus,

$$P(\bar{X} > 1.1) \approx 0.201.$$

Conclusion. The probability of triggering an internal review is approximately 0.20 (CLT), and the exact value is about 0.198.

Figure 81



Practice Question 15-1

A population is normally distributed with mean $\mu = 50$ and standard deviation $\sigma = 10$. A random sample of size $n = 25$ is taken, and the sample mean is denoted by $\bar{X}$. Find the probability that the sample mean exceeds 54, i.e.,

$$P(\bar{X} > 54).$$

Solution. Because the population is normally distributed, the sample mean is exactly normal:

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right) = N\left(50, \frac{10}{\sqrt{25}}\right) = N(50, 2).$$

Standardize:

$$P(\bar{X} > 54) = P\left(Z > \frac{54 - 50}{2}\right) = P(Z > 2) = 1 - \Phi(2).$$

Using $\Phi(2) \approx 0.9772$,

$$P(\bar{X} > 54) \approx 1 - 0.9772 = 0.0228.$$

Practice Question 15-2

A population has mean $\mu = 20$ and standard deviation $\sigma = 8$ (the population distribution is not necessarily normal). A random sample of size $n = 64$ is taken, and the sample mean is denoted by $\bar{X}$. Using the Central Limit Theorem, approximate the probability

$$P(\bar{X} < 18).$$

Solution. Because the population distribution is not necessarily normal, we use the Central Limit Theorem (CLT). For $n = 64$, the sampling distribution of the sample mean is approximately

$$\bar{X} \approx N\left(\mu, \frac{\sigma}{\sqrt{n}}\right) = N\left(20, \frac{8}{\sqrt{64}}\right) = N(20, 1).$$

Standardize:

$$P(\bar{X} < 18) \approx P\left(Z < \frac{18 - 20}{1}\right) = P(Z < -2) = \Phi(-2).$$

Therefore,

$$P(\bar{X} < 18) \approx 0.0228.$$

Practice Question 15-3

An insurance company knows that in the entire population of millions of apartment owners, the mean annual loss from damage is $\mu = \$150$, and the standard deviation of the loss is $\sigma = \$300$. The distribution of losses is strongly right-skewed: most policies have \$0 loss, but a few have large losses. If the company sells 10,000 policies, can it safely base its rates on the assumption that its average loss will be no greater than \$160?

Solution. Let $X_1, \dots, X_{10000}$ be the annual losses for the $n = 10,000$ policies, and let

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

We are told the population mean and standard deviation are $\mu = 150$ and $\sigma = 300$, and the distribution is strongly right-skewed. Because $n = 10,000$ is very large (and the variance is finite), the Central Limit Theorem implies the sampling distribution of $\bar{X}$ is approximately normal:

$$\bar{X} \approx N\left(\mu, \frac{\sigma}{\sqrt{n}}\right) = N\left(150, \frac{300}{\sqrt{10000}}\right) = N(150, 3).$$

To assess whether the average loss will be greater than \$160, compute

$$P(\bar{X} > 160) \approx 1 - \Phi\left(\frac{160 - 150}{3}\right) = 1 - \Phi\left(\frac{10}{3}\right).$$

Since $\frac{10}{3} \approx 3.33$ and $\Phi(3.33) \approx 0.99957$,

$$P(\bar{X} > 160) \approx 1 - 0.99957 = 0.00043.$$

Conclusion. Under the CLT assumptions (e.g., losses are independent across policies), the probability that the company's average loss exceeds \$160 is about 0.00043 (roughly 0.043%). So it is very reasonable to assume the average loss will be no greater than \$160.

CH 16. Confidence Intervals: The Basics

- **Statistical Inference**

- Statistical inference provides methods for drawing conclusions about a population from sample data
- Because a different sample might lead to different conclusions, we can't be certain that our conclusions are correct
- Statistical inference uses the language of probability to say how trustworthy our conclusions are

- **Simple Conditions for Inference About a Mean**

1. We have a simple random sample (SRS) from the population of interest. There is no nonresponse or other practical difficulty. The population is large compared to the size of the sample
2. The variable we measure has an exactly Normal Distribution $N(\mu, \sigma)$ in the population
3. We don't know the population mean μ . But we do know the population standard deviation σ

The Reasoning of Statistical Estimation

Example

A government report gives data for 936 men aged 20-29 years. The mean Body Mass Index (BMI) of these 936 men was $\bar{X} = 27.2$. On the basis of this sample, we want to estimate the mean BMI μ in the population of all 23.2 million American men in this age group.

- **Simple Condition (Assumption)**

- This sample is SRS and from a Normal Population
- Standard deviation of population $\sigma = 11.6$

- **Reasoning of statistical estimation**

1. To estimate μ , we use $\bar{X} = 27.2$ of the random sample with $n = 936$. We don't expect $\bar{X}$ to be exactly equal to μ , so we want to say how accurate this estimate is
2. We know that sampling distribution of $\bar{X}$ has the Normal distribution $N(\mu, \sigma/\sqrt{n})$. So the this sampling distribution has standard deviation

$$\frac{\sigma}{\sqrt{n}} = \frac{11.6}{\sqrt{936}} = 0.4$$

3. Based on the 68-95-99.7 rule for Normal Distribution, we know that $\bar{X}$ is within 2 standard deviation of the mean μ in 95% of all samples
 - Thus, for 95% of all samples of size 936, the distance between the sample mean $\bar{X}$ and the population mean μ is less than 0.8.
 - If we estimate that μ lies somewhere in the interval from $\bar{X} - 0.8$ to $\bar{X} + 0.8$, we'll be right for 95% of all possible samples

$$\bar{X} - 0.8 = 27.2 - 0.8 = 26.4$$

$$\bar{X} + 0.8 = 27.2 + 0.8 = 28.0$$

4. Because we got the interval 26.4 to 28.0 from a method that captures the population mean for 95% of all possible samples, we say that we are 95% confident that the mean μ of all young men is some value in that interval - no lower than 26.4 and no higher than 28.0

Practice Question 16-1

An SRS of 400 high school seniors gained an average of $\bar{X} = 40$ points in their second attempt at the SAT Mathematics exam. Assume that the change in score has a Normal distribution with standard deviation $\sigma = 25$. We want to estimate the mean change in score μ in the population of all high school seniors

1. Give a 95% confidence interval for μ based on this sample
2. Based on your confidence interval, how certain are you that the mean change in score μ in the population of all high school seniors is greater than 0?

Solution. We are given $\bar{X} = 40$, $n = 400$, and a known population standard deviation $\sigma = 25$. Because the population is Normal (and σ is known), a 95% confidence interval for μ is

$$\bar{X} \pm z^* \frac{\sigma}{\sqrt{n}}, \quad \text{where } z^* = 1.96.$$

Compute the standard error:

$$\frac{\sigma}{\sqrt{n}} = \frac{25}{\sqrt{400}} = \frac{25}{20} = 1.25.$$

Thus the margin of error is

$$1.96(1.25) = 2.45.$$

So the 95% confidence interval is

$$40 \pm 2.45 = (37.55, 42.45).$$

1. A 95% confidence interval for μ is (37.55, 42.45).
2. Since the entire 95% confidence interval is above 0, we are very confident that the population mean change μ is greater than 0. In fact, at the 5% significance level, we would reject $H_0 : \mu = 0$ in favor of $H_a : \mu > 0$.

Margin of Error and Confidence Level

- **Margin of Error**

- The margin of error is a number that is added to, and subtracted from, a statistical estimate to define a confidence interval at a given confidence level

- **Confidence Interval**

- A level C confidence interval for a parameter has two parts:
 1. An interval calculated from the data, usually of the form

$$\text{estimate} \pm \text{margin of error}$$

2. A confidence level C , which gives the probability that the interval will capture the true parameter value in repeated samples. That is, the confidence level is the success rate for the method

- **Interpreting a Confidence Level**

- The confidence level is the success rate of the method that produces the interval. We don't know whether the 95% confidence interval from a particular sample is one of the 95% that capture μ or one of the unlucky 5% that miss
- To say that we are **95% confident** that the unknown μ lies between the confidence interval is shorthand for “**We got these numbers using a method that gives correct results 95% of the time**”

Example (continued)

From the BMI example, we found that the confidence interval using the sample with size $n = 936$ and sample mean $\bar{X} = 27.2$ is

$$\bar{X} \leq \mu + 2 \cdot 0.4 \rightarrow 27.2 - 0.8 \leq \mu$$

$$\mu - 2 \cdot 0.4 \leq \bar{X} \rightarrow \mu \leq 27.2 + 0.8$$

As a result, the confidence interval with 95% confidence level is $26.4 \leq \mu \leq 28$

In this case, the point **estimate** is $\bar{X} = 27.2$ and the **margin of error** is $2 \cdot 0.4 = 0.8$

Interpretation: “We are 95% confident that the true mean BMI of all U.S. men aged 20-29 lies between 26.4 and 28.0”

In other words, in repeated samples, 95% of these intervals capture μ . For example, if we take 25 independent SRSs and construct a 95% confidence interval from each sample, then the expected number of intervals that do *not* contain μ is

$$25(0.05) = 1.25.$$

Figure 82

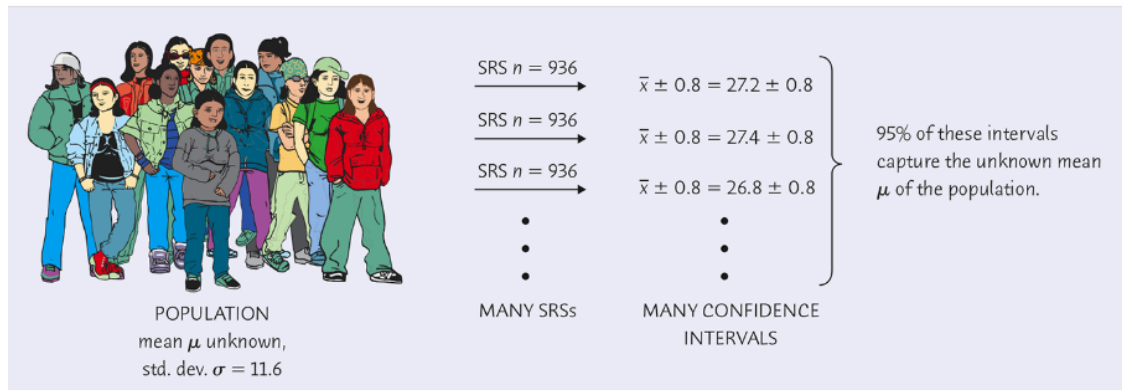
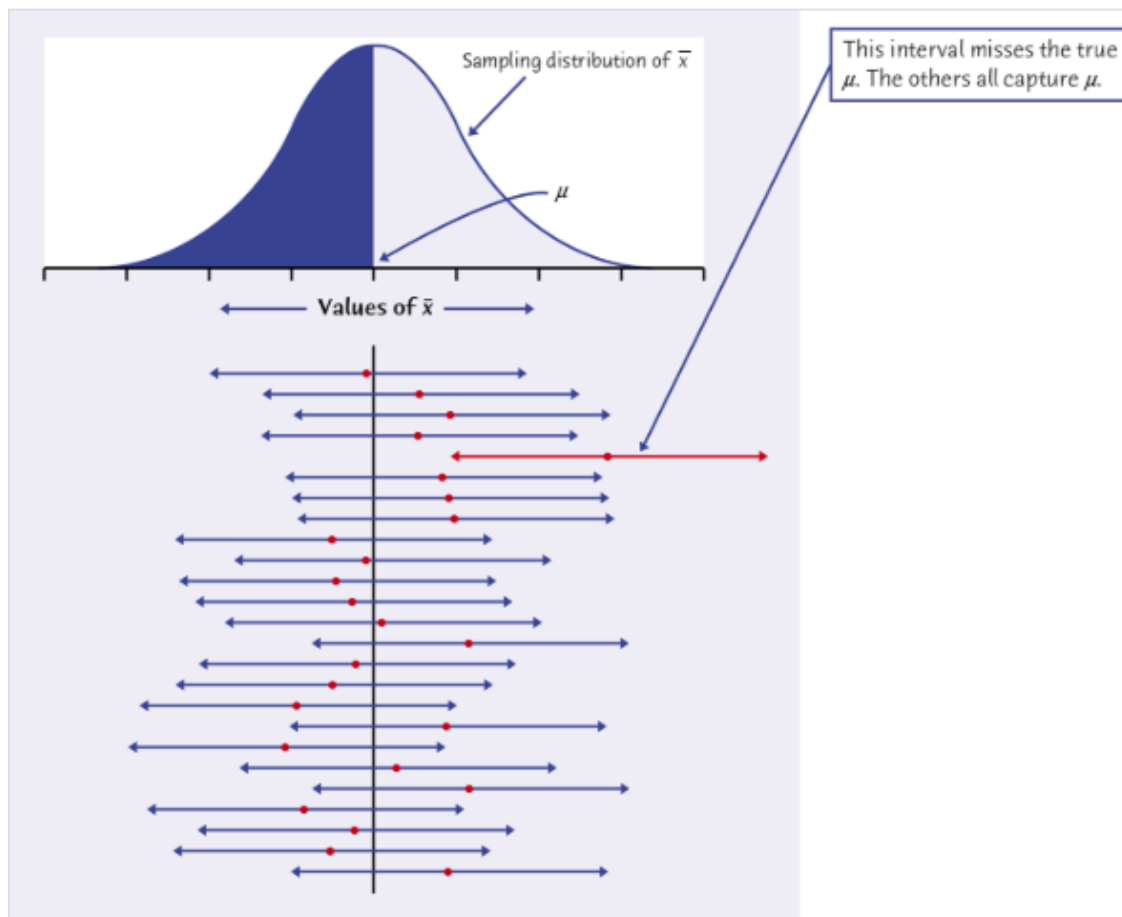


Figure 83



Practice Question 16-2

A Gallup Poll in December 2019 reported that 46% of the 5120 adults aged 18 and older in the sample said both their mental and their physical health was good or excellent. Gallup announced, “For results based on the combined sample of 5120 adults, aged 18 and older...the margin of sampling error is 2 percentage point at the 95% confidence level.”

1. What is the 95% confidence interval?
2. What does it mean to have 95% confidence in this interval?

Solution. The reported sample proportion is $\hat{p} = 0.46$, and Gallup reports a margin of sampling error of 0.02 (2 percentage points) at the 95% confidence level.

1. The 95% confidence interval is

$$\hat{p} \pm 0.02 = 0.46 \pm 0.02 = (0.44, 0.48),$$

i.e., from 44% to 48%.

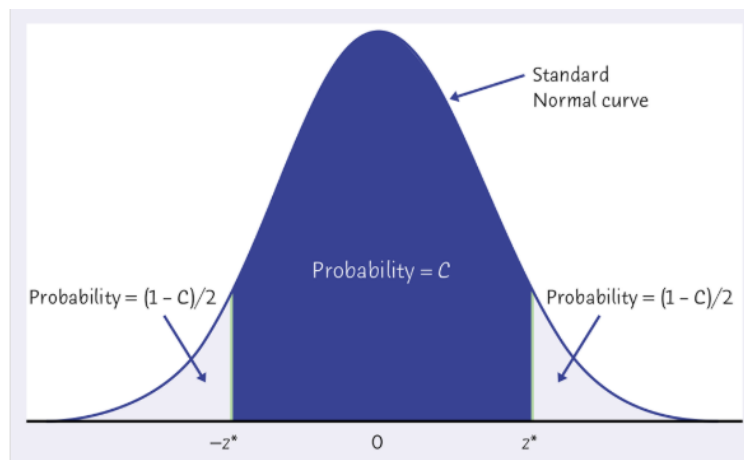
2. Having 95% confidence means that if Gallup were to repeat this polling method many times (taking many similar random samples of adults and constructing the interval $\hat{p} \pm 0.02$ each time), then about 95% of those intervals would contain the true population proportion p of adults who would say both their mental and physical health are good or excellent.

Confidence Intervals for a Population Mean

- Because all Normal distributions are the same in the standard scale, we can obtain everything we need from the standard Normal curve
- **Critical Value**
 - The critical value is a number z^* chosen so that the standard Normal curve has a prespecified area C between z^* and $-z^*$
- Confidence Interval for the Mean of a Normal Population
 - Draw an SRS of size n from a Normal population having unknown mean μ and known standard deviation σ . A level C confidence interval for μ is

$$\bar{X} \pm z^* \times \frac{\sigma}{\sqrt{n}}$$

Figure 84



How Confidence Intervals Behave

- We would like high confidence and also a small margin of error
 - High confidence: our method almost always gives correct answers
 - Small margin of error: we have pinned down the parameter quite precisely

- How do we get a small margin of error? $z^* \cdot \frac{\sigma}{\sqrt{n}}$
 - Critical value z^* and population standard deviation σ in the numerator and $\sqrt{n}$ in the denominator
 - Therefore, the margin of error gets smaller when
 1. z^* gets smaller. However, smaller z^* is the same as lower confidence level C . So, it has trade-off between the confidence level and the margin of error. To obtain a smaller margin of error from the same data, you must be willing to accept lower confidence level
 2. σ is smaller. The standard deviation σ measures the variation in the population. It is easier to pin down μ when σ is small
 3. n gets larger. Increasing the sample size n reduces the margin of error for any confidence level. Larger samples thus allow more precise estimates.

Practice Question 16-3

The NHANES surveyed the body mass index (BMI) of 936 young men. The mean BMI in the sample was $\bar{X} = 27.2$. We treated these data as an SRS from a normally distributed population with standard deviation $\sigma = 11.6$

- (a) Give three confidence intervals for the mean BMI μ in this population using 90%, 95%, and 99%
- (b) What are the margins of error for 90%, 95%, and 99% confidence? How does increasing the confidence level change the margin of error of a confidence interval when the sample size and population standard deviation remain the same?

Solution

- 90% CI: $z^* = 1.645$, Margins of error = $1.645(0.3792) \approx 0.62$

$$27.2 \pm 0.62 \Rightarrow (26.58, 27.82).$$

- 95% CI: $z^* = 1.960$, Margins of error = $1.960(0.3792) \approx 0.74$

$$27.2 \pm 0.74 \Rightarrow (26.46, 27.94).$$

- 99% CI: $z^* = 2.576$, Margins of error = $2.576(0.3792) \approx 0.98$

$$27.2 \pm 0.98 \Rightarrow (26.22, 28.18).$$

- **Margins of error and interpretation**

- Since $ME = z^* \times \frac{\sigma}{\sqrt{n}}$, when n and σ stay the same, increasing the confidence level increases z^* , so the margin of error increases (and the confidence interval becomes wider).

Practice Question 16-3 (continued)

- (c) Suppose that we had an SRS of just 100 young men. What would be the margin of error for 95% confidence?
- (d) Find the margins of error for 95% confidence based on SRSs of 400 young men and 1600 young men
- (e) Compare the three margins of error. How does increasing the sample size change the margin of error of a confidence interval when the confidence level and population standard deviation remain the same?

CH 17. Tests of Significance: The Basics

- The goal and the idea of Significant Tests:
 - **Goal:** to assess the evidence provided by data about some prior claim concerning a population parameter
 - **Idea:** an outcome that would rarely happen if a claim were true is good evidence that the claim is not true

Stating Hypotheses

- Null and Alternative Hypotheses
 1. **Null hypothesis:** H_0
 - The claim tested by a statistical test of significance
 - The test is designed to assess the strength of the evidence **against** the null hypothesis
 - Usually, the null hypothesis is a statement of “no effect” or “no difference”
 2. **Alternative hypothesis:** H_a
 - The claim about the population that we are trying to find evidence **for** is the alternative hypothesis
 - **Two-sided:** if it states that the parameter is different from the null value
 - **One-sided:** if it states that a parameter is larger than or smaller than the null hypothesis value

Example of Two-sided test:

A bottling machine is supposed to fill 500 mL on average. The company wants to know if it is off in either direction (overfilling or underfilling).

Let μ be the true mean fill amount (mL).

$$H_0 : \mu = 500$$

$$H_a : \mu \neq 500$$

- **Null hypothesis** $H_0 : \mu = 500$: The bottling machine is correctly calibrated on average; the true mean fill amount is exactly 500 mL (no systematic overfilling or underfilling).
- **Alternative hypothesis** $H_a : \mu \neq 500$: The machine is miscalibrated on average; the true mean fill amount is different from 500 mL (it may be underfilling, $\mu < 500$, or overfilling, $\mu > 500$).

Example of One-sided test (left-tailed):

A nutrition label claims that a brand of protein bars contains *at least* 20 grams of protein on average. A consumer group is concerned the company may be *overstating* the protein content, so they test whether the true mean protein content is actually *less than* 20 grams.

Let μ be the true mean protein content per bar (grams).

$$H_0 : \mu = 20$$

$$H_a : \mu < 20$$

- **Null hypothesis** $H_0 : \mu = 20$: The label claim holds at the boundary; the true mean protein content is 20 grams (not less than 20 grams on average).
- **Alternative hypothesis** $H_a : \mu < 20$: The company is overstating the protein content; the true mean protein content is less than 20 grams on average.

Example of One-sided test (right-tailed):

A public health agency investigates microplastic contamination in bottled water. Regulations state that the average concentration should not exceed 10 particles per liter. The agency wants to test whether the true mean concentration is *higher than* the safety limit.

Let μ be the true mean microplastic concentration in bottled water (particles per liter).

$$H_0 : \mu = 10$$

$$H_a : \mu > 10$$

- **Null hypothesis** $H_0 : \mu = 10$: The bottled water meets the safety limit at the boundary; the true mean concentration is 10 particles per liter (not higher than 10 on average).
- **Alternative hypothesis** $H_a : \mu > 10$: The bottled water exceeds the safety limit; the true mean concentration is greater than 10 particles per liter on average.

Practice Question 17-1: set up H_0 and H_a

A smartphone manufacturer claims that their new battery lasts *at least* 20 hours on average under standard testing conditions. A consumer testing lab suspects the true mean battery life is *less than* 20 hours. Let μ be the true mean battery life (in hours) for this smartphone model under the standard test. **Solution:**

$$H_0 : \mu = 20$$

$$H_a : \mu < 20$$

Practice Question 17-2

The average income of American women who work full-time and have only a high school diploma is \$37,616. You wonder whether the mean income of female graduates from your local high school who work full-time but have only a high school diploma is different from the national average. You obtain income information from an SRS of 62 female graduates of your high school who work full-time and have only a high school diploma, and find that $\bar{X} = \$36,453$. What are your null and alternative hypotheses?

$$H_0 : \mu = 37616$$

$$H_a : \mu \neq 37616$$

P-Value and Statistical Significance

- **Test Statistic and P-Value**

- **Test Statistic:** calculated from the sample data measures how far the data diverge from what we would expect if the null hypothesis H_0 were true. Unusually large values of the statistic show that the data are not consistent with H_0
- **P-value:** the probability, computed assuming that H_0 is true, that the test statistic would take a value as extreme as or more extreme than that actually observed
 - * Small p-values: evidence against H_0 because they say that the observed result would be unlikely to occur if H_0 were true
 - * Large p-values: fail to give evidence against H_0 .

- **Statistical Significance**

- **significance level(α):**, denoted by α , is the chosen cutoff for a hypothesis test.
- If the P-value is as small as or smaller than α (p-value $\leq \alpha$) , we say that the data are statistically significant at level α .

Example

- **Null hypothesis** $H_0 : \mu = 10$: The bottled water meets the safety limit at the boundary; the true mean concentration is 10 particles per liter (not higher than 10 on average).
- **Alternative hypothesis** $H_a : \mu > 10$: The bottled water exceeds the safety limit; the true mean concentration is greater than 10 particles per liter on average.
- A random sample of $n = 36$ bottles is tested. The sample mean is $\bar{X} = 10.66$. Assume the population is Normal and the population standard deviation is known to be $\sigma = 2$.

Test statistic (Z-test).

$$Z = \frac{\bar{X} - 10}{\sigma/\sqrt{n}}.$$

Plugging in $\bar{X} = 10.66$, $\sigma = 2$, and $n = 36$:

$$Z = \frac{10.66 - 10}{2/\sqrt{36}} = \frac{0.66}{2/6} = \frac{0.66}{0.3333} \approx 1.98.$$

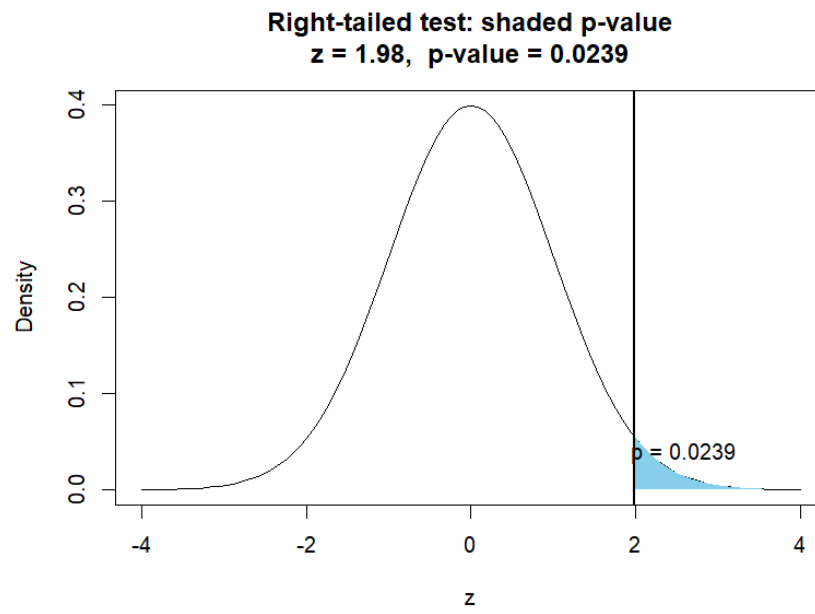
P-value (right-tailed). Because $H_a : \mu > 10$, the p-value is the right-tail probability:

$$p = P(Z \geq 1.98) = 1 - \Phi(1.98) \approx 0.024.$$

Interpretation (in words). Under $H_0 : \mu = 10$ (with $\sigma = 2$ and $n = 36$), a sample mean as high as 10.66 would occur only about 2.4% of the time. Since we observed $\bar{X} = 10.66$, this outcome is quite rare if H_0 were true. Therefore, the result provides evidence against H_0 (and in favor of $H_a : \mu > 10$).

Conclusion (at $\alpha = 0.05$). Using the significance level $\alpha = 0.05$, we compare the p-value to α . Since $p \approx 0.024 < 0.05$, we reject H_0 at the 5% significance level. Therefore, there is sufficient evidence (at $\alpha = 0.05$) to conclude that the true mean microplastic concentration exceeds 10 particles per liter.

Figure 85: P-value and Statistical Significance



Example.

Suppose that, for any cola, the sweetness-loss scores vary from taster to taster according to a Normal distribution with known standard deviation $\sigma = 1$. A positive score indicates that the cola *lost sweetness* after one month.

We wish to test whether the cola loses sweetness on average. Let μ be the population mean sweetness-loss score for this cola. Using data from $n = 10$ tasters, we test

$$H_0 : \mu = 0 \quad \text{vs.} \quad H_a : \mu > 0.$$

Case 1 (10 tasters).

1.6, 0.4, 0.5, -2.0, 1.5, -1.1, 1.3, -0.1, -0.3, 1.2

The sample mean is

$$\bar{x} = 0.30.$$

Since $\sigma = 1$ is known and $n = 10$, the Z test statistic is

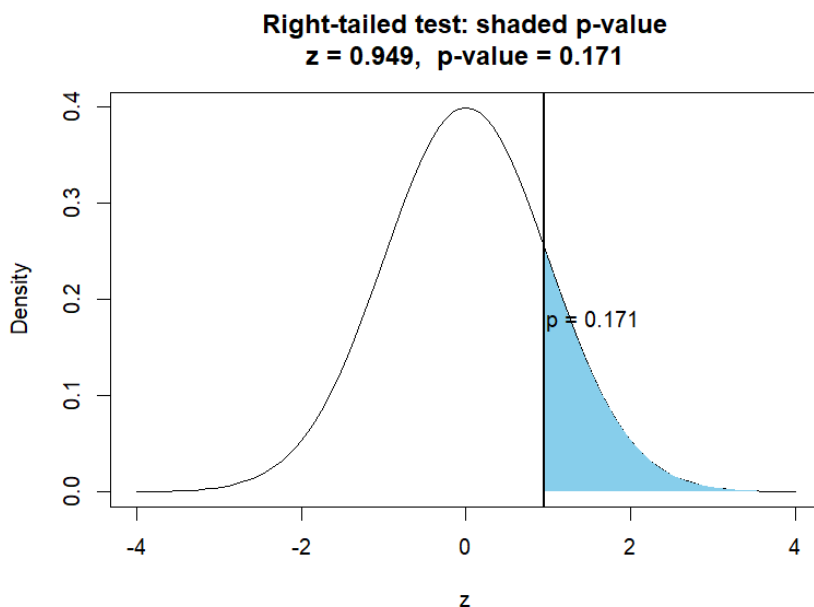
$$Z = \frac{\bar{X} - 0}{1/\sqrt{10}} = \bar{x}\sqrt{10} = 0.30\sqrt{10} = 0.949.$$

Because the alternative is right-tailed ($H_a : \mu > 0$), the p-value is

$$p = P(Z \geq 0.949) = 1 - \Phi(0.949) \approx 0.171.$$

Interpretation: If $H_0 : \mu = 0$ were true, then there is about a 17.1% chance of obtaining a sample mean at least as large as 0.30 (equivalently, a Z value at least as large as 0.949). This p-value is not small, so the data do not provide strong evidence that $\mu > 0$ for this cola.

Figure 86: P-value and Statistical Significance



Case 2 (10 tasters).

2.0, 0.4, 0.7, 2.0, -0.4, 2.2, -1.3, 1.2, 1.1, 2.3

The sample mean is

$$\bar{x} = 1.02.$$

Since $\sigma = 1$ is known and $n = 10$, the Z test statistic is

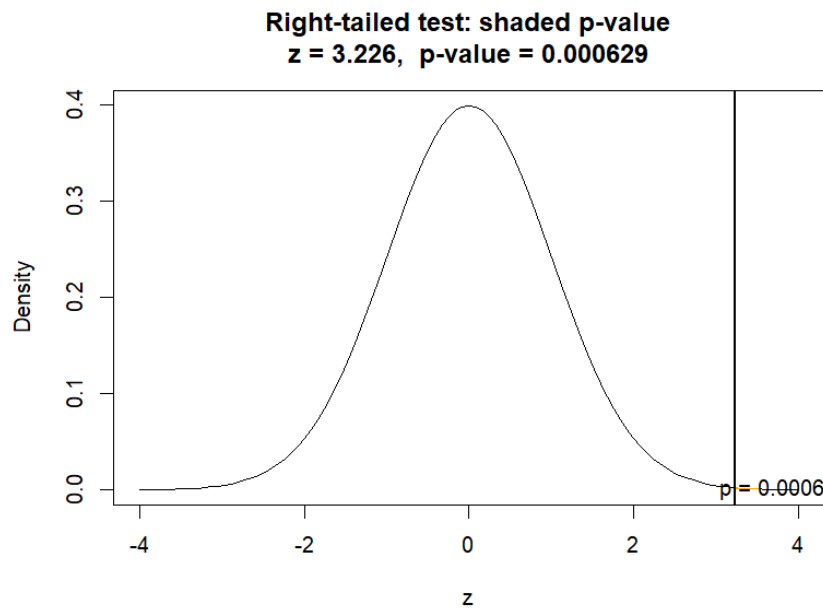
$$Z = \frac{\bar{X} - 0}{1/\sqrt{10}} = \bar{x}\sqrt{10} = 1.02\sqrt{10} = 3.226.$$

Because the alternative is right-tailed ($H_a : \mu > 0$), the p-value is

$$p = P(Z \geq 3.226) = 1 - \Phi(3.226) \approx 0.00063.$$

Interpretation: If $H_0 : \mu = 0$ were true, then there is about a 0.063% chance of obtaining a sample mean at least as large as 1.02 (equivalently, a Z value at least as large as 3.226). This p-value is extremely small, so the data provide strong evidence that $\mu > 0$ for this cola.

Figure 87: P-value and Statistical Significance



Test for a Population Mean

- Again the reasoning of a test:
 - sample data that would rarely occur if the null hypothesis H_0 were true provide evidence that H_0 is not true.
- Test of Significance: The four-Step Process
 1. What is the practical question that requires a statistical test?
 2. Identify the parameter, state null and alternative hypotheses, and choose the type of test that fits your situation
 3. Carry out the test in three phases:
 - Check the conditions for the test you plan to use (SRS, Normal distribution, know σ ...)
 - Calculate the test statistic
 - Find the P-value
 4. Return to the practical question to describe your result in this setting

• One-sample z Test for a Population Mean

- Draw an SRS of size n from a Normal distribution population that has unknown mean μ and known standard deviation σ . To test the null hypothesis that μ has a specified value,

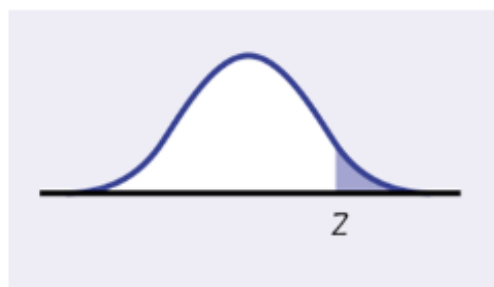
$$H_0 : \mu = \mu_0$$

- Calculate the **one-sample z test statistic**:

$$z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

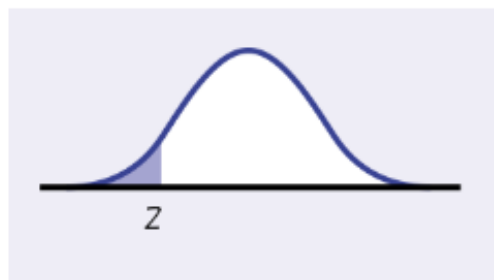
- Test statistic z measures how far the observed sample mean $\bar{X}$ deviates from the hypothesized population value μ_0
 - * The measurement is in the familiar standard scale obtained by dividing by the standard deviation of $\bar{X}$
- In terms of a variable Z having the standard Normal Distribution, the P-value for a test of H_0 against
 1. Right-tail one-sided test $H_a : \mu > \mu_0$ is $P(Z \geq z)$
 2. Left-tail one-sided test $H_a : \mu < \mu_0$ is $P(Z \leq z)$
 3. Two-sided test $H_a : \mu \neq \mu_0$ is $P(Z \leq -|z|) + P(Z \geq |z|) = 2P(Z \geq |z|)$

$$H_a: \mu > \mu_0 \text{ is } P(Z \geq z)$$



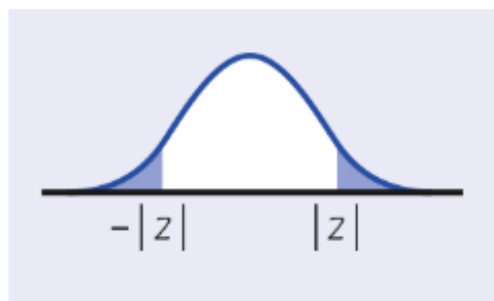
Moore/Notz, The Basic
Practice of Statistics, 9e,
© 2021 W. H. Freeman
and Company

$$H_a: \mu < \mu_0 \text{ is } P(Z \leq z)$$



Moore/Notz, The Basic
Practice of Statistics, 9e,
© 2021 W. H. Freeman
and Company

$$H_a: \mu \neq \mu_0 \text{ is } P(Z \leq -|z|) + P(Z \geq |z|) = 2P(Z \geq |z|)$$



Moore/Notz, The Basic
Practice of Statistics, 9e,
© 2021 W. H. Freeman
and Company

Figure 88: Z-test

Example (Two-sided).

The National Center for Health Statistics reports that the LDL cholesterol for adults has mean 130 and standard deviation 40 with normal distribution. The medical director of a large pharmaceutical company looks at the medical records of 72 executives and finds that the mean LDL in this sample is $\bar{x} = 124.86$. Is this evidence that the company's executives have different mean LDL from the general population?

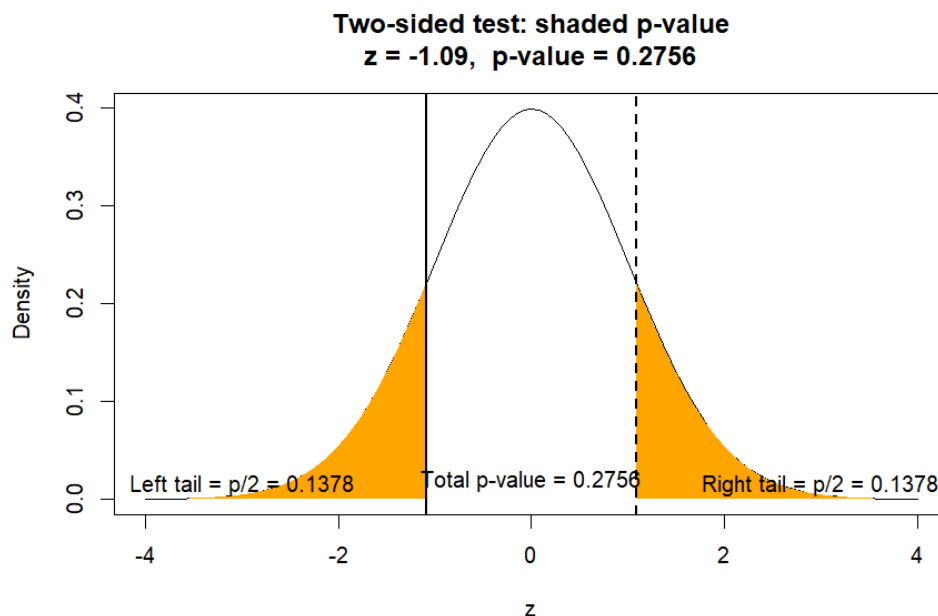
$$H_0 : \mu = 130$$

$$H_a : \mu \neq 130$$

z - statistic

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{124.86 - 130}{40/\sqrt{72}} = -1.09$$

Figure 89: Two-sided Test



Interpretation: *If the true mean LDL level for the company's executives were the same as the general population mean of 130 mg/dL, then there is about a 27.6% chance that a random sample of 72 executives would have a sample mean LDL at least as far from 130 as the observed mean 124.86 mg/dL (in either direction).* Because this probability is fairly large, the observed sample mean is not unusual under H_0 , so the data do not provide strong evidence that executives have a different mean LDL than the general population.

Practice Question 17-3

The average SAT math score for high school students nationwide is 520 points. A school district administrator wonders whether students in a newly established prep program have a different average SAT math score compared to the national average. The administrator collects score information from an SRS of 81 students in the prep program and finds that $\bar{X} = 535$ points. Assume the population standard deviation is $\sigma = 45$ points.

- (a) (3 points) Write the null and alternative hypotheses.

Answer:

Let μ denote the true mean SAT math score for students in the prep program.

$$H_0 : \mu = 520 \quad (\text{the program mean equals the national average})$$

$$H_a : \mu \neq 520 \quad (\text{the program mean differs from the national average})$$

This is a **two-sided** test because we are asking whether the mean is *different* (either higher or lower) from the national average.

- (b) (3 points) Find the 90% confidence interval for the true mean SAT math score of students in the prep program.

Answer:

The 90% confidence interval is given by:

$$\bar{X} \pm z^* \cdot \frac{\sigma}{\sqrt{n}}$$

where $z^* = 1.645$ for a 90% two-sided confidence interval.

Compute the standard error:

$$\frac{\sigma}{\sqrt{n}} = \frac{45}{\sqrt{81}} = \frac{45}{9} = 5.00$$

Compute the margin of error:

$$1.64 \times 5.00 = 8.20$$

Compute the confidence interval:

$$90\% \text{ CI} = 535 \pm 8.20 = (\mathbf{526.8}, \mathbf{543.2})$$

We are 90% confident that the true mean SAT math score for students in the prep program is between 526.8 and 543.2 points.

- (c) (3 points) Find the z -statistic.

Answer:

The z -statistic is computed as:

$$z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{535 - 520}{45/\sqrt{81}} = \frac{15}{5.00} = \mathbf{3.00}$$

- (d) (4 points) Using the P -value, test whether we can reject H_0 at the 10% significance level.

Answer:

Since this is a two-sided test, the P -value is:

$$P\text{-value} = 2 \times P(Z > |z|) = 2 \times P(Z > 3.00)$$

From the standard normal table:

$$P(Z > 3.00) = 1 - 0.9987 = 0.0013$$

$$P\text{-value} = 2 \times 0.0013 = \mathbf{0.0026}$$

Decision: Since $P\text{-value} = 0.0026 < \alpha = 0.10$, we **reject** H_0 at the 10% significance level.

Conclusion: There is sufficient evidence at the 10% significance level to conclude that the true mean SAT math score of students in the prep program differs from the national average of 520 points. Note also that the 90% confidence interval (526.78, 543.23) does not include $\mu_0 = 520$, which is consistent with this conclusion.

CH 18. Inference in Practice

- Up to this point, we use “**z-procedures**”
 - Under this z-procedure, we assume the “**simple conditions**”:
 1. the data are an SRS
 2. the population has a Normal distribution
 3. we know standard deviation σ of the population
 - In this z-procedure, a confidence interval (CI) for the mean μ is

$$\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}}$$

z^* : critical value required for a given significance level

- To test $H_0 : \mu = \mu_0$, we use the one-sample z statistic:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

- In later chapters, we will modify these procedures for inference about a population mean to make them useful in practice

Conditions for Inference in Practice

- Simple Conditions for Inference about a Mean
 1. We have a simple random sample (SRS) from the population of interest
 - There is no nonresponse or other practical difficulty
 - The population is large compared to the size of the sample
 2. The variable we measured has an exactly Normal distribution $N(\mu, \sigma)$ in the population
 3. We don't know the population mean μ . But we do know the population standard deviation σ
 - *This is rarely satisfied in practice. But we can replace this (Chapter 20)*

- Where the Data Come from Matters
 - The most important requirement for any inference procedure is that the data come from a process to which the laws of probability apply
 - Inference is most reliable when the data come from a **random sample** or a **randomized comparative experiment** (unbiased)
 - * Random Sample: use chance to choose respondents
 - * Randomized Comparative Experiments: use chance to assign subjects to treatments
 - If your data don't come from a random sample or a randomized comparative experiment, your conclusions may be challenged
 1. Practical problems such as nonresponse in sample or dropouts from an experiment can hinder inference even from a well-designed study
 2. There is no cure for fundamental flaws like voluntary response surveys or uncontrolled experiments
- What is the shape of the population distribution?
 - This condition is less essential than where the data come from
 - The **central limit theorem (CLT)** tells us that
 - * the sampling distribution of $\bar{x}$ is more Normal than the distribution of individual observations
 - * the sampling distribution of $\bar{x}$ becomes more Normal as the size of the sample increases ($n \uparrow$)
 - Therefore, in practice, if the sample is large, the sampling distribution $\bar{x}$ will be close to Normal even if individual measurements are strongly skewed

Practice Question 18-1

A survey of licensed drivers inquired about running red lights. One question asked, “of every 10 motorists who run a red light, about how many do you think will be caught?” The mean result for 880 respondents was $\bar{x} = 1.92$, and the standard deviation was $s = 1.83$. For this large sample, s will be close to the population standard deviation σ , so suppose we know that $\sigma = 1.83$.

1. **Give a 95% confidence interval for the mean opinion in the population of all licensed drivers.**

We have $n = 880$, $\bar{x} = 1.92$, and $\sigma = 1.83$. The standard error of $\bar{x}$ is

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{1.83}{\sqrt{880}} \approx \frac{1.83}{29.665} \approx 0.06169.$$

For a 95% confidence interval, the critical value is $z^{=1.96}$, so the margin of error is

$$ME = z \cdot SE = 1.96(0.06169) \approx 0.12091.$$

Therefore, the 95% confidence interval for μ is

$$\bar{x} \pm ME = 1.92 \pm 0.12091 = (1.799, 2.041).$$

Rounded: $\boxed{(1.80, 2.04)}$.

2. **The distribution of responses is skewed to the right rather than Normal. This will not strongly affect the z confidence interval for this sample. Why not?**

Although the population distribution of individual responses is right-skewed, the confidence interval is based on the sample mean $\bar{X}$. Because the sample size is large ($n = 880$), the Central Limit Theorem implies that the sampling distribution of $\bar{X}$ is approximately Normal (assuming finite variance). Hence, using the z critical value is appropriate and the skewness of the raw responses has little effect on the interval.

3. **The 880 respondents are an SRS from completed calls among 45,956 calls to randomly chosen residential telephone numbers listed in telephone directories. Only 5029 of the calls were completed. This information gives two reasons to suspect that the sample may not represent all licensed drivers. What are these reasons?**

- (a) **Undercoverage:** The sampling frame includes only residential telephone numbers listed in directories. This can exclude people with unlisted numbers, cell-phone-only households, or others not appearing in directories, who may have different opinions.
- (b) **Nonresponse bias:** Only 5029 of 45,956 calls were completed. People who answer and complete telephone surveys may differ systematically from those who do not (e.g., in availability, willingness to participate, demographics, or attitudes), so the completed-call sample may be biased.

Cautious about Confidence Intervals

- The Margin of Error (ME) Doesn't Cover all Errors
 - The margin of error in a confidence interval covers only random sampling errors
 - * Random Sampling Error: $\bar{X} - \mu$
 - Practical difficulties such as undercoverage and nonresponse are often more serious than random sampling error
 - The margin of error does not take such difficulties into account

Cautions about Significance Tests

1. How Small a P (p-value) is Convincing?

- How plausible is H_0 ?:
 - If H_0 represents an assumption that the people you must convince have believed for years, strong evidence (small P) will be needed to persuade them
- What are the consequences of rejecting H_0 ?
 - If rejecting H_0 in favor of H_a would trigger a costly change in procedure or policy, then the decision should be supported by strong evidence, reflected in a very small P -value.
- There is no sharp border between “significant” and “not significant”, only increasingly strong evidence as the P -value decreases
 - In practice, common significance levels include 10%, 5%, and 1%

2. Significance depends on the alternative hypothesis (H_a)

- The two-sided p-value combines two equal areas, one in each tail of a Normal Curve
- The one-sided p-value is just one of these areas, in the direction specified by the alternative hypothesis
- Therefore, use a two-sided alternative hypothesis unless you have the information or reason that led the researcher to choose a one-sided test

Figure 90: One-sided Test

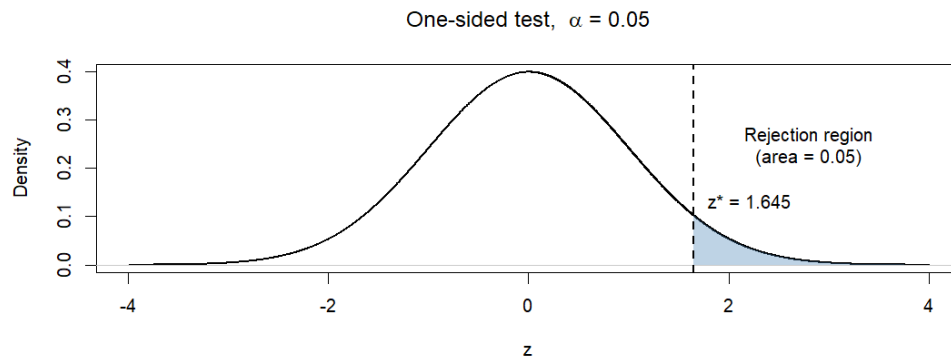
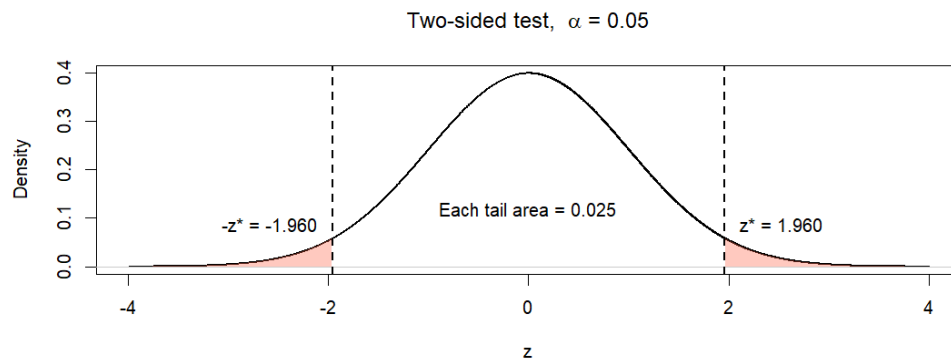


Figure 91: Two-sided Test



3. Significance depends on sample size

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

- numerator: how far the sample mean deviates from the hypothesized mean μ_0
- denominator: the standard deviation of $\bar{x}$. It measures how much random variation we expect.
- z gets larger (more significant) when the estimated effect $\bar{x} - \mu_0$ gets larger or when the number of observations n gets larger

- Sample size affects Statistical Significance
 - Because large random samples have small chance variation, very small population effects can be highly significant if the sample is large
 - Because small random samples have a lot of chance variation, even large population effects can fail to be significant if the sample is small
 - Statistical significance does not tell us whether an effect is large enough to be important.
 - That is, statistical significance is not the same thing as practical significance

Practice Question 18-2

In the absence of special preparation, SAT Mathematics scores in 2019 varied Normally with mean $\mu = 528$ and $\sigma = 117$. Fifty students go through a rigorous training program designed to raise their scores by improving their mathematics skills.

$$H_0 : \mu = 528$$

$$H_a : \mu > 528$$

Carry out a test in each of the following situations:

1. The students' average score is $\bar{x} = 555$. Is this result significant at the 5% level?
2. The average score is $\bar{x} = 556$. Is this result significant at the 5% level?

The Power of a Statistical Test of Significance

- Power
 - The **power** of a test against a specific alternative is the probability that the test will reject H_0 at a chosen significance level α when the specified alternative value of the parameter is true
- Type I error Type II error
 - Type I error: If we reject H_0 when in fact H_0 is true
 - Type II error: If we fail to reject H_0 when in fact H_a is true

Example

We test

$$H_0 : \mu = 50 \quad \text{vs.} \quad H_a : \mu > 50$$

at significance level $\alpha = 0.05$, with $\sigma = 10$ and $n = 25$.

- Test statistic.

$$Z = \frac{\bar{X} - 50}{\sigma/\sqrt{n}} = \frac{\bar{X} - 50}{10/\sqrt{25}} = \frac{\bar{X} - 50}{2}.$$

- Rejection rule.

For a right-tailed test at $\alpha = 0.05$, the critical value is $z_{0.95} = 1.645$. Reject H_0 if

$$Z \geq 1.645 \iff \frac{\bar{X} - 50}{2} \geq 1.645 \iff \bar{X} \geq 50 + 1.645(2) = 53.29.$$

- Power at a specific alternative.

Suppose the true mean is $\mu = 55$. Then $\bar{X} \sim N(55, (\sigma/\sqrt{n})) = N(55, 2)$. The power is

$$\text{Power} = P(\text{reject } H_0 \mid \mu = 55) = P(\bar{X} \geq 53.29 \mid \mu = 55).$$

Standardizing,

$$P(\bar{X} \geq 53.29 \mid \mu = 55) = P\left(\frac{\bar{X} - 55}{2} \geq \frac{53.29 - 55}{2}\right) = P(Z \geq -0.855) = \Phi(0.855) \approx 0.803.$$

So the power is approximately 0.80.

Part IV

Inference about Variables

CH 20. Inference about a Population Mean

- Up to this point, we included the unrealistic assumption that we knew the population standard deviation (σ)
- In this chapter, we discard the unrealistic condition that we know the population standard deviation (σ) and present procedures for practical use

Conditions for Inference about a Mean

- Conditions for Inference about a Mean
 - We can regard our data as a **simple random sample (SRS)** from the population. This condition is very important
 - Observations from the population have a Normal distribution with mean μ and standard deviation σ .
 - * In practice, it is enough that the distribution be symmetric and single-peaked unless the sample is very small
 - * Both μ and σ are unknown *parameters*
- **Standard Error (SE)**
 - When the standard deviation of a statistic is estimated from data, the result is called the **standard error (SE)** of the statistic
 - The standard error of the sample mean $\bar{x}$ is

$$SE(\bar{X}) = s/\sqrt{n},$$

s = standard deviation of the sample

- When the simple conditions for inference are satisfied, the sample mean $\bar{x}$ has the Normal distribution with mean μ and standard deviation $\sigma/\sqrt{n}$.
- However, because we don't know σ , we estimate it by the sample standard deviation s

$$\sigma/\sqrt{n} \rightarrow s/\sqrt{n}$$

Example.

If a sample of a size $n = 20$ from a population has standard deviation $s = 8$, the standard error of the sample mean would be

$$s/\sqrt{n} = 8/\sqrt{20} = 1.789$$

It means, the sampling distribution will be

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t_{19}.$$

- Do not confuse the standard error (SE) with the sample standard deviation (s)
 - s : the sample standard deviation is an estimate of the standard deviation of the population from which the sample is selected
 - SE: the standard error is not an estimate of the population standard deviation

Practice Question 20-1

A study of commuting times reports the one-way travel times to work of a random sample of 1000 employed adults in Seattle. The mean is $\bar{x} = 30.1$ minutes, and the standard deviation is $s = 27.2$ minutes. What is the standard error (SE) of the mean?

$$SE(\bar{X}) = \frac{s}{\sqrt{n}} = \frac{27.2}{\sqrt{1000}} \approx 0.860 \text{ minutes}$$

The t Distributions

- If we knew the value of σ , we would base confidence intervals and tests for μ on one-sample z statistic

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

- This z statistic has the standard normal distribution $Z \sim N(0, 1)$

- In practice, we don't know σ , so we **substitute** the standard error $s/\sqrt{n}$ of $\bar{x}$ for its standard deviation $\sigma/\sqrt{n}$

- The statistic that results does not have a Normal distribution
- It has a new distribution called “**t-distribution**”

- **The One-Sample t Statistic and the t Distributions**

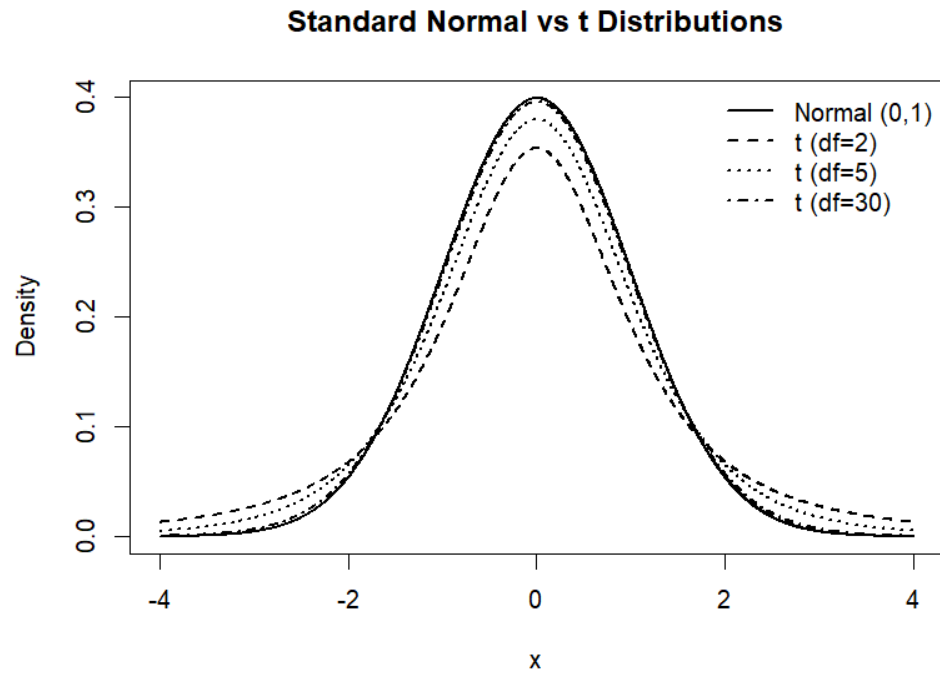
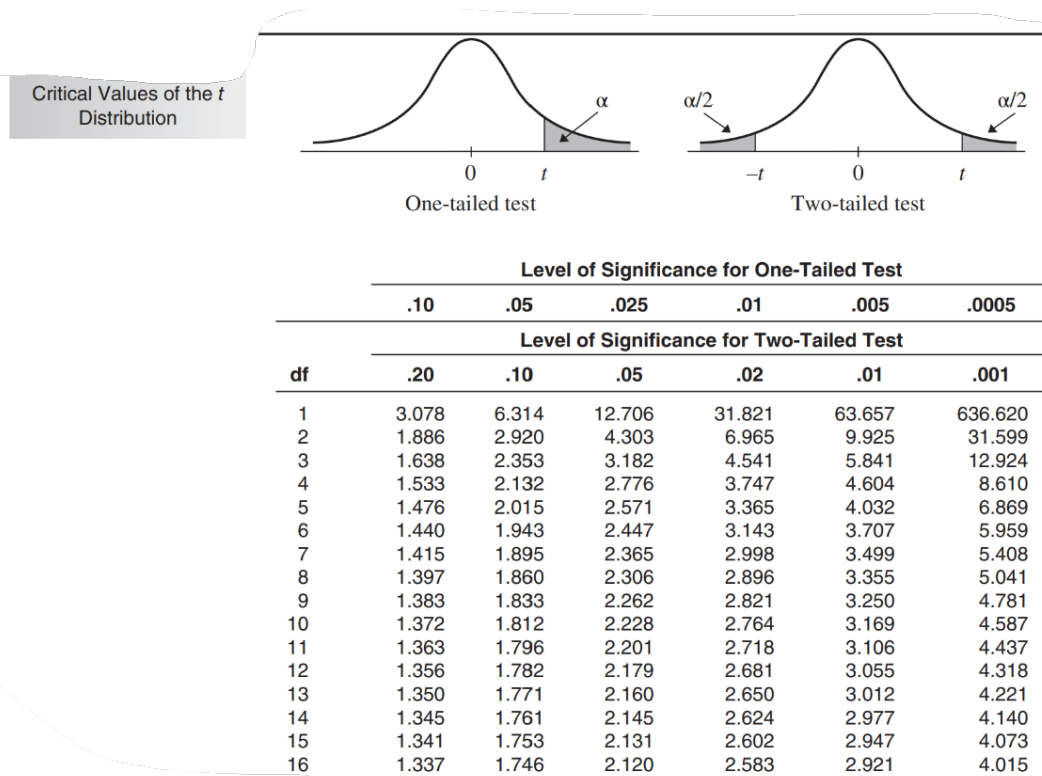
- Draw an SRS of size n from a large population that has the Normal Distribution with mean μ and standard deviation σ .
- **The one-sample t statistic**

$$t = \frac{\bar{x} - \mu}{SE} = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

- * It has the t -distribution with $n - 1$ degrees of freedom

- **Facts about the t distribution**

- The density curves of the t distributions are similar in shape to the standard Normal curve
 - * Symmetric about zero, single-peaked, and bell-shaped
- The variability of the t distributions a bit greater than of the standard Normal distribution
 - * More probability in the tails and less in the center
 - * This is because substituting the estimate s for the fixed parameter σ introduces more variation into the statistic
- As the degrees of freedom increase ($n - 1 \uparrow$), the t density curve approaches the $N(0, 1)$ curve ever more closely
 - * This is because s estimates σ more accurately as the sample size increases
 - * So when the sample is large, using s causes little extra variation in the statistic

Figure 92: Standard normal distribution and t -distributionsFigure 93: Table of t -distribution critical values

Practice Question 20-2

Use t -distribution critical values table to find the critical value for a one-sided test with level $\alpha = 0.01$ based on the t_{10} .

$$t^* = 2.764$$

Look up t_{10} row and $\alpha = 0.01$ (one-sided) column in the t -table.

Practice Question 20-3

You have an SRS of size 15 and calculate the one-sample t -statistic. What is the critical value t^* such that

1. t has probability 0.025 to the right of t^* One-sided, upper tail: $\alpha = 0.025$, $df = 14$

$$\Rightarrow t^* = 2.145$$

2. t has probability 0.01 to the left of t^* One-sided, lower tail: $\alpha = 0.01$, $df = 14$. By

$$\text{symmetry, } t^* \text{ is negative} \Rightarrow t^* = -2.624$$

3. t has probability 0.005 to the right of t^* and 0.005 to the left of $-t^*$ Two-sided: total

$$\alpha = 0.01, \text{ each tail} = 0.005, df = 14 \Rightarrow t^* = 2.977$$

The One-sample t Confidence Interval

- The One-Sample t confidence Interval
 - Draw an SRS of size n from a large population having unknown mean μ . A level C confidence interval for μ is

$$\bar{x} \pm t^* \times \frac{s}{\sqrt{n}}$$

t^* : the critical value for the t_{n-1} density curve area C between $-t^*$ and t^*

$$\text{SE: } \frac{s}{\sqrt{n}}, \text{ ME: } t^* \times \frac{s}{\sqrt{n}}$$

- The interval is exact when the population distribution is Normal and is approximately correct for large n in other cases

Practice Question 20-4

You take an SRS of size $n = 16$ from a population with unknown σ . The sample mean is $\bar{x} = 12.4$ and the sample standard deviation is $s = 3.1$. Construct a 95% confidence interval for the population mean μ .

Solution. Step 2: Find the Critical Value

$$t^* = 2.131 \quad (t_{15}, \text{ two-sided, } \alpha = 0.05)$$

Step 3: Compute the Margin of Error

$$m = t^* \times \text{SE} = 2.131 \times 0.775 \approx 1.652$$

Step 4: Construct the Interval

$$\bar{x} \pm m = 12.4 \pm 1.652$$

$$(10.748, 14.052)$$

We are 95% confident that the true population mean μ lies between 10.748 and 14.052 (based on this sample).

Practice Question 20-5

A coffee shop owner wants to estimate the average amount of time customers spend waiting in line during the morning rush. On a randomly selected set of $n = 30$ mornings, the owner records the average waiting time (in minutes) for that morning. The sample mean is $\bar{x} = 4.8$ minutes and the sample standard deviation is $s = 1.6$ minutes. Construct a 90% confidence interval for the true mean morning waiting time μ .

Solution. Step 2: Find the Critical Value

$$t^* = 1.699 \quad (t_{29}, \text{ two-sided}, \alpha = 0.10)$$

Step 3: Compute the Margin of Error

$$m = t^* \times \text{SE} = 1.699 \times 0.292 \approx 0.496$$

Step 4: Construct the Interval

$$\bar{x} \pm m = 4.8 \pm 0.496$$

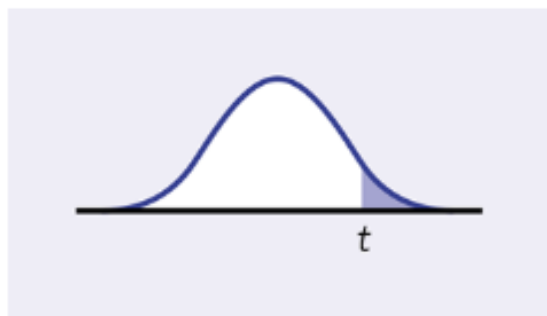
$$\boxed{(4.304, 5.296)}$$

The One-Sample t Test

- The One-Sample t -Test
 - Draw an SRS of size n from a large population having unknown mean μ .
 - To test the hypothesis $H_0 : \mu = \mu_0$, compute the one-sample t statistic

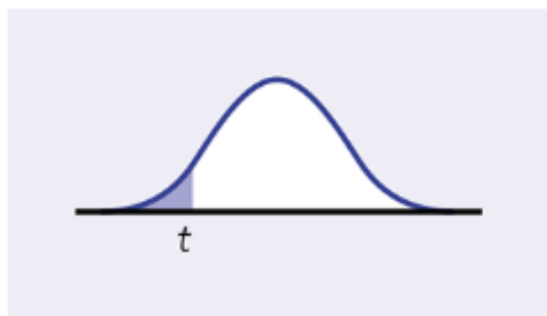
$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$
 - In terms of a variable t having the t_{n-1} distribution the P-value for a test of H_0 against
 1. Right-tail one-sided test $H_a : \mu > \mu_0$ is $P(T \geq t)$
 2. Left-tail one-sided test $H_a : \mu < \mu_0$ is $P(T \leq t)$
 3. Two-sided test $H_a : \mu \neq \mu_0$ is $P(T \leq -|t|) + P(T \geq |t|) = 2P(T \geq |t|)$

$$H_a: \mu > \mu_0 \text{ is } P(T \geq t)$$



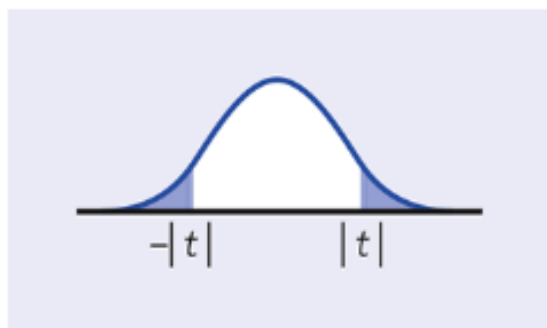
Moore/Notz, *The Basic Practice of Statistics*, 9e,
© 2021 W. H. Freeman
and Company

$$H_a: \mu < \mu_0 \text{ is } P(T \leq t)$$



Moore/Notz, *The Basic Practice of Statistics*, 9e,
© 2021 W. H. Freeman
and Company

$$H_a: \mu \neq \mu_0 \text{ is } 2P(T \geq |t|)$$



Moore/Notz, *The Basic Practice of Statistics*, 9e,
© 2021 W. H. Freeman
and Company

Practice Question 20-6

The one sample t statistic for testing

$$H_0 : \mu = 0$$

$$H_a : \mu > 0$$

from a sample of $n = 101$ observations has the value $t = 3.00$.

1. What are the degrees of freedom for this statistic?

$$df = n - 1 = 101 - 1 = 100$$

2. Is the value $t = 3.00$ significant at the 10% level? Is it significant at the 5% level? Is it significant at the 1% level?

This is a **one-sided** test ($H_a : \mu > 0$), so we compare $t = 3.00$ against upper-tail critical values for t_{100} :

Level α	Critical value t^*	$t = 3.00 > t^*$?	Significant?
10%	1.290	Yes	✓
5%	1.660	Yes	✓
1%	2.364	Yes	✓

$t = 3.00$ is significant at all three levels.

	one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
	two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df												
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62	
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599	
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924	
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610	
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869	
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959	
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408	
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041	
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781	
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587	
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437	
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318	
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221	
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140	
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073	
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015	
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965	
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922	
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883	
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850	
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819	
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792	
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768	
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745	
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725	
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707	
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690	
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674	
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659	
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646	
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551	
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460	
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416	
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390	
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300	

Practice Question 20-6

A cereal manufacturer claims that the average amount of cereal in its “20 oz” boxes is $\mu = 20$ ounces. A consumer group takes an SRS of $n = 81$ boxes and measures the amount of cereal in each box. They compute the one-sample t statistic for testing

$$H_0 : \mu = 20$$

$$H_a : \mu \neq 20$$

and obtain the value $t = -2.30$.

1. What are the degrees of freedom for this statistic?

$$df = n - 1 = 81 - 1 = 80$$

2. Is the result statistically significant at the 10% level? At the 5% level? At the 1% level?

This is a **two-sided** test ($H_a : \mu \neq 20$), so we compare $|t| = 2.30$ against two-sided critical values for t_{80} :

Level α	Critical value t^*	$ t = 2.30 > t^*$?	Significant?
10%	1.664	Yes	✓
5%	1.990	Yes	✓
1%	2.639	No	×

$t = -2.30$ is significant at the 10% and 5% levels, but NOT at the 1% level.

Robustness of t Procedures

- Robust Procedures
 - A confidence interval or significance test is called **robust** if the confidence level or P-value does not change very much when the conditions for use of the procedure are violated
 - As the size of the sample increases, central limit theorem ensures that the distribution of the sample mean $\bar{x}$ becomes more nearly normal and the t distribution becomes more accurate for critical values and P-values of the t procedures

- Using the t Procedures
 - Always make a plot to check for skewness and outliers before you use the t -procedures
 - Except in the case of small samples, the condition that the data are an SRS from the population of interest is more important than the condition that the population distribution is Normal
 1. Sample size less than 15: t procedures if the data appear to be close to Normal (roughly symmetric, single-peaked, no outliers). If the data are clearly skewed or if outliers are present, do not use t
 2. Sample size at least 15: The t procedures can be used except in the presence of outliers or strong skewness
 3. Large samples: The t procedures can be used even for clearly skewed distributions when the sample is large, roughly $n \geq 40$

CH 21. Comparing Two Means

- In this chapter, we discuss t procedures for comparing the means of two Normal populations when we have **independent** samples from these two populations

Two-Sample Problems

- **Two-Sample Problems**
 - The goal of inference is to compare the responses to two treatments or to compare the characteristics of two populations
 - We have a separate sample from each treatment or each population

Example: A Two-sample problem from a randomized comparative experiment

Does financial literacy training improve household saving behavior? A randomized experiment assigned individuals from low-income households to two groups. A total of 105 participants received a brochure containing general information about budgeting and saving. Another 108 participants attended a structured financial literacy workshop that included weekly sessions for one month. After six months, each participant's savings rate (the percentage of income saved) was measured using bank account records by a researcher who did not know which program the participant had received.

Example: A Two-sample problem from random samples separately selected from two populations

Do urban and rural households differ in average monthly electricity use? An energy researcher separately selected a random sample of 120 urban households and a random sample of 115 rural households from a state utility database. For each household, the monthly electricity consumption (in kilowatt-hours) during July was recorded. The researcher then compared the mean electricity use for the two populations.

Comparing Two Population Means

- Comparing two populations or the responses to two treatments starts with data analysis
 - boxplots
 - histogram
 - compare the shapes, centers, and variabilities of the two samples
- The most common goal of inference is to **compare the average or typical responses in the two populations**

- **Conditions for Inference Comparing Two Means**

1. We have two SRSs, from two distinct populations. The samples are ***independent***.
 - That is, one sample has no influence on the other
 - We measure the same response variable (outcome variable) for both samples
2. Both populations are ***Normally Distributed***.
 - The means (μ_1, μ_2) and standard deviations (σ_1, σ_2) are unknown
 - In practice, it is enough that the distributions have similar shapes and that the data have no strong outliers

- **Two populations and Samples**

- Let's assume we have data come from a randomized comparative experiment that randomly divides subjects into two groups and exposes each group to a different treatment

$$* H_0 : \mu_1 = \mu_2 \rightarrow \mu_1 - \mu_2 = 0$$

$$* H_a : \mu_1 \neq \mu_2 \rightarrow \mu_1 - \mu_2 \neq 0$$

- Two Populations

Population	Variable	Mean	Standard Deviation
1	x_1	μ_1	σ_1
2	x_2	μ_2	σ_2

- Two Samples

Population	Sample Size	Sample Mean	Sample Standard Deviation
1	n_1	$\bar{x}_1$	s_1
2	n_2	$\bar{x}_2$	s_2

Example

An economist studies whether a financial literacy workshop increases household saving rates. Households are randomly assigned to two groups: one group receives a savings brochure, and the other attends a financial literacy workshop. After six months, the savings rate is recorded

Population	Variable	Mean	Standard Deviation
1	Saving rate (brochure)	μ_1	σ_1
2	Saving rate (workshop)	μ_2	σ_2

Population	Sample Size	Sample Mean	Sample Standard Deviation
1	$n_1 = 60$	$\bar{x}_1 = 5.8$	$s_1 = 2.0$
2	$n_2 = 65$	$\bar{x}_2 = 7.1$	$s_2 = 2.3$

- $H_0 : \mu_1 = \mu_2$
- $H_a : \mu_1 \neq \mu_2$
- To make inferences about the difference $\mu_1 - \mu_2$, we use the difference $\bar{x}_1 - \bar{x}_2$

Two-Sample t Procedures

- To assess the significance of the observed difference between the means of our two samples, we follow a familiar path
- Whether an observed difference is unusual depends **not only on the difference between the two means but also on the variability in the observations**
 - This is because widely different means can arise just by chance if the individual observations vary a great deal
 - To take variation into account, we standardize the observed difference $(\bar{x}_1 - \bar{x}_2)$ by dividing by its standard deviation

$$SD_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

- * SD gets larger as σ_1 or σ_2 increases
- * SD gets smaller as the sample sizes n_1 and n_2 increase
- Because we don't know the population standard deviations (σ_1, σ_2) , we **estimate** them from the sample standard deviations from our two samples

$$SE_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

- When we standardize the estimate by dividing it by its standard error, the result is the **two-sample t statistic**:

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$= \frac{\text{Estimate of parameter} - \text{Null hypothesis value of the parameter}}{\text{Standard error of estimate}}$$

- * The statistic has the same interpretation as any z or t statistic: says how far $\bar{x}_1 - \bar{x}_2 = 0$ is from $\mu_1 - \mu_2 = 0$ in standard error units

• The Two-Sample t Procedures

1. Draw an SRS of n_1 from a large Normal population with unknown mean μ_1 and draw an independent SRS of size n_2 from another large Normal population with unknown mean μ_2 .
2. A level C confidence interval for $\mu_1 - \mu_2$ is given by

$$(\bar{x}_1 - \bar{x}_2) \pm t^* \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

- t^* : the critical value for confidence level C for the t distribution

3. To test the hypothesis $H_0 : \mu_1 = \mu_2$, calculate the **two-sample t statistic**

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

4. Find the p -value from the t distribution with degrees of freedom

$$df = \min(n_1 - 1, n_2 - 1)$$

- Locate the row corresponding to df in the t table
- Find where the observed $|t|$ falls among the critical values t^* in that row
- The p -value for a **two-sided test** satisfies $p = 2 \times P(T \geq |t|)$, where $T \sim t(df)$
- If $|t|$ exceeds the critical value t^* at significance level α , then $p < \alpha$ and we reject H_0
- Note: the t table gives *upper-tail* critical values, so a two-sided p -value is found by doubling the tail probability

Example

A retail company wants to test whether a promotional campaign increases weekly sales. A random sample of stores is selected: some ran the promotion, others did not. The following data were collected:

	Sample size	Sample mean	Sample SD
With promotion	$n_1 = 7$	$\bar{x}_1 = 390$	$s_1 = 42$
Without promotion	$n_2 = 7$	$\bar{x}_2 = 320$	$s_2 = 38$

We test the hypotheses:

$$H_0 : \mu_1 = \mu_2$$

$$H_a : \mu_1 \neq \mu_2$$

Step 1: Compute the t -statistic

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{390 - 320}{\sqrt{\frac{42^2}{7} + \frac{38^2}{7}}} = \frac{70}{\sqrt{252 + 206.3}} = \frac{70}{\sqrt{458.3}} = \frac{70}{21.41} = 3.27$$

Step 2: Determine degrees of freedom

$$df = \min(n_1 - 1, n_2 - 1) = \min(6, 6) = 6$$

Step 3: Look up the t -table

With $df = 6$, the critical values from the t -table are:

df	t^* at $\alpha = 0.10$	t^* at $\alpha = 0.05$	t^* at $\alpha = 0.01$
6	1.943	2.447	3.707

Two-sided critical values for $t(df = 6)$

Since $2.447 < |t| = 3.27 < 3.707$, we have $0.01 < p < 0.05$.

Step 4: Conclusion

Since $0.01 < p < 0.05$, we can draw the following conclusions:

- Since $p < 0.05$, we **reject** H_0 at the 5% significance level. There is statistically significant evidence that the promotional campaign increases weekly sales.
- Since $p > 0.01$, we **fail to reject** H_0 at the 1% significance level. The evidence is not strong enough to conclude significance at this stricter threshold.

This illustrates that the conclusion of a hypothesis test depends on the chosen significance level α .

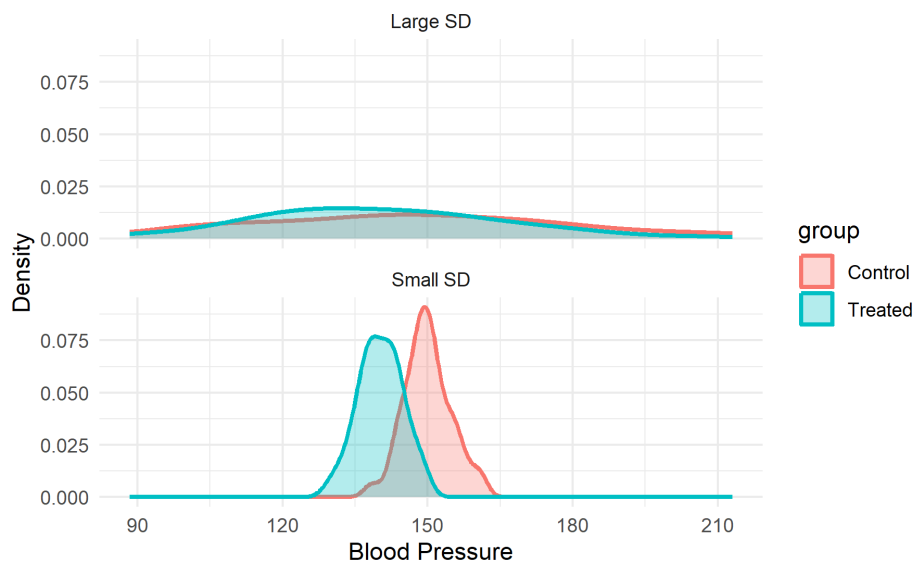
Example: Small Variance vs. Large Variance

Consider a clinical study examining whether a new drug reduces blood pressure. We compare two groups: a **treated group** (received the drug) and a **control group** (no drug). Both groups have the same sample size ($n = 40$), and the true mean difference in blood pressure is 10 mmHg (Treated mean = 140, Control mean = 150).

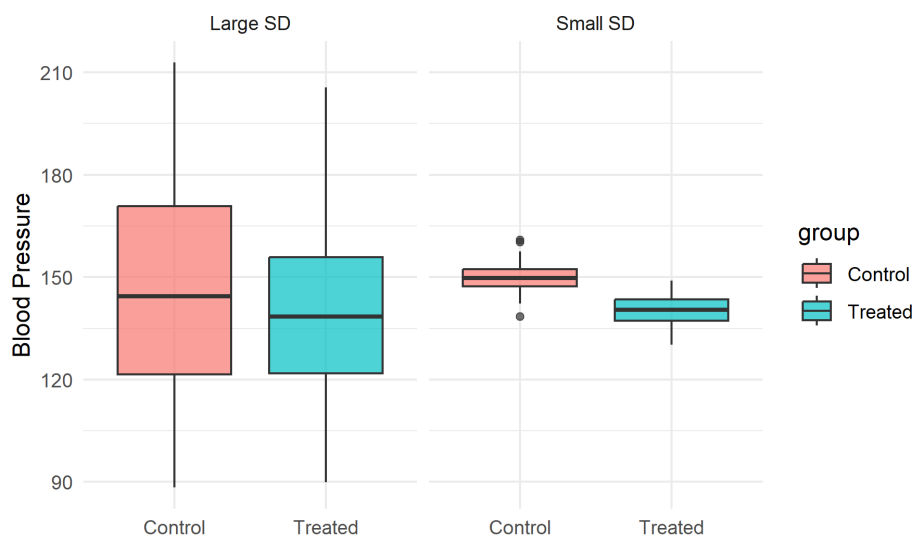
$$H_0 : \mu_{\text{Treated}} = \mu_{\text{Control}}$$

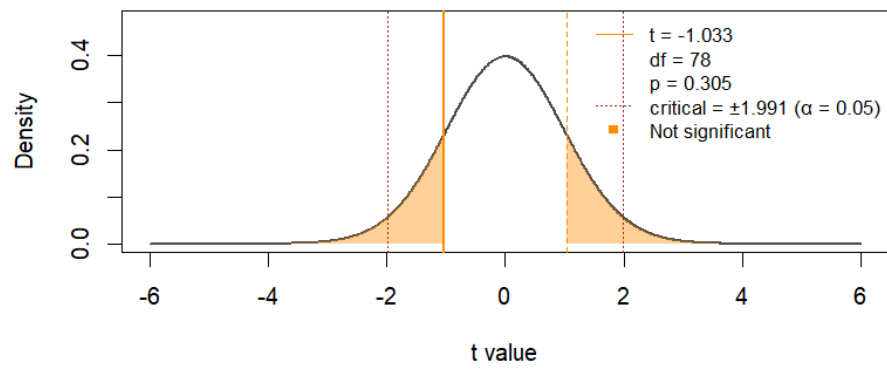
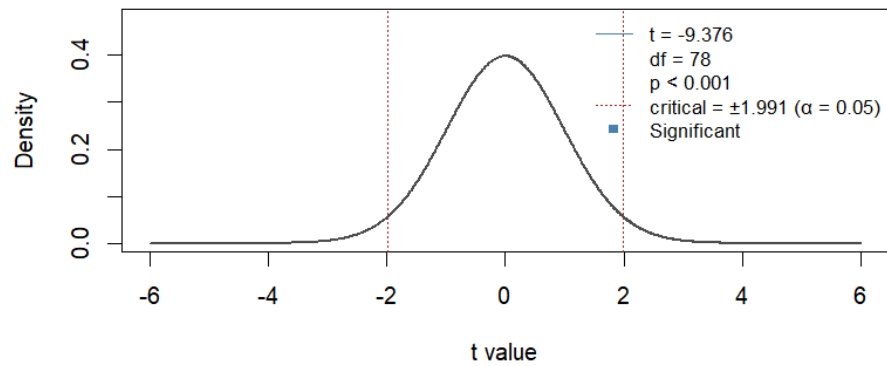
$$H_a : \mu_{\text{Treated}} \neq \mu_{\text{Control}}$$

Density Comparison: Large vs Small Variance



Blood Pressure by Treatment Group



Large SD (sd = 30) [Not significant, $p = 0.305$]**Small SD (sd = 5) [Significant, $p < 0.001$]**

Example: Small n vs. Large n

A pharmaceutical company conducts a clinical trial to test whether Wegovy (semaglutide) leads to greater weight loss compared to a control group. The true mean weight loss is 12 lbs for the Wegovy group and 5 lbs for the control group, with a standard deviation of 15 lbs in both groups. We test the hypotheses:

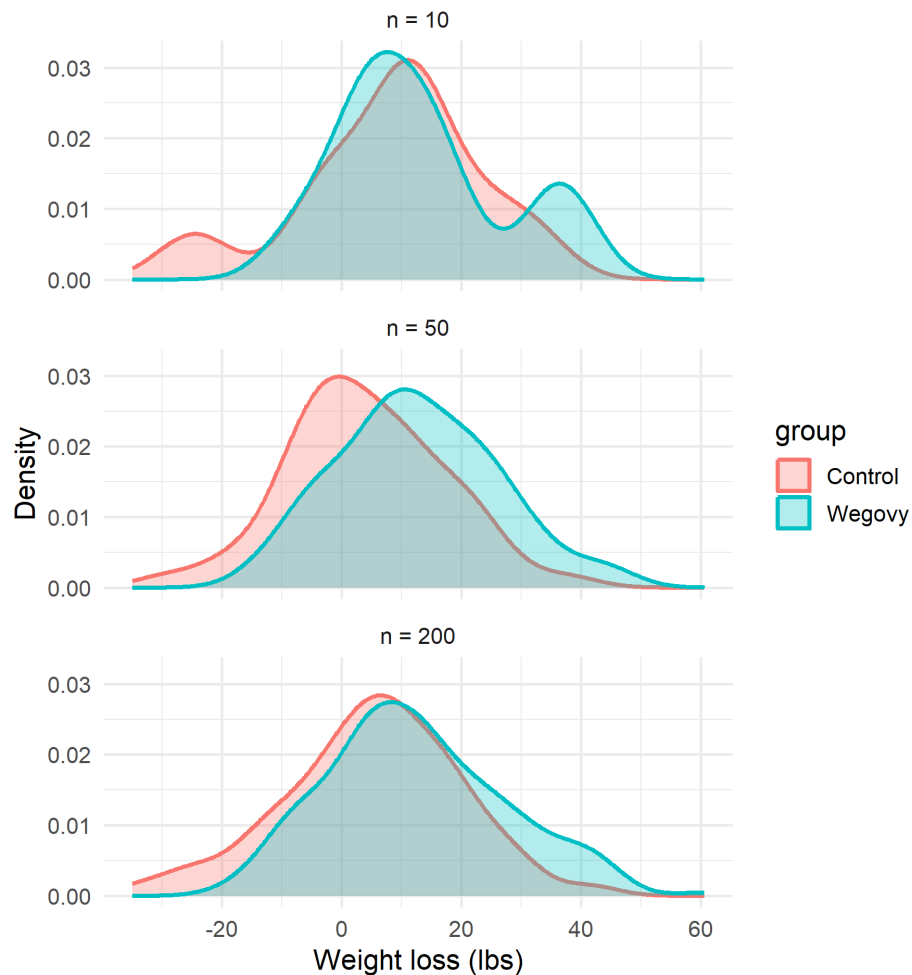
$$H_0 : \mu_{\text{Wegovy}} = \mu_{\text{Control}}$$

$$H_1 : \mu_{\text{Wegovy}} \neq \mu_{\text{Control}}$$

We examine three scenarios with increasing sample sizes ($n = 10, 50, 200$ per group), holding the mean difference and standard deviation constant.

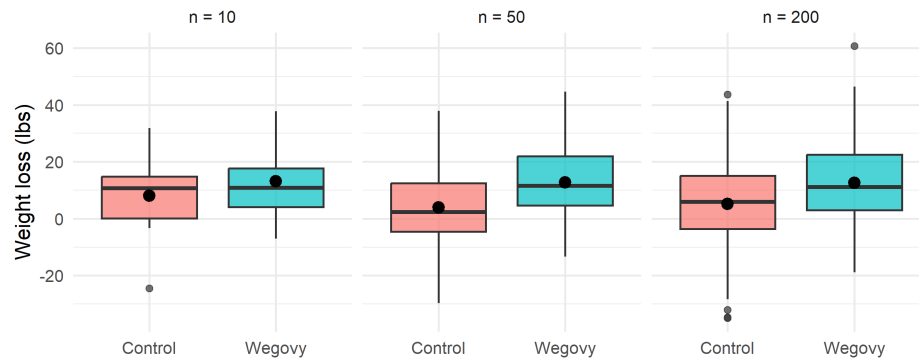
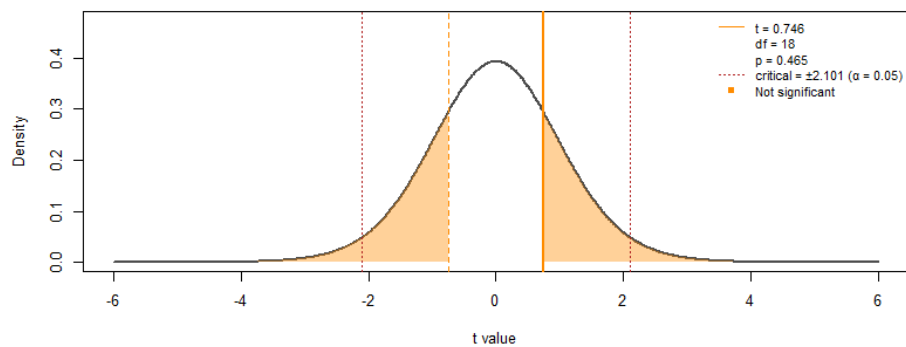
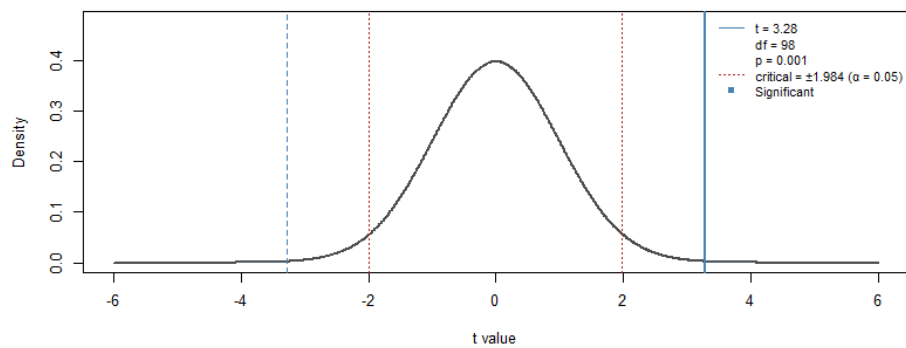
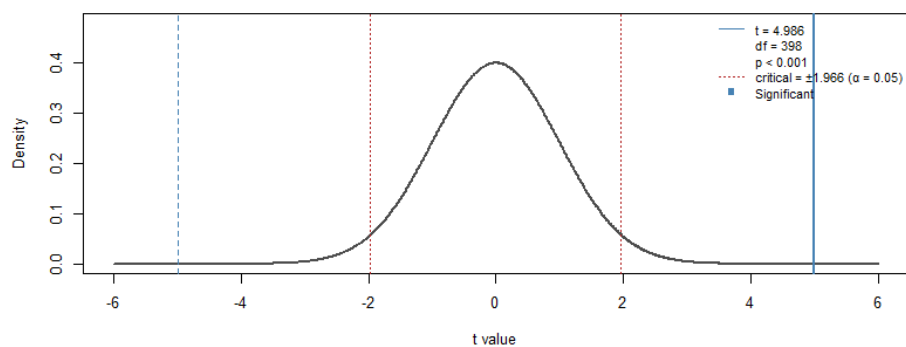
Weight loss: same mean difference, increasing sample :

Wegovy mean = 12 lbs, Control mean = 5 lbs, SD = 15 lbs



Weight loss by treatment group

Black dot = group mean

**n = 10 per group [Not significant, $p = 0.465$]****n = 50 per group [Significant, $p = 0.001$]****n = 200 per group [Significant, $p < 0.001$]**

Practice Question 21-1

A city health department wants to test whether a new exercise program reduces resting heart rate. Participants were randomly assigned to either the exercise program or a control group. The following data were collected:

	Sample size	Sample mean	Sample SD
Exercise program	$n_1 = 10$	$\bar{x}_1 = 68$	$s_1 = 8$
Control group	$n_2 = 6$	$\bar{x}_2 = 75$	$s_2 = 9$

We expect the exercise program to *reduce* resting heart rate, so we conduct a one-sided test:

$$H_0 : \mu_1 = \mu_2$$

$$H_a : \mu_1 < \mu_2$$

Step 1: Compute the t -statistic

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{68 - 75}{\sqrt{\frac{8^2}{10} + \frac{9^2}{6}}} = \frac{-7}{\sqrt{6.4 + 13.5}} = \frac{-7}{\sqrt{19.9}} = \frac{-7}{4.461} = -1.569$$

Step 2: Determine degrees of freedom

$$df = \min(n_1 - 1, n_2 - 1) = \min(9, 5) = 5$$

Step 3: Look up the t -table

For a **one-sided** test, we look up the critical value in the *left tail* only. With $df = 5$, the critical values are:

df	t^* at $\alpha = 0.10$	t^* at $\alpha = 0.05$	t^* at $\alpha = 0.01$
5	-1.476	-2.015	-3.365

One-sided (left-tail) critical values for $t(df = 5)$

Since $-1.476 > t = -1.569 > -2.015$, we have $0.05 < p < 0.10$.

Step 4: Conclusion

Since $0.05 < p < 0.10$:

- Since $p > 0.05$, we **fail to reject** H_0 at the 5% significance level. There is not sufficient evidence that the exercise program reduces resting heart rate.
- Since $p < 0.10$, we **reject** H_0 at the 10% significance level. There is marginal evidence that the exercise program reduces resting heart rate.

Note: For a one-sided test, the p -value is the area in *one tail only*, which is half the p -value of the corresponding two-sided test. This makes it easier to reject H_0 when we have a clear directional hypothesis — but the direction must be specified *before* collecting data.

Practice Question 21-2

An e-commerce company wants to test whether a new website design changes the average time customers spend on the site. A random sample of customers is observed under each design. The following data were collected:

	Sample size	Sample mean (minutes)	Sample SD
New design	$n_1 = 9$	$\bar{x}_1 = 12.4$	$s_1 = 3.2$
Old design	$n_2 = 7$	$\bar{x}_2 = 10.1$	$s_2 = 2.8$

Since the new design could either increase or decrease time on site, we conduct a two-sided test:

$$H_0 : \mu_1 = \mu_2$$

$$H_a : \mu_1 \neq \mu_2$$

Step 1: Compute the t -statistic

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{12.4 - 10.1}{\sqrt{\frac{3.2^2}{9} + \frac{2.8^2}{7}}} = \frac{2.3}{\sqrt{1.138 + 1.120}} = \frac{2.3}{\sqrt{2.258}} = \frac{2.3}{1.503} = 1.530$$

Step 2: Determine degrees of freedom

$$df = \min(n_1 - 1, n_2 - 1) = \min(8, 6) = 6$$

Step 3: Look up the t -table

For a **two-sided** test with $df = 6$, the critical values are:

df	t^* at $\alpha = 0.10$	t^* at $\alpha = 0.05$	t^* at $\alpha = 0.01$
6	1.943	2.447	3.707

Two-sided critical values for $t(df = 6)$

Since $|t| = 1.530 < 1.943$, we have $p > 0.10$.

Step 4: Conclusion

Since $p > 0.10$, we **fail to reject** H_0 at all conventional significance levels ($\alpha = 0.10, 0.05, 0.01$). There is not sufficient evidence that the new website design changes the average time customers spend on the site.

Note: The two-sided test is appropriate here because the company has no prior expectation about the direction of the change — the new design could increase or decrease time on site. Choosing a two-sided test requires a larger $|t|$ to reject H_0 compared to a one-sided test, making it a more conservative and generally safer choice when direction is uncertain.

CH 22. Inference about a Population Proportion

- So far we have focused on inference about population *means*. Now we turn to questions about the **proportion** of some outcome in a population.
- Even when the original response is a quantitative variable — such as total cholesterol — we might be more interested in whether or not someone has cholesterol greater than 200 mg/dl, which converts the variable into a **binary outcome**: high or not high.

The Sample Proportion $\hat{p}$

- We are interested in the unknown proportion p of a population that has some outcome
- The statistic that estimates the parameter p is the **sample proportion**:

$$\hat{p} = \frac{\text{number of success in the sample}}{\text{total number of individuals in the sample}}$$

- How good is the statistic $\hat{p}$ as an estimate of the parameter p ?
 - The sampling distribution of $\hat{p}$ answers this question

- **Sampling Distribution of a Sample Proportion**

- Draw an SRS of size n from a large population that contains proportion p of successes.
- Let $\hat{p}$ be the **sample proportion** of successes,

$$\hat{p} = \frac{\text{number of successes in the sample}}{n}$$

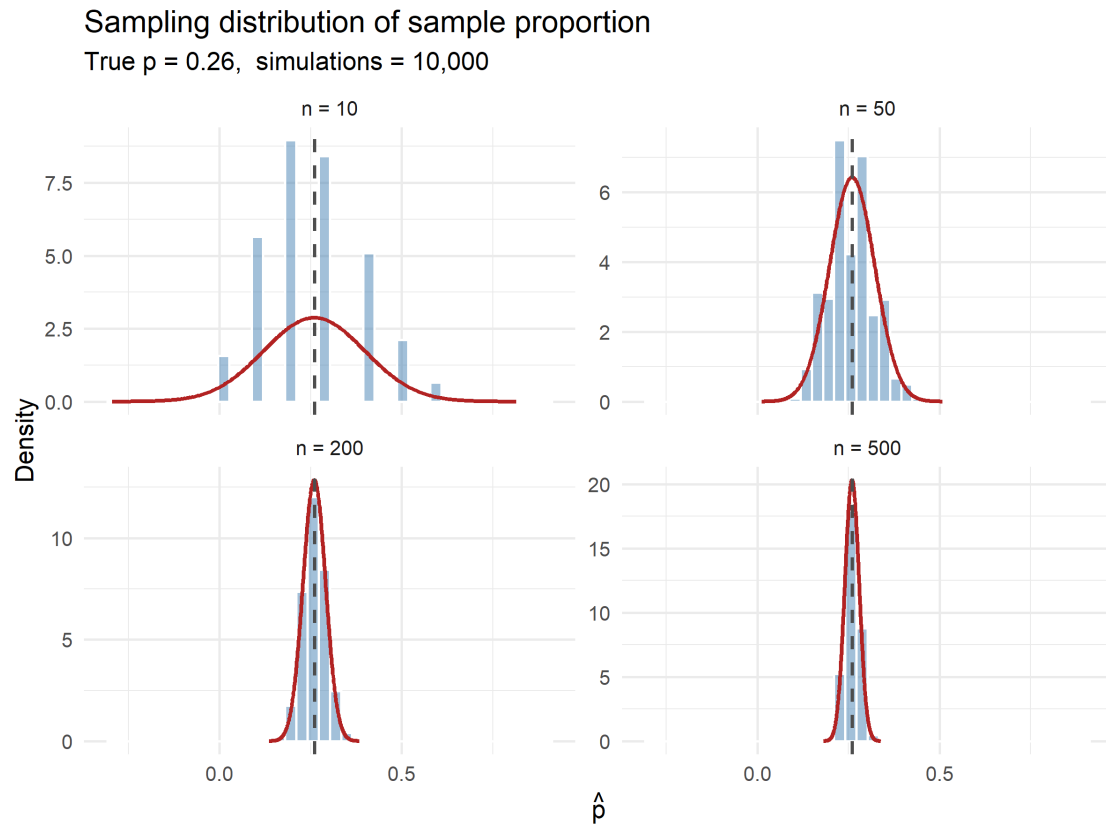
Then

1. The mean of the sampling distribution is p
2. The standard deviation of the sampling distribution is

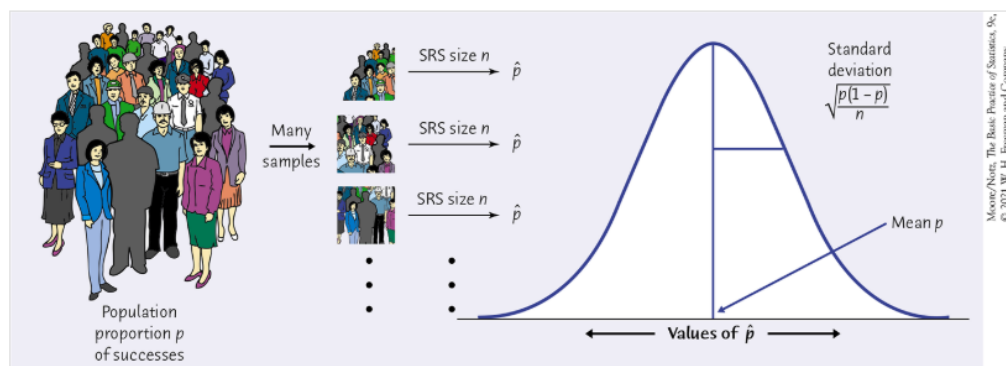
$$\sqrt{\frac{p(1-p)}{n}}$$

3. As the sample size increases, the sampling distribution of $\hat{p}$ becomes approximately Normal. That is, for large n

$$\hat{p} \sim N(p, \sqrt{p(1-p)/n})$$



- The behavior of sample proportions $\hat{p}$ is similar to the behavior of sample means $\bar{x}$
 - $\hat{p}$ is an unbiased estimator of p
 - The standard deviation of $\hat{p}$ gets smaller as the sample size n gets larger, so that estimation is likely to be more accurate when the sample is larger



Practice Question 22-1

A researcher is studying credit card delinquency among American households. In a random sample of 4,532 households, 813 reported being at least 30 days late on a credit card payment in the past year.

1. Describe the population and explain in words what the parameter p represents.
2. Give the numerical value of the statistic $\hat{p}$ that estimates p .
3. If n is large enough, what kind of distribution will $\hat{p}$ follow? State the conditions that must be satisfied.

Answer

- **Population:** All American households with a credit card. The parameter p is the true proportion of households that were at least 30 days late on a credit card payment in the past year.
- **Sample proportion:**

$$\hat{p} = \frac{813}{4532} = 0.1794$$

That is, approximately 17.9% of households in the sample reported a late credit card payment.

- **Sampling distribution:** If n is large enough, $\hat{p}$ follows an approximately Normal distribution:

$$\hat{p} \approx N\left(p, \sqrt{\frac{p(1-p)}{n}}\right) \approx N(0.1794, 0.0057)$$

Large-Sample Confidence Intervals for a Proportion

- We can follow the same path from sampling distribution to confidence interval as we did for $\bar{x}$ in Chapter 15
- The confidence interval is

$$\hat{p} \pm z^* \sqrt{\frac{p(1-p)}{n}}$$

- This won't do because we don't know the value of p . So we replace the standard deviation with the standard error of $\hat{p}$

$$SE_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

- So the confidence interval will be

$$\text{estimate} \pm z^* SE_{\text{estimate}}$$

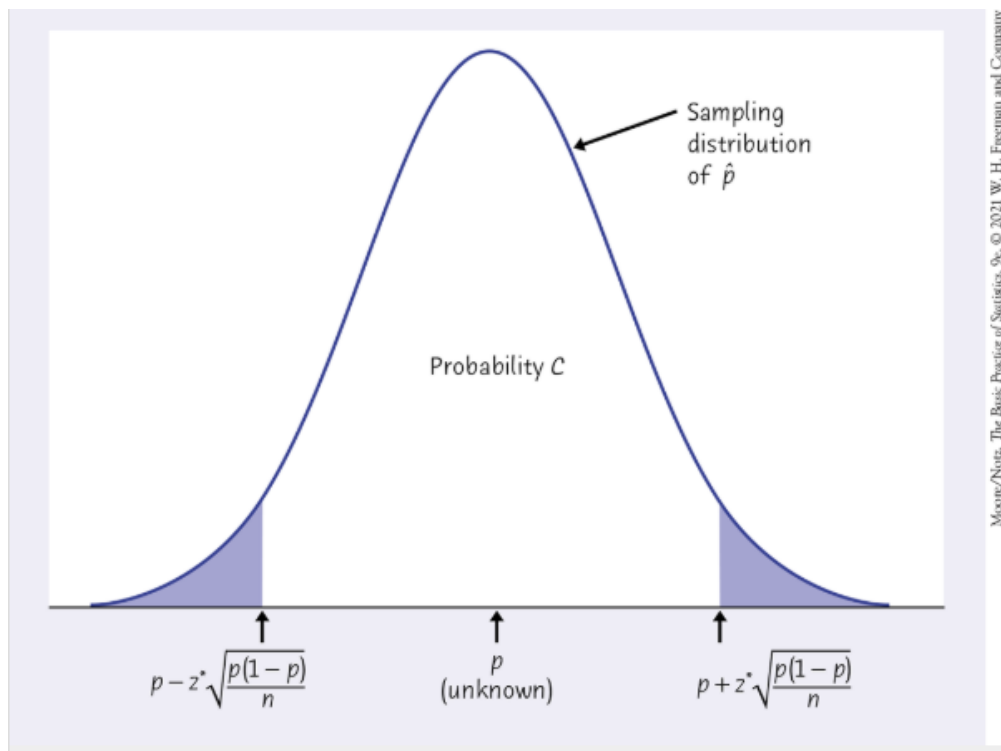
$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

• Large-Sample Confidence Interval for a Population Proportion

- Draw an SRS of size n from a large population that contains an unknown proportion p successes.
- An approximate level C confidence interval for p is

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

- * z^* : the critical value for the standard Normal density curve with area C between $-z^*$ and z^*
- * Use this interval only when the numbers of successes and failures in the sample are both at least 15



- Why not t ?

- We do not change z^* to t^* when we replace the standard deviation with the standard error for sample proportion $\hat{p}$.
- For the sample mean $\bar{x}$, we used t^* because we had to estimate *two* unknown quantities — the population mean μ and the population standard deviation σ — introducing extra uncertainty that the t -distribution accounts for.
- For the sample proportion $\hat{p}$, there is only *one* unknown parameter: p . Once we know p , the standard deviation is completely determined:

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$$

There is no separate unknown variance to estimate — the variance is a direct function of p itself.

- Therefore, substituting $\hat{p}$ for p in the standard error introduces much less additional uncertainty than substituting s for σ in the mean case. For large n , the Normal approximation remains appropriate and we use z^* instead of t^* .
- As a rule of thumb, the Normal approximation is valid when:

$$n\hat{p} \geq 10 \quad \text{and} \quad n(1 - \hat{p}) \geq 10$$

Practice Question 22-2

A financial analyst wants to estimate the proportion of small businesses that applied for a bank loan in the past year. In a random sample of 1,200 small businesses, 312 reported applying for a bank loan.

1. Construct a 95% confidence interval for the true proportion p of small businesses that applied for a bank loan.
2. Interpret the confidence interval in context.
3. Construct a 90% confidence interval and compare it to the 95% interval.

Answer

1. **95% confidence interval:**

The general formula is:

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

First compute the standard error:

$$SE(\hat{p}) = \sqrt{\frac{0.260 \times 0.740}{1200}} = \sqrt{\frac{0.1924}{1200}} = \sqrt{0.000160} = 0.01265$$

For a 95% confidence interval, $z^* = 1.960$:

$$0.260 \pm 1.960 \times 0.01265 = 0.260 \pm 0.0248 = (0.235, 0.285)$$

2. **Interpretation:** We are 95% confident that the true proportion of small businesses that applied for a bank loan in the past year is between 23.5% and 28.5%.
3. **90% confidence interval:**

For a 90% confidence interval, $z^* = 1.645$:

$$0.260 \pm 1.645 \times 0.01265 = 0.260 \pm 0.0208 = (0.239, 0.281)$$

Comparing the two intervals:

Confidence level	Interval	Width
90%	(0.239, 0.281)	0.042
95%	(0.235, 0.285)	0.050

The 95% interval is wider than the 90% interval. This illustrates the trade-off between confidence and precision — higher confidence requires a wider interval to guarantee capturing the true parameter p .

Significance Tests for a Proportion

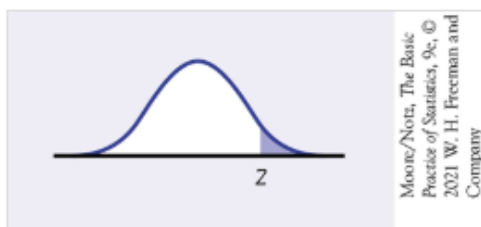
• Significance Tests for a Proportion

- Draw an SRS of size n from a large population that contains an unknown proportion p of successes.
- To test the hypothesis $H_0 : p = \hat{p}$, compute the z **statistic**

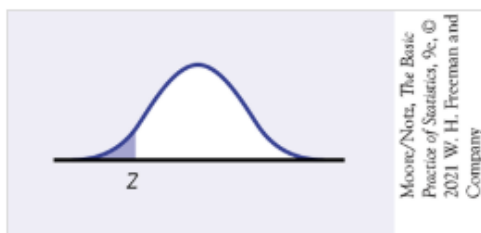
$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

- In terms of a variable Z having the standard Normal distribution, the approximate P-value for a test of H_a against
 - * $H_a : p > p_0$ is $P(Z \geq z)$
 - * $H_a : p < p_0$ is $P(Z \leq z)$
 - * $H_a : p \neq p_0$ is $2P(Z \geq |z|)$

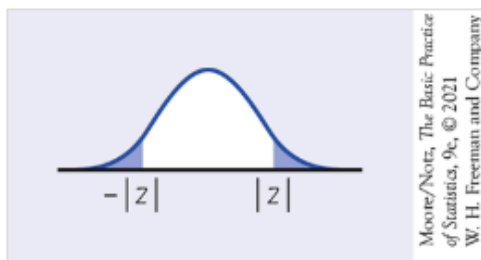
$$H_a : p > p_0 \text{ is } P(Z \geq z)$$



$$H_a : p < p_0 \text{ is } P(Z \leq z)$$



$$H_a : p \neq p_0 \text{ is } 2P(Z \geq |z|)$$



Practice Question 22-3

A national survey conducted in 2020 found that 34% of American adults held credit card debt. An economist suspects that this proportion has changed since then. In a recent random sample of 850 American adults, 312 reported holding credit card debt.

1. State the null and alternative hypotheses.
2. Compute the z -statistic.
3. Find the p -value and state your conclusion at $\alpha = 0.05$.

Answer

1. **Hypotheses:**

$$H_0 : p = 0.34$$

$$H_a : p \neq 0.34$$

The two-sided test is appropriate because the economist suspects the proportion has *changed* — it could have increased or decreased since 2020.

2. **Compute the z -statistic:**

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}} = \frac{0.367 - 0.34}{\sqrt{\frac{0.34 \times 0.66}{850}}} = \frac{0.027}{\sqrt{0.000264}} = \frac{0.027}{0.01625} = 1.661$$

3. **p -value and conclusion:**

For a two-sided test, the p -value is:

$$p\text{-value} = 2P(Z \geq |z|) = 2P(Z \geq 1.661) = 2 \times 0.0484 = 0.0968$$

Since $p\text{-value} = 0.0968 > 0.05$, we **fail to reject** H_0 at the 5% significance level. There is not sufficient evidence that the proportion of American adults holding credit card debt has changed since 2020.

However, since $p\text{-value} < 0.10$, we would reject H_0 at the 10% significance level — the result is marginally significant and may warrant further investigation with a larger sample.

Chapter 23. Comparing Two Proportions

- When a comparison involves the *means* of two populations, we use the two-sample t methods of Chapter 21
- In this chapter, we consider two-sample problems in which the measurement on the individual can be categorized as either a success or a failure (binary outcome)
- Our goal is to compare the **proportions** of successes in the two populations

Two-Sample Problems: Proportions

Example: Two Sample Problems: Proportions

- Does the proportion of households receiving government assistance differ between urban and rural areas?
- Does the proportion of individuals with health insurance differ between high-income and low-income groups?
- Does the proportion of households that hold stock investments differ between married and single individuals?
- We will use notation similar to that used in our study of two-sample t statistics

Population	Population Proportion	Sample Size	Sample Proportion
1	p_1	n_1	$\hat{p}_1$
2	p_2	n_2	$\hat{p}_2$

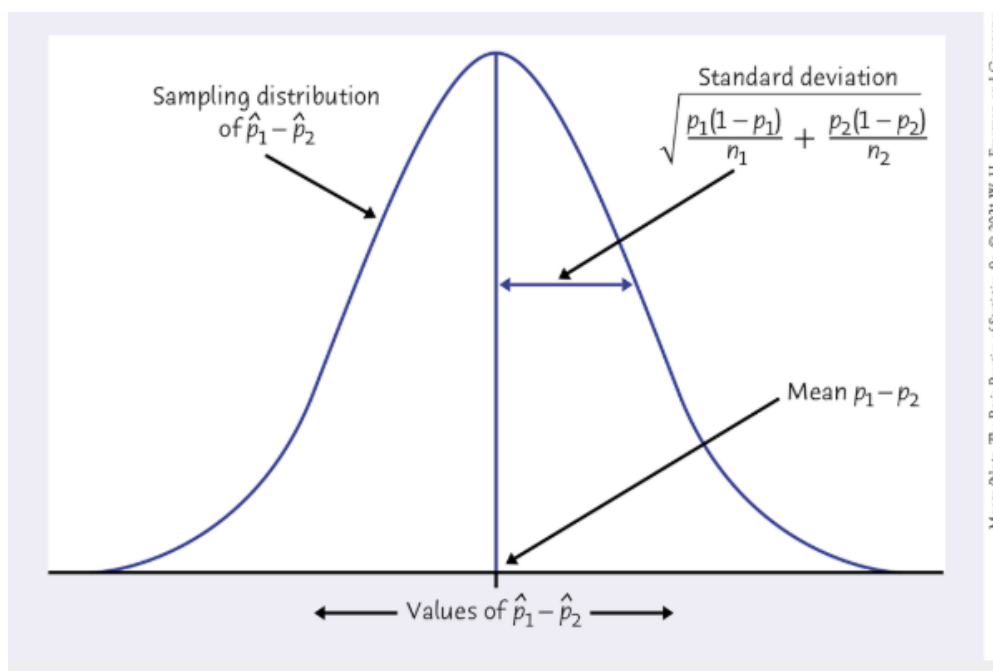
- We compare the populations by doing inference about the difference $p_1 - p_2$ between the population proportions
- The statistic that estimates this difference is the difference between the two sample proportions, $\hat{p}_1 - \hat{p}_2$

The Sampling Distribution of a Difference between Proportions

- To use $\hat{p}_1 - \hat{p}_2$ for inference, we must know its **sampling distribution**. Here are the facts we need:
 1. When the samples are large, the distribution of $\hat{p}_1 - \hat{p}_2$ is **approximately Normal**
 2. The mean of the sampling distribution is $p_1 - p_2$. That is, the difference between sample proportions is an **unbiased estimator** of the difference between population proportions
 3. The standard deviation of the distribution is

$$\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$$

- The variability in the difference depends on the **combined variability** of the two samples



Large-Sample Confidence Intervals for Comparing Proportions

- To obtain a confidence interval, **replace** the population proportions p_1 and p_2 in the standard deviation with the sample proportions
- The result is the **standard error** of the statistic $\hat{p}_1 - \hat{p}_2$:

$$SE_{\hat{p}_1 - \hat{p}_2} = \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$$

- The confidence interval has the same form we met in Chapter 22:

$$\text{estimate} \pm z^* SE_{\text{estimate}}$$

- **Large-Sample Confidence Intervals for Comparing Two proportions**

- Draw an SRS of size n_1 from a large population having proportion p_1 of successes and draw an independent SRS of size n_2 from another large population having proportion p_2 of successes
- When n_1 and n_2 are large, an approximate level C confidence interval for $p_1 - p_2$ is

$$(\hat{p}_1 - \hat{p}_2) \pm z^* SE_{\hat{p}_1 - \hat{p}_2}, \text{ where } SE_{\hat{p}_1 - \hat{p}_2} = \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$$

- z^* : the critical value for that standard Normal density curve with area C between $-z^*$ and z^*

- **Confidence Intervals provide evidence**

- If the confidence interval for the difference $p_1 - p_2$ includes 0
 - * We should not rule out the possibility that the difference is negligible
- If all the values in the interval for $p_1 - p_2$ are positive
 - * This provides evidence that $p_1 > p_2$
- If all the values in the interval for $p_1 - p_2$ are negative
 - * This provides evidence that $p_1 < p_2$
- Narrow confidence intervals for $p_1 - p_2$
 - * Suggest greater certainty about the magnitude of the difference
- Wide confidence intervals for $p_1 - p_2$
 - * indicate large uncertainty about the difference

Practice Question 23-1

Does the proportion of households receiving government assistance differ between urban and rural areas?

A random sample of households was collected from urban and rural areas. The results are summarized below.

Population	Description	Sample Size	Number of Successes	Sample Proportion
1	urban	$n_1 = 100$	$x_1 = 75$	$\hat{p}_1 = \frac{75}{100} = 0.75$
2	rural	$n_2 = 100$	$x_2 = 56$	$\hat{p}_2 = \frac{56}{100} = 0.56$

Let p_1 be the true proportion of urban households receiving government assistance, and let p_2 be the true proportion of rural households receiving government assistance.

1. Compute the point estimate $\hat{p}_1 - \hat{p}_2$.
 2. Find the standard error of $\hat{p}_1 - \hat{p}_2$.
 3. Construct a 95% confidence interval for $p_1 - p_2$.
 4. Interpret the confidence interval in the context of the problem.
1. Compute the point estimate $\hat{p}_1 - \hat{p}_2$.

$$\hat{p}_1 - \hat{p}_2 = 0.75 - 0.56 = 0.19$$

So, the point estimate of the difference in population proportions is

$$\hat{p}_1 - \hat{p}_2 = 0.19.$$

2. Find the standard error of $\hat{p}_1 - \hat{p}_2$.

The standard error for the difference in two sample proportions is

$$SE = \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}.$$

Substituting the sample values:

$$\begin{aligned} SE &= \sqrt{\frac{0.75(1 - 0.75)}{100} + \frac{0.56(1 - 0.56)}{100}} \\ &= \sqrt{\frac{0.75(0.25)}{100} + \frac{0.56(0.44)}{100}} \\ &= \sqrt{\frac{0.1875}{100} + \frac{0.2464}{100}} \end{aligned}$$

$$= \sqrt{0.001875 + 0.002464}$$

$$= \sqrt{0.004339} \approx 0.0659.$$

Therefore,

$$SE \approx 0.0659.$$

3. Construct a 95% confidence interval for $p_1 - p_2$.

For a 95% confidence interval, the critical value is

$$z^* = 1.96.$$

The margin of error is

$$ME = z^* \cdot SE = 1.96(0.0659) \approx 0.1292.$$

Thus, the confidence interval is

$$(\hat{p}_1 - \hat{p}_2) \pm ME$$

$$= 0.19 \pm 0.1292$$

$$= (0.0608, 0.3192).$$

So, the 95% confidence interval for $p_1 - p_2$ is

$$(0.061, 0.319)$$

after rounding to three decimal places.

4. Interpret the confidence interval in the context of the problem.

We are 95% confident that the true proportion of urban households receiving government assistance is between 0.061 and 0.319 higher than the true proportion of rural households receiving government assistance.

In other words, urban households appear to have a higher proportion receiving government assistance than rural households, by about 6.1% to 31.9%.

Significance Tests for Comparing Proportions

- An observed difference between two sample proportions can reflect an actual difference between the populations, or it may just be due to chance variation in random sampling
- Significance tests help us decide if the effect we see in samples is really there in the populations
- The null hypothesis says that there is no difference between the two populations: $H_0 : p_1 = p_2$ The alternative hypothesis says what kind of difference we expect

$$- H_a : p_1 \neq p_2$$

$$- H_a : p_1 > p_2$$

$$- H_a : p_1 < p_2$$

- **Pooled Sample Proportion: $\hat{p}$**

- To do a hypothesis test, we standardize the difference between the sample proportions $\hat{p}_1 - \hat{p}_2$ to get a z -statistic
- If $H_0 : p_1 = p_2$ is true, both samples come from populations in which the **same** unknown proportion p
- Therefore, we take advantage of this by combining the two samples to estimate this single p instead of estimating p_1 and p_2 separately
- Pooled Sample Proportion:

$$\hat{p} = \frac{\text{number of successes in both samples combined}}{\text{number of individuals in both samples combined}} = \frac{x_1 + x_2}{n_1 + n_2}$$

where x_1, x_2 = number of successes

- Thus, for significant test, we use $\hat{p}$ for estimating the variability instead of $\hat{p}_1$ and $\hat{p}_2$:

$$\begin{aligned} SE_{\hat{p}_1 - \hat{p}_2} &= \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}} = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n_1} + \frac{\hat{p}(1 - \hat{p})}{n_2}} \\ &= \sqrt{\hat{p}(1 - \hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)} \end{aligned}$$

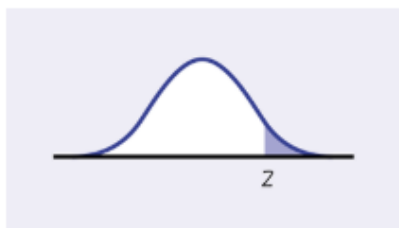
• Significance Test for Comparing Two Proportions

- Draw an SRS of size n_1 from a large population having proportion p_1 of successes and draw an independent SRS of size n_2 from another large population having proportion p_2 of successes
- To test the hypothesis $H_0 : p_1 = p_2$, first find the **pooled proportion $\hat{p}$ of successes in both samples combined**
- Then compute the z -statistic

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

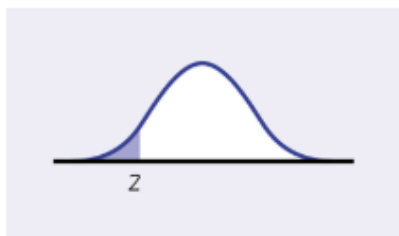
- In terms of a variable Z having the standard Normal distribution, the P-value for a test of H_0 against

$$H_a: p_1 > p_2 \text{ is } P(Z \geq z)$$



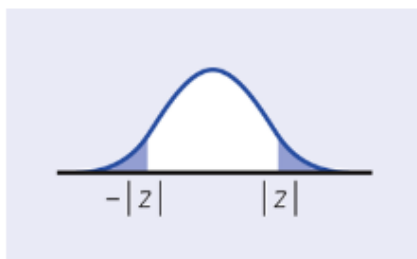
Moore/Notz, The Basic
Practice of Statistics, 9e, ©
2021 W. H. Freeman and
Company

$$H_a: p_1 < p_2 \text{ is } P(Z \leq z)$$



Moore/Notz, The Basic
Practice of Statistics, 9e, ©
2021 W. H. Freeman and
Company

$$H_a: p_1 \neq p_2 \text{ is } 2P(Z \geq |z|)$$



Moore/Notz, The Basic Practice
of Statistics, 9e, © 2021
W. H. Freeman and Company

Example

An economist wants to investigate whether the proportion of households that own stock investments differs between married and single individuals. A random sample was collected from each group. The results are summarized below. Let p_1 be the true proportion of married individuals who own stock investments, and let p_2 be the true proportion of single individuals who own stock investments.

Population	Description	Sample Size	Number of Successes
1	Married	$n_1 = 180$	$x_1 = 81$
2	Single	$n_2 = 140$	$x_2 = 42$

1. Compute the sample proportions $\hat{p}_1$ and $\hat{p}_2$.

$$\hat{p}_1 = \frac{x_1}{n_1} = \frac{81}{180} = 0.450 \quad \hat{p}_2 = \frac{x_2}{n_2} = \frac{42}{140} = 0.300$$

2. State the null and alternative hypotheses.

$$H_0 : p_1 = p_2$$

$$H_a : p_1 \neq p_2$$

The two-sided test is appropriate because the economist wants to detect a difference in either direction — married individuals could be more or less likely to own stocks than single individuals.

3. Compute the pooled sample proportion $\hat{p}$.

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{81 + 42}{180 + 140} = \frac{123}{320} = 0.384$$

4. Compute the z -statistic.

First compute the standard error using the pooled proportion:

$$\begin{aligned} SE &= \sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)} \\ &= \sqrt{0.384 \times 0.616 \times \left(\frac{1}{180} + \frac{1}{140} \right)} \\ &= \sqrt{0.2366 \times 0.01270} \\ &= \sqrt{0.003005} \\ &= 0.05482 \end{aligned}$$

Then compute the z -statistic:

$$z = \frac{\hat{p}_1 - \hat{p}_2}{SE} = \frac{0.450 - 0.300}{0.05482} = \frac{0.150}{0.05482} = 2.736$$

5. Find the p -value and state your conclusion at $\alpha = 0.05$.

$$p\text{-value} = 2P(Z \geq |2.736|) = 2 \times 0.0031 = 0.0062$$

Since $p\text{-value} = 0.0062 < 0.05$, we **reject** H_0 at the 5% significance level. There is strong evidence that the proportion of married individuals who own stock investments differs from the proportion of single individuals who own stock investments.

6. Construct a 95% confidence interval for $p_1 - p_2$ and interpret it.

For the confidence interval we use the **unpooled** standard error:

$$\begin{aligned} SE &= \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}} \\ &= \sqrt{\frac{0.450 \times 0.550}{180} + \frac{0.300 \times 0.700}{140}} \\ &= \sqrt{0.001375 + 0.001500} \\ &= \sqrt{0.002875} \\ &= 0.05362 \end{aligned}$$

The 95% confidence interval is:

$$\begin{aligned} (\hat{p}_1 - \hat{p}_2) \pm z^* \times SE &= 0.150 \pm 1.96 \times 0.05362 \\ &= 0.150 \pm 0.105 \\ &= (0.045, 0.255) \end{aligned}$$

Interpretation: We are 95% confident that the true proportion of married individuals who own stock investments is between 4.5% and 25.5% higher than the true proportion of single individuals who own stock investments.

Since the entire interval is **positive** and does not include zero, this confirms our earlier conclusion that married individuals are significantly more likely to own stock investments than single individuals.

latex

Answer

1. Sample proportions:

$$\hat{p}_1 = \frac{x_1}{n_1} = \frac{102}{120} = 0.850 \quad \hat{p}_2 = \frac{x_2}{n_2} = \frac{94}{150} = 0.627$$

2. Hypotheses:

$$H_0 : p_1 = p_2$$

$$H_a : p_1 \neq p_2$$

The two-sided test is appropriate because the researcher wants to detect a difference in either direction.

3. Pooled sample proportion:

$$\hat{p}_{pool} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{102 + 94}{120 + 150} = \frac{196}{270} = 0.726$$

4. z -statistic:

First compute the standard error using the pooled proportion:

$$\begin{aligned} SE &= \sqrt{\hat{p}_{pool}(1 - \hat{p}_{pool}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)} \\ &= \sqrt{0.726 \times 0.274 \times \left(\frac{1}{120} + \frac{1}{150} \right)} \\ &= \sqrt{0.1989 \times 0.01500} \\ &= \sqrt{0.002983} \\ &= 0.05462 \end{aligned}$$

Then compute the z -statistic:

$$z = \frac{\hat{p}_1 - \hat{p}_2}{SE} = \frac{0.850 - 0.627}{0.05462} = \frac{0.223}{0.05462} = 4.083$$

5. p -value and conclusion:

$$p\text{-value} = 2P(Z \geq |4.083|) = 2 \times 0.0000222 = 0.0000444$$

Since $p\text{-value} < 0.05$, we **reject** H_0 at the 5% significance level. There is strong evidence that the proportion of individuals with health insurance differs between high-income and low-income groups.

6. 95% confidence interval for $p_1 - p_2$:

For the confidence interval we use the **unpooled** standard error:

$$\begin{aligned} SE &= \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}} \\ &= \sqrt{\frac{0.850 \times 0.150}{120} + \frac{0.627 \times 0.373}{150}} \\ &= \sqrt{0.001063 + 0.001559} \\ &= \sqrt{0.002621} \\ &= 0.05120 \end{aligned}$$

The 95% confidence interval is:

$$\begin{aligned} (\hat{p}_1 - \hat{p}_2) \pm z^* \times SE &= 0.223 \pm 1.96 \times 0.05120 \\ &= 0.223 \pm 0.100 \\ &= (0.123, 0.323) \end{aligned}$$

Interpretation: We are 95% confident that the true proportion of high-income individuals with health insurance is between 12.3% and 32.3% higher than the true proportion of low-income individuals with health insurance.

Since the entire interval is **positive** and does not include zero, this confirms our earlier conclusion that high-income individuals are significantly more likely to have health insurance than low-income individuals.

Chapter 26. Inference for Regression

- Inference for regression allows us to determine whether the relationship we observe in a scatterplot is valid for some larger population
- When the data are a sample from a larger population, we need statistical inference to answer questions like these about the population
 - Is there really a linear relationship between x and y in the population, or might the pattern we see in the scatterplot plausibly arise just by chance?
 - What is the slope (rate of change) that relates y to x in the population, including a margin of error for our estimate of the slope?
 - If we use the least-squares line to predict y for a given value of x , how accurate is our prediction (again, with a margin of error)?

Conditions for Regression Inference

- **Conditions for Regression Inference**

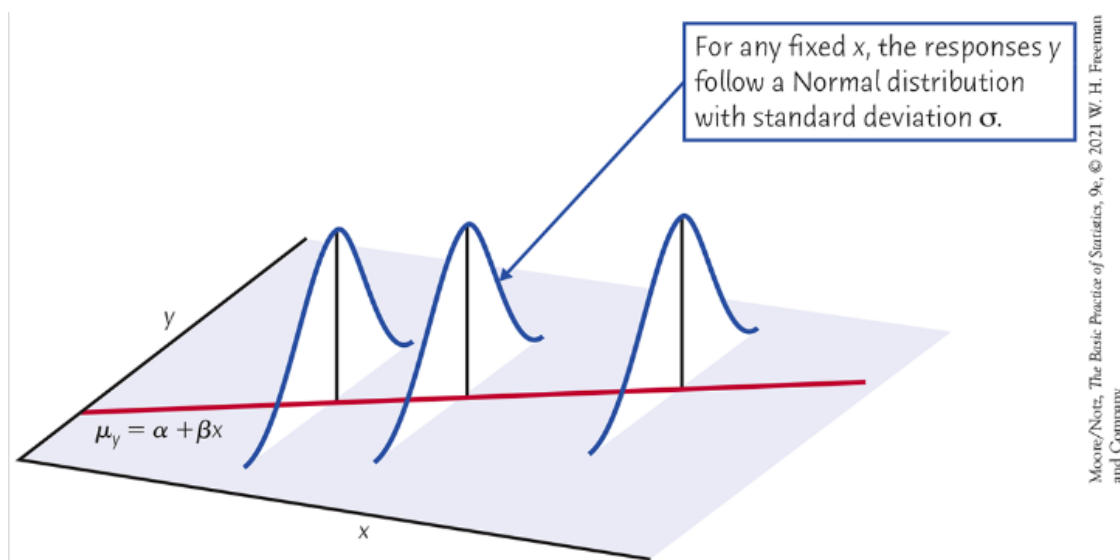
- We have n observations on an explanatory variable x and a response variable y . Our goal is to study or predict the behavior of y for given values of x
 1. **Normality and Independence:** For any fixed value of x , the response y varies according to a Normal distribution. Repeated responses y are independent of each other
 2. **Linearity:** The mean response μ_y , has a straight-line relationship with x given by a population **regression line**

$$\mu_y = \alpha + \beta x$$

α, β are unknown parameters

3. **Constant Variance (Homoscedasticity):** The standard deviation of y (call it σ) is the same for all values of x . The value of σ is unknown
- There are thus three population parameters that we must estimate from the data: α , β and σ

- Conditions for regression inference in picture form
 - The line in the figure is the **population regression line**
 - The mean of the response y moves along this line as the explanatory variable x takes different values
 - The Normal curves show how y will vary when x is held fixed at different values
 - All the curves have the same σ , so the variability of y is the same for all values of x
- We can never directly verify the conditions — we can only check them approximately using sample



Example: Wage and Education

We want to study how years of education (x) affects hourly wage (y). The population regression model is:

$$\text{wage}_i = \alpha + \beta \times \text{education}_i + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2)$$

where α , β , and σ are unknown population parameters to be estimated from the data.

1. Normality and Independence

For any fixed level of education — say, exactly 16 years (college graduates) — wages are not all identical. Some earn more, some earn less, but the distribution of wages among all college graduates follows a Normal distribution centered on the average wage for that education level.

Independence means that one person's wage does not influence another person's wage. If you randomly sample workers, knowing that worker A earns \$25/hour tells you nothing about what worker B earns.

- *What violates this?* If you sampled workers from the same firm, their wages might be correlated due to firm-specific pay scales — violating independence.

2. Linearity

The average wage increases by a fixed amount β for each additional year of education, regardless of where on the education scale you are. Going from 8 to 9 years of education increases average wage by the same amount as going from 15 to 16 years:

$$\mu_{\text{wage}} = \alpha + \beta \times \text{education}$$

- *What violates this?* If the return to education is much higher at the college level than at the high school level — a curved relationship — then a straight line is not appropriate and we need a nonlinear model.

3. Constant Variance (Homoscedasticity)

The spread of wages around the average is the same for workers with 8 years of education as for workers with 16 years. The standard deviation σ does not change as education increases.

- *What violates this?* In reality, wages tend to be more spread out at higher education levels — a college graduate might earn anywhere from \$20 to \$200/hour, while a high school dropout has a much narrower range. This is called **heteroscedasticity** and is very common in wage data.

Condition	What it says	Common violation
Normality & Independence	y is Normal at each x ; observations do not affect each other	Clustered sampling, time series data
Linearity	Mean of y is a straight line in x	Diminishing returns, threshold effects
Constant variance	Spread of y is the same at all x	Wages, income, financial returns

Summary of regression inference conditions

Estimating the Parameters

• Estimating the Population Regression Line

- Check the conditions for regression are met
- Calculate the least-squares line $\hat{y} = a + bx$
 - * b : slope of the least-squares line is an unbiased estimator of the population slope β
 - * a : the intercept of the least-squares line is an unbiased estimator of the population intercept α

• Regression Standard Error

- The parameter σ is the standard deviation, which describes the variability of the response y about the population regression line
- **Residuals** estimate how much y varies about the population line

$$\text{residual} = y - \hat{y}$$

- There are n residuals, one for each data point.
- The residuals from a least-squares line always have **mean zero**
- We estimate σ by a sample standard deviation of the residuals: **Regression Standard Error**

$$s = \sqrt{\frac{1}{n-2} \sum \text{residual}^2} = \sqrt{\frac{1}{n-2} \sum (y - \hat{y})^2}$$

- * $\sum (y - \hat{y})^2$: the sum of the squared deviations of the data points from the regression line
- * $n - 2$: degrees of freedom. This is because, to compute standard deviation, we must two parameters, both the intercept and slope.

Testing the Hypothesis of No Linear Relationship

- Test hypotheses about the slope β of the population regression line:

$$H_0 : \beta = 0$$

$$H_a : \beta \neq 0$$

- If $\beta = 0$, then the regression line with slope 0 is horizontal. The mean of y does not change at all when x changes
- So H_0 says there is ***no linear relationship*** between x and y in the population

- Significance Test for Regression Slope

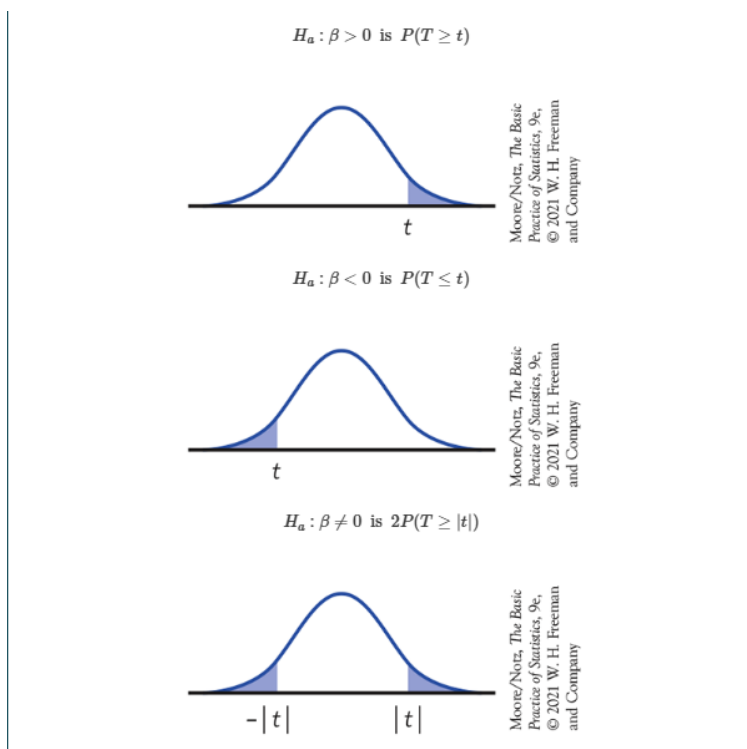
- To test the hypothesis $H_0 : \beta = 0$, compute the t-statistic

$$t = \frac{b}{SE_b}$$

- In this formula, the standard error of the least-squares slope b is

$$SE_b = \frac{s}{\sqrt{\sum (x - \bar{x})^2}}$$

- The sum runs over all observations on the explanatory variable x
- In terms of a random variable T having the t_{n-2} distribution, the P-value for a test of H_0 against



Standard Deviation of $\hat{\beta}$

What does $SD(\hat{\beta})$ mean?

$\hat{\beta}$ is a **random variable** — every time you draw a different sample from the same population, you get a different value of $\hat{\beta}$. The standard deviation of $\hat{\beta}$ measures how much $\hat{\beta}$ varies across those repeated samples.

This is exactly the same concept as the standard deviation of $\bar{y}$ — it measures how much the sample mean varies from sample to sample.

Example: Wage and Education

Suppose the true population model is:

$$\text{wage}_i = 5 + 2 \times \text{education}_i + \varepsilon_i, \quad \varepsilon_i \sim N(0, 100)$$

So the true slope is $\beta = 2$ — each additional year of education increases average wage by \$2/hour — and $\sigma = 10$.

Now imagine drawing **many different random samples** of $n = 50$ workers from this population and estimating $\hat{\beta}$ each time:

Sample	$\hat{\beta}$
1	2.34
2	1.87
3	2.15
4	1.92
$\vdots$	$\vdots$

Each sample gives a slightly different slope estimate. The **standard deviation of $\hat{\beta}$** measures how spread out these estimates are around the true $\beta = 2$.

Formula

$$SD(\hat{\beta}) = \frac{\sigma}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2}}$$

Three things affect $SD(\hat{\beta})$:

1. Larger $\sigma \rightarrow$ larger $SD(\hat{\beta})$

More noise in y makes it harder to pin down the slope. If wages are very unpredictable even after accounting for education, different samples will give very different slope estimates.

2. More spread in $x \rightarrow$ smaller $SD(\hat{\beta})$

If education ranges from 8 to 20 years, you have a long lever arm to estimate the slope precisely. If everyone has between 12 and 13 years of education, you can barely distinguish any slope at all.

3. Larger $n \rightarrow$ smaller $SD(\hat{\beta})$

More observations means $\sum(x_i - \bar{x})^2$ grows, which shrinks $SD(\hat{\beta})$.

From standard deviation to standard error

In practice, σ is unknown so we replace it with the estimated regression standard deviation s :

$$\underbrace{\frac{\sigma}{\sqrt{\sum(x_i - \bar{x})^2}}}_{\text{standard deviation of } \hat{\beta}} \xrightarrow{\text{replace } \sigma \text{ with } s} \underbrace{\frac{s}{\sqrt{\sum(x_i - \bar{x})^2}}}_{\text{standard error of } \hat{\beta}}$$

This follows the same pattern as in Chapter 21 for the sample mean:

$$\underbrace{\frac{\sigma}{\sqrt{n}}}_{\text{standard deviation of } \bar{y}} \xrightarrow{\text{replace } \sigma \text{ with } s} \underbrace{\frac{s}{\sqrt{n}}}_{\text{standard error of } \bar{y}}$$

Summary

	Formula	Meaning	Known?
$SD(\hat{\beta})$	$\frac{\sigma}{\sqrt{\sum(x_i - \bar{x})^2}}$	Spread of $\hat{\beta}$ across repeated samples	No — uses true σ
$SE(\hat{\beta})$	$\frac{s}{\sqrt{\sum(x_i - \bar{x})^2}}$	Estimated spread of $\hat{\beta}$	Yes — uses estimated s
σ	$\sqrt{\frac{\sum \epsilon_i^2}{n}}$	Noise in individual y around the line	No — population parameter
s	$\sqrt{\frac{\sum \epsilon_i^2}{n-2}}$	Estimated noise in individual y	Yes — computed from data

The key insight is that $SD(\hat{\beta})$ describes the **estimator** $\hat{\beta}$ across hypothetical repeated samples, while σ describes **individual observations** around the regression line. They are related but measure fundamentally different things.

Confidence Intervals for the Regression Slope

- The slope β of the population regression line is usually the most important parameter in a regression problem
 - The slope is the rate of change of the mean response as the explanatory variable increases
 - The slope b of the least-squares line is an unbiased estimator of β
- The confidence interval is more useful because it shows how accurate the estimate b is likely to be

$$\begin{aligned} &\text{estimate} \pm t^* SE_{\text{estimate}} \\ &b \pm t^* SE_b \end{aligned}$$

• Confidence Interval for Regression Slope

- A level C confidence interval for the slope β of the population regression line is

$$b \pm t^* SE_b$$

- t^* : the critical value for the t_{n-2} distribution density curve with area C between $-t^*$ and t^*

Example

A researcher uses a random sample of 27 workers from the CPS1988 dataset to study the relationship between education (years) and weekly wage (USD). The least-squares regression line is:

$$\text{wage} = 142.3 + 28.6 \times \text{education}$$

The standard error of the slope is $SE_b = 8.4$. Construct a 95% confidence interval for the slope β .

Step 1: Identify the values

	Value	Description
n	27	Sample size
b	28.6	Estimated slope
SE_b	8.4	Standard error of slope
df	$n - 2 = 25$	Degrees of freedom
C	95%	Confidence level

Step 2: Find the critical value t^*

For a 95% confidence interval with $df = n - 2 = 25$, the critical value from the t -table is:

$$t^* = 2.060$$

Step 3: Compute the margin of error

$$ME = t^* \times SE_b = 2.060 \times 8.4 = 17.3$$

Step 4: Construct the confidence interval

$$b \pm t^* SE_b = 28.6 \pm 17.3 = (11.3, 45.9)$$

Interpretation: We are 95% confident that each additional year of education is associated with an increase in weekly wage of between \$11.3 and \$45.9, on average.

Since the entire confidence interval is **positive** and does not include zero, this provides evidence that education has a positive effect on weekly wage in the population. Formally, this is consistent with rejecting the null hypothesis:

$$H_0 : \beta = 0 \quad (\text{education has no effect on wage})$$

$$H_a : \beta \neq 0 \quad (\text{education has an effect on wage})$$

Practice Question 26-1

An economist uses the PSID1976 dataset to study the relationship between work experience (years) and hourly wage (USD) among married women in paid employment. The following regression is estimated in R:

```
fit <- lm(wage ~ experience, data = df)
summary(fit)
```

The output is as follows:

Call:

```
lm(formula = wage ~ experience, data = df)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-3.6046	-1.8369	-0.6553	0.9954	22.5281

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.6148	0.3294	10.973	< 2e-16 ***
experience	0.0307	0.0164	1.874	0.0614 .

Residual standard error: 3.107 on 426 degrees of freedom

Multiple R-squared: 0.00818

Adjusted R-squared: 0.00585

F-statistic: 3.511 on 1 and 426 DF, p-value: 0.06139

Answer the following questions:

1. Write out the estimated regression equation.
2. Interpret the estimated slope coefficient in context.
3. Interpret the estimated intercept coefficient in context. Does it make practical sense?
4. What is the residual standard error s ? What does it measure?
5. Test the hypothesis $H_0 : \beta = 0$ vs $H_a : \beta \neq 0$ at $\alpha = 0.05$. State the t-statistic, p-value, and conclusion.
6. Construct a 95% confidence interval for the slope β using $t^* = 1.97$ and interpret it.

Answer**1. Estimated regression equation:**

$$\widehat{\text{wage}} = 3.615 + 0.031 \times \text{experience}$$

2. Slope interpretation:

Each additional year of work experience is associated with an increase in hourly wage of \$0.031, on average. The effect of experience on wages is very small compared to the effect of education.

3. Intercept interpretation:

The intercept $\hat{\alpha} = 3.615$ suggests that a woman with zero years of work experience earns \$3.62 per hour on average. This is plausible — it represents the starting wage for a new entrant to the labor market with no prior experience.

4. Residual standard error:

$s = 3.107$ dollars. This estimates σ — the typical deviation of an individual worker's hourly wage from the regression line. On average, the actual hourly wage differs from the predicted wage by about \$3.11.

5. Hypothesis test for the slope:

$$H_0 : \beta = 0 \quad (\text{experience has no effect on wage})$$

$$H_a : \beta \neq 0 \quad (\text{experience has an effect on wage})$$

From the output:

$$t = \frac{b}{SE_b} = \frac{0.0307}{0.0164} = 1.874 \quad p\text{-value} = 0.0614$$

Since $p\text{-value} = 0.0614 > 0.05$, we **fail to reject** H_0 at the 5% significance level. There is not sufficient evidence that work experience has a statistically significant effect on hourly wage at $\alpha = 0.05$. However, the result is marginally significant — we would reject H_0 at the 10% significance level since $p\text{-value} < 0.10$.

6. 95% confidence interval for β :

$$b \pm t^* \times SE_b = 0.0307 \pm 1.97 \times 0.0164 = 0.0307 \pm 0.0323 = (-0.0016, 0.0630)$$

We are 95% confident that each additional year of work experience is associated with a change in hourly wage of between $-\$0.002$ and $\$0.063$, on average. Since the interval **includes zero**, this confirms our earlier conclusion that we fail to reject $H_0 : \beta = 0$ at $\alpha = 0.05$.

Practice Question 26-2

An urban economist uses the Boston Housing dataset ($n = 506$ census tracts) to study the relationship between the percentage of lower-status population (`lstat`, %) and median home value (`medv`, in \$1,000s). The following regression is estimated in R:

```
fit <- lm(medv ~ lstat, data = Boston)
summary(fit)
```

The output is as follows:

Call:

```
lm(formula = medv ~ lstat, data = Boston)
```

Residuals:

Min	1Q	Median	3Q	Max
-15.168	-3.990	-1.318	2.034	24.500

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	34.5538	0.5626	61.415	0.000 ***
lstat	-0.9500	0.0387	-24.528	0.000 ***

Residual standard error: 6.216 on 504 degrees of freedom

Multiple R-squared: 0.5441

Adjusted R-squared: 0.5432

F-statistic: 601.6 on 1 and 504 DF, p-value: 0.000

Answer the following questions:

1. Write out the estimated regression equation.
2. Interpret the estimated slope coefficient in context.
3. Interpret the estimated intercept coefficient in context. Does it make practical sense?
4. What is the residual standard error s ? What does it measure?
5. Test the hypothesis $H_0 : \beta = 0$ vs $H_a : \beta \neq 0$ at $\alpha = 0.05$. State the t-statistic, p-value, and conclusion.
6. Construct a 95% confidence interval for the slope β using $t^* = 1.965$ and interpret it.

Answer**1. Estimated regression equation:**

$$\widehat{\text{medv}} = 34.554 - 0.950 \times \text{lstat}$$

2. Slope interpretation:

Each one percentage point increase in the lower-status population is associated with a decrease in median home value of \$950, on average. The negative slope confirms that neighborhoods with higher poverty rates tend to have lower home values.

3. Intercept interpretation:

The intercept $\hat{\alpha} = 34.554$ suggests that a neighborhood with zero lower-status population would have a median home value of \$34,554 on average. This is plausible — it represents the baseline home value in a neighborhood with no lower-status residents. However, no census tract in the dataset has $\text{lstat} = 0$, so this is an extrapolation beyond the range of the data.

4. Residual standard error:

$s = 6.216$ (in \$1,000s). This estimates σ — the typical deviation of an individual census tract's median home value from the regression line. On average, actual home values differ from predicted values by about \$6,216.

5. Hypothesis test for the slope:

$$H_0 : \beta = 0 \quad (\text{poverty rate has no effect on home value})$$

$$H_a : \beta \neq 0 \quad (\text{poverty rate has an effect on home value})$$

From the output:

$$t = \frac{b}{SE_b} = \frac{-0.950}{0.0387} = -24.528 \quad p\text{-value} \approx 0.000$$

Since $p\text{-value} < 0.05$, we **reject** H_0 at the 5% significance level. There is very strong evidence that the poverty rate has a statistically significant negative effect on median home value.

6. 95% confidence interval for β :

$$b \pm t^* \times SE_b = -0.950 \pm 1.965 \times 0.0387 = -0.950 \pm 0.076 = (-1.026, -0.874)$$

We are 95% confident that each one percentage point increase in the lower-status population is associated with a decrease in median home value of between \$874 and \$1,026, on average. Since the entire interval is **negative** and does not include zero, this confirms our earlier conclusion that we reject $H_0 : \beta = 0$ at $\alpha = 0.05$.

Chapter 27. Random Variables

- A **random variable** is a numerical outcome of a random phenomenon
- When we observe a random process — rolling a die, measuring a person’s height, recording a stock price — the result is a number that varies from trial to trial
- We use random variables to build precise mathematical models for uncertainty
 - How do we describe the distribution of possible values?
 - What is the “average” outcome we expect in the long run?
 - How much do outcomes vary around that average?

1. Random Variables

Definition

- **Formal Definition**

- Let Ω be the **sample space** — the set of all possible outcomes of a random experiment
- A **random variable** X is a function that assigns a real number to each outcome:

$$X : \Omega \rightarrow \mathbb{R}$$

- We use capital letters X, Y, Z for random variables, and lowercase x, y, z for specific **realized values**

- **Example**

- Flip a coin twice. The sample space is $\Omega = \{HH, HT, TH, TT\}$
- Define X = number of heads. Then:

$$X(HH) = 2, \quad X(HT) = 1, \quad X(TH) = 1, \quad X(TT) = 0$$

- X maps each outcome to a number. When we say “ $X = 1$,” we mean the event $\{HT, TH\}$ occurred

- **Two Types of Random Variables**

- **Discrete:** X takes a countable set of values (e.g., $0, 1, 2, \dots$)
- **Continuous:** X can take any value in an interval of $\mathbb{R}$ (e.g., height, weight, time)

2. PMF, PDF, and CDF

Discrete Random Variables: Probability Mass Function

- For a discrete random variable X , the **probability mass function (PMF)** specifies the probability of each possible value:

$$p(x) = P(X = x)$$

- The PMF must satisfy two conditions:

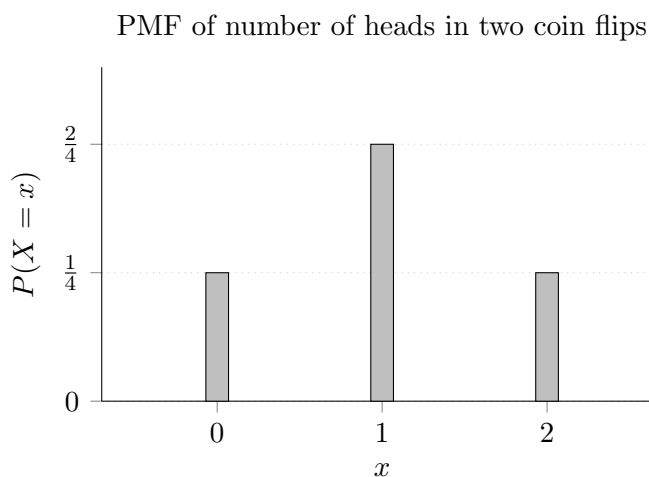
1. $p(x) \geq 0$ for all x

2. $\sum_{\text{all } x} p(x) = 1$

- Example:** Let X = number of heads in two coin flips

x	0	1	2
$P(X = x)$	1/4	2/4	1/4

The probabilities sum to $1/4 + 2/4 + 1/4 = 1$. ✓



Each bar height equals $P(X = x)$. The bars sum to 1.

Continuous Random Variables: Probability Density Function

- For a continuous random variable X , the **probability density function (PDF)** $f(x)$ describes how probability is distributed over an interval
- A key difference from discrete: $P(X = x) = 0$ for any single value x

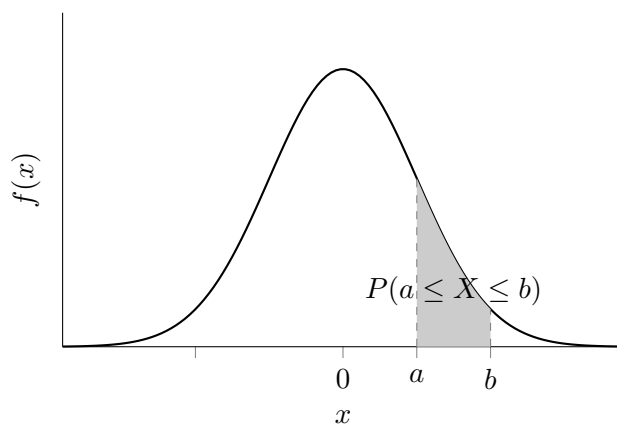
- Instead, we compute probabilities over **intervals**:

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

- The PDF must satisfy:

1. $f(x) \geq 0$ for all x
2. $\int_{-\infty}^{\infty} f(x) dx = 1$

PDF: probability equals area under the curve



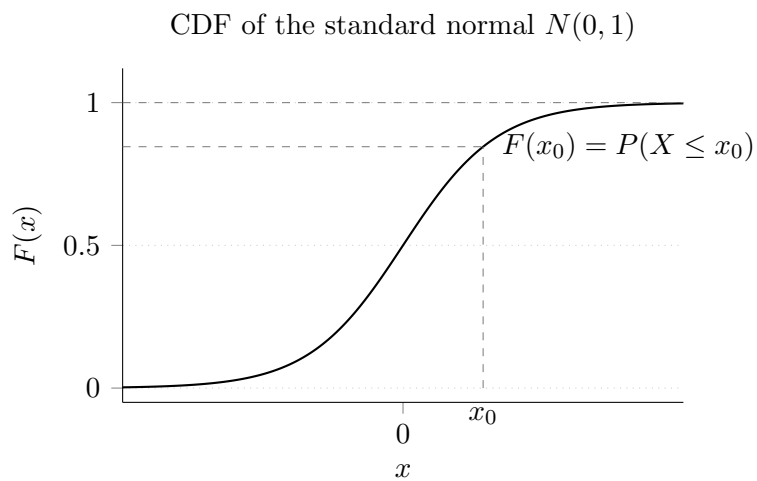
The shaded area equals $P(a \leq X \leq b) = \int_a^b f(x) dx$. The total area under the curve equals 1.

Cumulative Distribution Function

- The **cumulative distribution function (CDF)** $F(x)$ gives the probability that X takes a value *at most* x :

$$F(x) = P(X \leq x)$$

- For discrete X : $F(x) = \sum_{t \leq x} P(X = t)$
- For continuous X : $F(x) = \int_{-\infty}^x f(t) dt$
- The CDF always satisfies: $0 \leq F(x) \leq 1$, and is non-decreasing



$F(x)$ is non-decreasing, approaches 0 as $x \rightarrow -\infty$, and approaches 1 as $x \rightarrow +\infty$.

3. Joint Distributions

Joint PMF and Joint PDF

- When we study two random variables X and Y together, we need their **joint distribution**, which describes the probability of all combinations of values of X and Y simultaneously.
- Joint PMF** (discrete): $P(X = x, Y = y)$ gives the probability that X takes value x and Y takes value y .

Example: Roll a fair die twice. Let X = result of the first roll and Y = result of the second roll. Each pair (x, y) has probability $1/36$. Now let X = number of heads in the first two flips of a fair coin, and Y = number of heads in the next two flips, independently. The joint PMF is:

XY	0	1	2	$P(X = x)$
0	$\frac{1}{16}$	$\frac{2}{16}$	$\frac{1}{16}$	$\frac{4}{16}$
1	$\frac{2}{16}$	$\frac{4}{16}$	$\frac{2}{16}$	$\frac{8}{16}$
2	$\frac{1}{16}$	$\frac{2}{16}$	$\frac{1}{16}$	$\frac{4}{16}$
$P(Y = y)$	$\frac{4}{16}$	$\frac{8}{16}$	$\frac{4}{16}$	1

Each interior cell is $P(X = x) \cdot P(Y = y)$, confirming independence. The row sums give the marginal PMF of X and the column sums give the marginal PMF of Y — each is just the familiar Binomial(2, 0.5) distribution. The grand total equals 1. ✓

- Joint PDF** (continuous): $f(x, y)$ such that probabilities over regions are computed as:

$$P(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f(x, y) dy dx$$

The joint PDF must satisfy $f(x, y) \geq 0$ for all x, y , and $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dy dx = 1$.

Example: Let X and Y be jointly continuous with joint PDF:

$$f(x, y) = 6xy^2, \quad 0 \leq x \leq 1, 0 \leq y \leq 1$$

Verify it is a valid PDF:

$$\int_0^1 \int_0^1 6xy^2 dy dx = \int_0^1 6x \left[\frac{y^3}{3} \right]_0^1 dx = \int_0^1 6x \cdot \frac{1}{3} dx = \int_0^1 2x dx = \left[x^2 \right]_0^1 = 1 \quad \checkmark$$

Compute a probability, e.g. $P(X \leq 1/2, Y \leq 1/2)$:

$$\begin{aligned} \int_0^{1/2} \int_0^{1/2} 6xy^2 dy dx &= \int_0^{1/2} 6x \left[\frac{y^3}{3} \right]_0^{1/2} dx = \int_0^{1/2} 6x \cdot \frac{1}{24} dx = \int_0^{1/2} \frac{x}{4} dx \\ &= \frac{1}{4} \left[\frac{x^2}{2} \right]_0^{1/2} = \frac{1}{4} \cdot \frac{1}{8} = \frac{1}{32} \end{aligned}$$

Find the marginal PDF of X by integrating out Y :

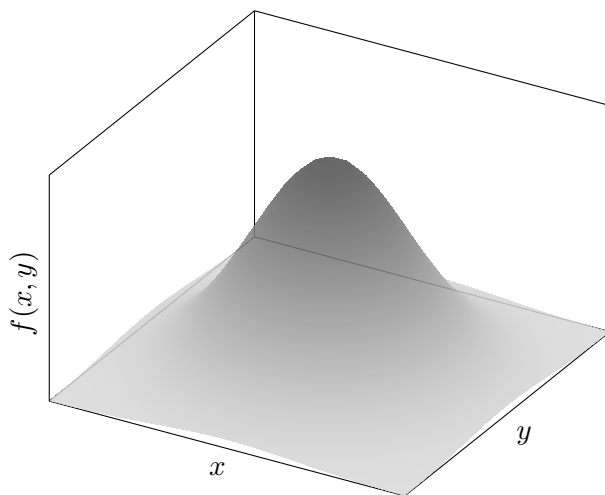
$$f_X(x) = \int_0^1 6xy^2 dy = 6x \left[\frac{y^3}{3} \right]_0^1 = 2x, \quad 0 \leq x \leq 1$$

Find the marginal PDF of Y by integrating out X :

$$f_Y(y) = \int_0^1 6xy^2 dx = 6y^2 \left[\frac{x^2}{2} \right]_0^1 = 3y^2, \quad 0 \leq y \leq 1$$

Check independence: since $f(x, y) = 6xy^2 = (2x)(3y^2) = f_X(x) \cdot f_Y(y)$, X and Y are **independent**. ✓

Joint PDF $f(x, y)$: a surface over the (x, y) plane



A bivariate normal joint PDF. The volume under the entire surface equals 1. $P(a \leq X \leq b, c \leq Y \leq d)$ is the volume above the rectangle $[a, b] \times [c, d]$.

Marginal Distributions

- The **marginal distribution** of X is obtained by integrating out Y from the joint PDF:

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

and similarly the marginal of Y :

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

- Intuitively, marginalizing means averaging over all possible values of the other variable. The marginal PDF of X tells us the distribution of X alone, ignoring Y .
- **Discrete case:** replace integrals with sums: $P(X = x) = \sum_y P(X = x, Y = y)$.

Independence of Random Variables

- X and Y are **independent** if and only if their joint distribution factors into the product of the marginals:

$$f(x, y) = f_X(x) \cdot f_Y(y) \quad \text{for all } x, y$$

- Intuitively, knowing the value of X gives no information about Y , and vice versa.
- An important consequence of independence is that the expected value of the product factors:

$$E[XY] = E[X] E[Y] \quad \text{when } X, Y \text{ independent}$$

Derivation: when $f(x, y) = f_X(x) f_Y(y)$,

$$\begin{aligned} E[XY] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f(x, y) dx dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_X(x) f_Y(y) dx dy \\ &= \int_{-\infty}^{\infty} x f_X(x) dx \cdot \int_{-\infty}^{\infty} y f_Y(y) dy \\ &= E[X] \cdot E[Y] \end{aligned}$$

This factoring is what drives the covariance-zero result under independence, which we will use extensively in Section 4.

4. Expected Value, Variance, and Moments

Expected Value

- The **expected value** (or **mean**) of a random variable X , denoted $E[X]$ or μ , is the long-run average value of X over many repetitions

- **Discrete case:**

$$E[X] = \mu = \sum_{\text{all } x} x \cdot P(X = x)$$

- **Continuous case:**

$$E[X] = \mu = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

Example: Expected Number of Heads

Let X = number of heads in two coin flips.

$$\begin{aligned} E[X] &= 0 \cdot P(X = 0) + 1 \cdot P(X = 1) + 2 \cdot P(X = 2) \\ &= 0 \cdot \frac{1}{4} + 1 \cdot \frac{2}{4} + 2 \cdot \frac{1}{4} \\ &= 0 + \frac{1}{2} + \frac{1}{2} = 1 \end{aligned}$$

On average, we expect 1 head in two flips. This matches intuition.

Properties of Expected Value

- **Linearity:**

$$E[aX + b] = a E[X] + b$$

for any constants a and b .

Derivation: Let $f(x)$ be the PDF of X . Then:

$$\begin{aligned} E[aX + b] &= \int_{-\infty}^{\infty} (ax + b) f(x) dx \\ &= \int_{-\infty}^{\infty} ax f(x) dx + \int_{-\infty}^{\infty} b f(x) dx \\ &= a \int_{-\infty}^{\infty} x f(x) dx + b \int_{-\infty}^{\infty} f(x) dx \\ &= a E[X] + b \cdot 1 = a E[X] + b \end{aligned}$$

The second integral equals 1 because any PDF integrates to 1 over its full support.

- **Expected value of a sum:**

$$E[X + Y] = E[X] + E[Y]$$

This holds for *any* X and Y , whether or not they are independent.

Derivation: Let $f(x, y)$ be the joint PDF of X and Y . Then:

$$\begin{aligned} E[X + Y] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x + y) f(x, y) dx dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f(x, y) dx dy + \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f(x, y) dx dy \\ &= \int_{-\infty}^{\infty} x \int_{-\infty}^{\infty} f(x, y) dy dx + \int_{-\infty}^{\infty} y \int_{-\infty}^{\infty} f(x, y) dx dy \\ &= \int_{-\infty}^{\infty} x f_X(x) dx + \int_{-\infty}^{\infty} y f_Y(y) dy \\ &= E[X] + E[Y] \end{aligned}$$

The inner integrals collapse to marginal PDFs: $\int_{-\infty}^{\infty} f(x, y) dy = f_X(x)$. No independence assumption is needed — the joint PDF always integrates to the marginal.

- **Expected value of a function** (Law of the Unconscious Statistician, LOTUS):

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f(x) dx$$

We do *not* need to find the distribution of $g(X)$ first — we apply g directly inside the integral.

Derivation: Let $Y = g(X)$ with PDF $f_Y(y)$. By definition, $E[Y] = \int_{-\infty}^{\infty} y f_Y(y) dy$. To connect this back to f_X , we use the change-of-variables formula. For a monotone differentiable g :

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \left| \frac{d}{dy} g^{-1}(y) \right|$$

Substituting $x = g^{-1}(y)$, so $y = g(x)$ and $dy = g'(x) dx$:

$$\begin{aligned} E[Y] &= \int_{-\infty}^{\infty} y f_Y(y) dy \\ &= \int_{-\infty}^{\infty} g(x) f_X(x) |g'(x)| \frac{dy}{|g'(x)|} \\ &= \int_{-\infty}^{\infty} g(x) f_X(x) dx \end{aligned}$$

The result extends to non-monotone g by partitioning the domain. LOTUS says we never need f_Y explicitly — we integrate $g(x)$ against the original density $f_X(x)$.

Variance and Standard Deviation

- The **variance** of X , denoted $\text{Var}(X)$ or σ^2 , measures how spread out the values of X are around the mean $\mu = E[X]$

- **Definition:**

$$\text{Var}(X) = E[(X - \mu)^2]$$

- Expanding using the continuous formula (LOTUS with $g(x) = (x - \mu)^2$):

$$\text{Var}(X) = \int_{-\infty}^{\infty} (x - \mu)^2 \cdot f(x) dx$$

- **Computational shortcut:**

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

Derivation:

$$\begin{aligned} \text{Var}(X) &= E[(X - \mu)^2] \\ &= E[X^2 - 2\mu X + \mu^2] \\ &= E[X^2] - 2\mu E[X] + \mu^2 \quad (\text{linearity of } E[\cdot]) \\ &= E[X^2] - 2\mu \cdot \mu + \mu^2 \quad (\text{since } E[X] = \mu) \\ &= E[X^2] - 2\mu^2 + \mu^2 \\ &= E[X^2] - \mu^2 \\ &= E[X^2] - (E[X])^2 \end{aligned}$$

- The **standard deviation** of X is:

$$\text{SD}(X) = \sigma = \sqrt{\text{Var}(X)}$$

σ is in the *same units* as X , making it easier to interpret than variance. Since variance squares the deviations $(x - \mu)^2$, the unit of $\text{Var}(X)$ is the square of the unit of X . Taking the square root restores the original unit.

Covariance

- **Covariance** measures the direction and magnitude of the linear relationship between two random variables X and Y :

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

- Expanding using the joint PDF (LOTUS in two variables):

$$\text{Cov}(X, Y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)(y - \mu_Y) f(x, y) dx dy$$

- **Computational shortcut:**

$$\text{Cov}(X, Y) = E[XY] - E[X] E[Y]$$

Derivation:

$$\begin{aligned} \text{Cov}(X, Y) &= E[(X - \mu_X)(Y - \mu_Y)] \\ &= E[XY - \mu_X Y - \mu_Y X + \mu_X \mu_Y] \\ &= E[XY] - \mu_X E[Y] - \mu_Y E[X] + \mu_X \mu_Y \quad (\text{linearity}) \\ &= E[XY] - \mu_X \mu_Y - \mu_Y \mu_X + \mu_X \mu_Y \\ &= E[XY] - \mu_X \mu_Y \\ &= E[XY] - E[X] E[Y] \end{aligned}$$

This shortcut always holds — it is a pure algebraic identity derived solely from expanding the product and applying linearity of expectation. No assumptions about independence, distribution type, or any other property of X and Y are required. The only condition is that $E[X]$, $E[Y]$, and $E[XY]$ are all finite.

A common source of confusion is the independence result $\text{Cov}(X, Y) = 0$. This is a *consequence* of independence, not a property of the shortcut itself. What independence does is force $E[XY] = E[X] E[Y]$, which makes the shortcut evaluate to zero. Without independence, $E[XY] \neq E[X] E[Y]$ in general, and the shortcut correctly captures the nonzero covariance. The shortcut holds either way.

- **Covariance under independence:** if X and Y are independent, their joint PDF factors as $f(x, y) = f_X(x) f_Y(y)$, so:

$$\begin{aligned} \text{Cov}(X, Y) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)(y - \mu_Y) f_X(x) f_Y(y) dx dy \\ &= \int_{-\infty}^{\infty} (x - \mu_X) f_X(x) dx \cdot \int_{-\infty}^{\infty} (y - \mu_Y) f_Y(y) dy \\ &= E[X - \mu_X] \cdot E[Y - \mu_Y] = 0 \cdot 0 = 0 \end{aligned}$$

The double integral factors into a product because of independence, and each factor equals zero since $E[X - \mu_X] = 0$.

- **Caution:** $\text{Cov}(X, Y) = 0$ does *not* imply independence in general — it only rules out a *linear* relationship.
- Note that $\text{Cov}(X, X) = E[(X - \mu)^2] = \text{Var}(X)$, so variance is a special case of covariance.

Properties of Variance

- **Scaling and shifting:**

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

Derivation: Let $\mu = E[X]$, so $E[aX + b] = a\mu + b$ by linearity. Then:

$$\begin{aligned}\text{Var}(aX + b) &= E\left[\left((aX + b) - (a\mu + b)\right)^2\right] \\ &= E\left[(aX - a\mu)^2\right] \\ &= E\left[a^2(X - \mu)^2\right] \\ &= a^2 E\left[(X - \mu)^2\right] \\ &= a^2 \text{Var}(X)\end{aligned}$$

The shift b cancels entirely in the second line — shifting a distribution left or right does not change how spread out it is. Only the scale factor a survives, and it enters squared because variance measures squared deviations.

- **General variance of a sum:**

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

Derivation: Let $\mu_X = E[X]$ and $\mu_Y = E[Y]$. Then:

$$\begin{aligned}\text{Var}(X + Y) &= E\left[\left((X + Y) - (\mu_X + \mu_Y)\right)^2\right] \\ &= E\left[\left((X - \mu_X) + (Y - \mu_Y)\right)^2\right] \\ &= E\left[(X - \mu_X)^2 + 2(X - \mu_X)(Y - \mu_Y) + (Y - \mu_Y)^2\right] \\ &= \text{Var}(X) + 2\text{Cov}(X, Y) + \text{Var}(Y)\end{aligned}$$

- **Independent case:** when X and Y are independent, $\text{Cov}(X, Y) = 0$ (shown above), so the formula simplifies to:

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

Independence is essential for this simplification — unlike for expected values, it cannot be dropped.

Moments

- **Moments** are summary statistics that describe the shape of a distribution
- The **r -th raw moment** of X about the origin is:

$$\mu'_r = E[X^r]$$

- The **r -th central moment** (about the mean μ) is:

$$\mu_r = E[(X - \mu)^r]$$

Central moments measure spread and shape relative to the center of the distribution.

Why raw moments for the mean, but central moments from the second order onward?

- **First moment:** the mean is used to locate the *center* of the distribution on the number line, so we naturally use the raw moment $E[X]$. The corresponding first central moment $E[X - \mu]$ is always exactly zero by definition — it carries no information.
- **Second moment and beyond:** once the center μ is established, all higher-order quantities describe the *shape* of the distribution *relative to that center*. Using raw moments here would conflate location with shape. For example, the raw second moment $E[X^2]$ grows large simply because the distribution is far from zero, even if it is tightly concentrated. Centering first — computing $E[(X - \mu)^2]$ — isolates spread from location entirely.
- The key property that makes central moments the right choice is **translation invariance**: for any constant c ,

$$E[(X + c - E[X + c])^r] = E[(X - \mu)^r]$$

Shifting the distribution left or right leaves all central moments unchanged. Raw moments do not have this property. Since variance, skewness, and kurtosis are meant to describe shape independently of location, central moments are the natural tool from the second order onward.

- The computational shortcut $\text{Var}(X) = E[X^2] - (E[X])^2$ is the bridge between the two: it expresses the second *central* moment in terms of *raw* moments, which is convenient for calculation while the conceptual definition remains centered.

The first four moments

- **First moment** ($r = 1$): the mean. The first central moment is always zero: $\mu_1 = E[X - \mu] = 0$.

- **Second central moment** ($r = 2$): the variance

$$\mu_2 = E[(X - \mu)^2] = \text{Var}(X) = \sigma^2$$

- **Third central moment** ($r = 3$): measures asymmetry, used to define **skewness**:

$$\text{Skewness} = \frac{\mu_3}{\sigma^3} = \frac{E[(X - \mu)^3]}{(\text{Var}(X))^{3/2}}$$

- Skewness > 0 : right-skewed (long right tail)
- Skewness < 0 : left-skewed (long left tail)
- Skewness $= 0$: symmetric (e.g., the normal distribution)

- **Fourth central moment** ($r = 4$): measures tail heaviness, used to define **kurtosis**:

$$\text{Kurtosis} = \frac{\mu_4}{\sigma^4}, \quad \text{Excess kurtosis} = \frac{\mu_4}{\sigma^4} - 3$$

Excess kurtosis subtracts 3 (the kurtosis of the normal) as a benchmark.

- Excess kurtosis > 0 : *leptokurtic* — heavier tails than the normal
- Excess kurtosis < 0 : *platykurtic* — lighter tails than the normal
- Excess kurtosis $= 0$: *mesokurtic* — normal-like tails

Moment Generating Function

- The **moment generating function (MGF)** of X is:

$$M_X(t) = E[e^{tX}]$$

defined for all t in some open interval around 0.

- The MGF *generates* all raw moments via differentiation:

$$E[X^r] = M_X^{(r)}(0) = \left. \frac{d^r}{dt^r} M_X(t) \right|_{t=0}$$

- **Uniqueness**: if two random variables have the same MGF, they have the same distribution.
- **Independence**: if X and Y are independent, then $M_{X+Y}(t) = M_X(t) \cdot M_Y(t)$.

Moment	Formula	Interpretation
Mean	$\mu'_1 = E[X]$	Center of the distribution
Variance	$\mu_2 = E[(X - \mu)^2]$	Spread around the mean
Skewness	μ_3/σ^3	Asymmetry
Kurtosis	μ_4/σ^4	Tail heaviness
MGF	$M_X(t) = E[e^{tX}]$	Encodes all moments

Summary of moments

Estimators and Unbiasedness

- In statistics, we rarely know the true parameters of a population — instead we **estimate** them from data. An **estimator** is a function of the observed sample that is used to approximate an unknown parameter.
- Formally, let θ be an unknown population parameter (such as a mean μ or a proportion p). An estimator $\hat{\theta}$ is itself a **random variable** — it varies across different samples drawn from the same population.
- **Definition:** an estimator $\hat{\theta}$ is **unbiased** for θ if:

$$E[\hat{\theta}] = \theta$$

That is, the expected value of the estimator equals the true parameter. On average, across all possible samples, the estimator hits the right target.

- **Bias:** if $E[\hat{\theta}] \neq \theta$, the estimator is **biased**, and the bias is defined as:

$$\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta$$

- Unbiasedness is a property about the *center* of the estimator's distribution — it does not say anything about how variable the estimator is across samples. An unbiased estimator can still be highly imprecise if its variance $\text{Var}(\hat{\theta})$ is large.

The bias–variance tradeoff

- The **mean squared error (MSE)** of an estimator combines both bias and variance into a single measure of overall accuracy:

$$\text{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$$

- *Derivation:* writing $(\hat{\theta} - \theta) = (\hat{\theta} - E[\hat{\theta}]) + (E[\hat{\theta}] - \theta)$ and expanding:

$$\begin{aligned} \text{MSE}(\hat{\theta}) &= E\left[(\hat{\theta} - E[\hat{\theta}]) + \text{Bias}(\hat{\theta})\right]^2 \\ &= E\left[(\hat{\theta} - E[\hat{\theta}])^2\right] + 2\text{Bias}(\hat{\theta}) E\left[\hat{\theta} - E[\hat{\theta}]\right] + \text{Bias}(\hat{\theta})^2 \\ &= \text{Var}(\hat{\theta}) + 0 + \text{Bias}(\hat{\theta})^2 \end{aligned}$$

The middle term vanishes because $E[\hat{\theta} - E[\hat{\theta}]] = 0$. This gives the fundamental decomposition:

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2$$

- When $\hat{\theta}$ is unbiased, $\text{Bias}(\hat{\theta}) = 0$, so $\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta})$. Minimizing MSE then reduces to minimizing variance.

Examples of unbiased estimators

- **Sample mean** $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ is unbiased for the population mean μ :

$$E[\bar{X}] = \frac{1}{n} \sum_{i=1}^n E[X_i] = \frac{1}{n} \cdot n\mu = \mu$$

- **Sample proportion** $\hat{p} = X/n$ is unbiased for the population proportion p :

$$E[\hat{p}] = \frac{E[X]}{n} = \frac{np}{n} = p$$

- **Sample variance** $s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ is unbiased for the population variance σ^2 :

$$E[s^2] = \sigma^2$$

The divisor is $n-1$ rather than n precisely to correct for a bias that arises from using $\bar{X}$ in place of the unknown μ . The full derivation follows.

Derivation: $E[s^2] = \sigma^2$

Let $X_1, \dots, X_n$ be i.i.d. with $E[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2$. Define the sample mean $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$.

Step 1: Rewrite the sum of squared deviations.

Add and subtract μ inside each squared term:

$$\begin{aligned} \sum_{i=1}^n (X_i - \bar{X})^2 &= \sum_{i=1}^n [(X_i - \mu) - (\bar{X} - \mu)]^2 \\ &= \sum_{i=1}^n (X_i - \mu)^2 - 2(\bar{X} - \mu) \sum_{i=1}^n (X_i - \mu) + \sum_{i=1}^n (\bar{X} - \mu)^2 \end{aligned}$$

Note that $\sum_{i=1}^n (X_i - \mu) = n(\bar{X} - \mu)$, so the middle term becomes $-2n(\bar{X} - \mu)^2$. The last term is $n(\bar{X} - \mu)^2$. Therefore:

$$\begin{aligned} \sum_{i=1}^n (X_i - \bar{X})^2 &= \sum_{i=1}^n (X_i - \mu)^2 - 2n(\bar{X} - \mu)^2 + n(\bar{X} - \mu)^2 \\ &= \sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X} - \mu)^2 \end{aligned}$$

Step 2: Take expectations of both sides.

$$E\left[\sum_{i=1}^n (X_i - \bar{X})^2\right] = E\left[\sum_{i=1}^n (X_i - \mu)^2\right] - n E[(\bar{X} - \mu)^2]$$

Step 3: Evaluate each expectation separately.

For the first term, since the X_i are identically distributed:

$$E\left[\sum_{i=1}^n (X_i - \mu)^2\right] = \sum_{i=1}^n E[(X_i - \mu)^2] = \sum_{i=1}^n \text{Var}(X_i) = n\sigma^2$$

For the second term, $(\bar{X} - \mu)^2$ is the squared deviation of $\bar{X}$ from its mean μ , so:

$$E[(\bar{X} - \mu)^2] = \text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

Step 4: Substitute back.

$$E\left[\sum_{i=1}^n (X_i - \bar{X})^2\right] = n\sigma^2 - n \cdot \frac{\sigma^2}{n} = n\sigma^2 - \sigma^2 = (n-1)\sigma^2$$

Step 5: Conclude.

$$E[s^2] = E\left[\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2\right] = \frac{1}{n-1} \cdot (n-1)\sigma^2 = \sigma^2 \quad \checkmark$$

The factor $(n-1)$ in the denominator of s^2 cancels exactly the $(n-1)\sigma^2$ that appears in the numerator's expectation. This is why we divide by $n-1$ and not n .

What goes wrong with n in the denominator?

If we instead defined $\tilde{s}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$, then:

$$E[\tilde{s}^2] = \frac{1}{n} \cdot (n-1)\sigma^2 = \frac{n-1}{n}\sigma^2 < \sigma^2$$

This estimator systematically *underestimates* σ^2 . The reason is intuitive: we are measuring deviations from $\bar{X}$, which is itself computed from the data and sits as close to the data as possible — closer than the true μ would. This makes the deviations $(X_i - \bar{X})$ artificially small on average, and dividing by $n-1$ instead of n corrects for exactly this shrinkage.

5. Common Discrete Distributions

5.1 Bernoulli Distribution

- Models a single trial with two outcomes: success ($X = 1$) or failure ($X = 0$)
- Parameter: p = probability of success, $0 \leq p \leq 1$
- PMF:

$$P(X = x) = p^x(1 - p)^{1-x}, \quad x \in \{0, 1\}$$

- We write $X \sim \text{Bernoulli}(p)$
- **Mean:**

$$E[X] = 1 \cdot P(X = 1) + 0 \cdot P(X = 0) = 1 \cdot p + 0 \cdot (1 - p) = p$$

- **Variance:** first compute $E[X^2]$. Since $X \in \{0, 1\}$, we have $X^2 = X$, so $E[X^2] = E[X] = p$. Then by the computational shortcut:

$$\text{Var}(X) = E[X^2] - (E[X])^2 = p - p^2 = p(1 - p)$$

Variance is maximized at $p = 1/2$ and equals zero when $p = 0$ or $p = 1$, which matches intuition: a coin that always lands heads has no randomness.

5.2 Binomial Distribution

- Models the number of successes in n independent Bernoulli trials, each with success probability p
- Parameters: n (number of trials), p (probability of success)
- PMF:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n$$

- We write $X \sim \text{Binomial}(n, p)$
- Since $X = X_1 + X_2 + \dots + X_n$ where each $X_i \sim \text{Bernoulli}(p)$ independently, we can derive the mean and variance directly from properties already established.
- **Mean:** by linearity of expectation,

$$E[X] = E\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n E[X_i] = \sum_{i=1}^n p = np$$

- **Variance:** since the X_i are independent, variances add:

$$\text{Var}(X) = \text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) = \sum_{i=1}^n p(1-p) = np(1-p)$$

Both results follow entirely from $E[X_i] = p$ and $\text{Var}(X_i) = p(1-p)$ — no direct computation with the Binomial PMF is needed.

5.3 Sample Proportions

- Suppose we observe n independent Bernoulli(p) trials. Let X = number of successes. Then $X \sim \text{Binomial}(n, p)$.
- The **sample proportion** is:

$$\hat{p} = \frac{X}{n}$$

$\hat{p}$ is a random variable — it varies from sample to sample.

- **Mean of $\hat{p}$:** using linearity of expectation,

$$E[\hat{p}] = E\left[\frac{X}{n}\right] = \frac{E[X]}{n} = \frac{np}{n} = p$$

$\hat{p}$ is an **unbiased estimator** of p , as shown in Section 4.

- **Variance of $\hat{p}$:**

$$\text{Var}(\hat{p}) = \text{Var}\left(\frac{X}{n}\right) = \frac{\text{Var}(X)}{n^2} = \frac{np(1-p)}{n^2} = \frac{p(1-p)}{n}$$

- **Standard error of $\hat{p}$:**

$$\text{SE}(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

In practice, we estimate this by plugging in $\hat{p}$ for the unknown p :

$$\widehat{\text{SE}}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Normal Approximation for $\hat{p}$

- By the **Central Limit Theorem**, when n is large, $\hat{p}$ is approximately normally distributed:

$$\hat{p} \approx N\left(p, \frac{p(1-p)}{n}\right)$$

The standardized version is:

$$Z = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \approx N(0, 1)$$

- **Rule of thumb:** the approximation is reliable when $np \geq 10$ and $n(1-p) \geq 10$.
- This approximation is the foundation of hypothesis tests and confidence intervals for proportions.

Confidence Interval for a Proportion

- A 95% confidence interval for p is:

$$\hat{p} \pm 1.96 \cdot \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

- More generally, a $(1 - \alpha) \times 100\%$ confidence interval uses the critical value z^* :

$$\hat{p} \pm z^* \cdot \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

Practice Question 27-A: Sample Proportions

In a random sample of 400 voters, 220 say they support a ballot measure.

1. Compute the sample proportion $\hat{p}$.
2. Compute $E[\hat{p}]$ and $SE(\hat{p})$.
3. Construct a 95% confidence interval for the true proportion p .
4. Check that the normal approximation conditions are met.

Answer

1. **Sample proportion:**

$$\hat{p} = \frac{220}{400} = 0.55$$

2. **Mean and standard error:**

$$\begin{aligned} E[\hat{p}] &= p \quad (\text{unbiased}) \\ \widehat{SE}(\hat{p}) &= \sqrt{\frac{0.55 \times 0.45}{400}} = \sqrt{\frac{0.2475}{400}} = \sqrt{0.000619} \approx 0.0249 \end{aligned}$$

3. **95% confidence interval:**

$$0.55 \pm 1.96 \times 0.0249 = 0.55 \pm 0.049 = (0.501, 0.599)$$

We are 95% confident the true proportion of supporters lies between 50.1% and 59.9%.

4. **Normal approximation check:**

$$\begin{aligned} n\hat{p} &= 400 \times 0.55 = 220 \geq 10 \quad \checkmark \\ n(1 - \hat{p}) &= 400 \times 0.45 = 180 \geq 10 \quad \checkmark \end{aligned}$$

Both conditions are satisfied.

5.4 Comparing Two Proportions

- We often want to compare proportions from two independent groups.
- Let group 1 have n_1 observations with $X_1 \sim \text{Binomial}(n_1, p_1)$, and group 2 have n_2 observations with $X_2 \sim \text{Binomial}(n_2, p_2)$.
- The sample proportions are $\hat{p}_1 = X_1/n_1$ and $\hat{p}_2 = X_2/n_2$.
- We are interested in the **difference** $\hat{p}_1 - \hat{p}_2$.

Mean and Variance of the Difference

- By linearity of expectation:

$$E[\hat{p}_1 - \hat{p}_2] = p_1 - p_2$$

The difference in sample proportions is an unbiased estimator of the difference in true proportions.

$$\text{Var}(\hat{p}_1 - \hat{p}_2) = \text{Var}(\hat{p}_1) + \text{Var}(\hat{p}_2) = \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}$$

- **Standard error of the difference:**

$$\text{SE}(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$$

Estimated by plugging in $\hat{p}_1$ and $\hat{p}_2$:

$$\widehat{\text{SE}}(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

Normal Approximation and Confidence Interval

- When both samples are large, the difference is approximately normal:

$$\hat{p}_1 - \hat{p}_2 \approx N\left(p_1 - p_2, \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}\right)$$

- **Rule of thumb:** all four quantities $n_1\hat{p}_1$, $n_1(1-\hat{p}_1)$, $n_2\hat{p}_2$, $n_2(1-\hat{p}_2)$ should be ≥ 10 .
- A 95% confidence interval for $p_1 - p_2$:

$$(\hat{p}_1 - \hat{p}_2) \pm 1.96 \cdot \widehat{\text{SE}}(\hat{p}_1 - \hat{p}_2)$$

Hypothesis Test for $H_0 : p_1 = p_2$

- Under H_0 , both groups share a common proportion p . We estimate it with the **pooled proportion**:

$$\hat{p}_{\text{pool}} = \frac{X_1 + X_2}{n_1 + n_2}$$

- The pooled standard error is:

$$\text{SE}_{\text{pool}} = \sqrt{\hat{p}_{\text{pool}}(1 - \hat{p}_{\text{pool}}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

- The test statistic is:

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\text{SE}_{\text{pool}}} \approx N(0, 1) \quad \text{under } H_0$$

- We reject H_0 at the 5% level when $|Z| > 1.96$.

Practice Question 27-B: Comparing Two Proportions

A company runs an A/B test on two versions of a checkout page. Of 500 users shown version A, 120 completed a purchase. Of 600 users shown version B, 168 completed a purchase.

1. Compute $\hat{p}_A$ and $\hat{p}_B$.
2. Compute $E[\hat{p}_A - \hat{p}_B]$ (as an estimate) and $\widehat{SE}(\hat{p}_A - \hat{p}_B)$.
3. Construct a 95% confidence interval for $p_A - p_B$.
4. Conduct a hypothesis test of $H_0 : p_A = p_B$ against $H_a : p_A \neq p_B$ at the 5% significance level.
5. Check the normal approximation conditions.

Answer**1. Sample proportions:**

$$\hat{p}_A = \frac{120}{500} = 0.24, \quad \hat{p}_B = \frac{168}{600} = 0.28$$

2. Estimated difference and standard error:

$$\begin{aligned} \hat{p}_A - \hat{p}_B &= 0.24 - 0.28 = -0.04 \\ \widehat{SE}(\hat{p}_A - \hat{p}_B) &= \sqrt{\frac{0.24 \times 0.76}{500} + \frac{0.28 \times 0.72}{600}} \\ &= \sqrt{\frac{0.1824}{500} + \frac{0.2016}{600}} \\ &= \sqrt{0.000365 + 0.000336} = \sqrt{0.000701} \approx 0.0265 \end{aligned}$$

3. 95% confidence interval:

$$-0.04 \pm 1.96 \times 0.0265 = -0.04 \pm 0.052 = (-0.092, 0.012)$$

Since this interval contains 0, we do not have strong evidence that the two conversion rates differ.

4. Hypothesis test:

Pooled proportion:

$$\hat{p}_{\text{pool}} = \frac{120 + 168}{500 + 600} = \frac{288}{1100} \approx 0.2618$$

Pooled standard error:

$$SE_{\text{pool}} = \sqrt{0.2618 \times 0.7382 \times \left(\frac{1}{500} + \frac{1}{600} \right)} = \sqrt{0.1933 \times 0.003\bar{6}} \approx \sqrt{0.00070\bar{6}} \approx 0.0266$$

Test statistic:

$$Z = \frac{-0.04}{0.0266} \approx -1.50$$

Since $|Z| = 1.50 < 1.96$, we **fail to reject** H_0 at the 5% level. There is insufficient evidence to conclude that version B has a different conversion rate than version A.

5. **Normal approximation check:**

$$\begin{array}{ll} n_A \hat{p}_A = 120 \geq 10 & \checkmark \\ n_B \hat{p}_B = 168 \geq 10 & \checkmark \end{array} \quad \begin{array}{ll} n_A(1 - \hat{p}_A) = 380 \geq 10 & \checkmark \\ n_B(1 - \hat{p}_B) = 432 \geq 10 & \checkmark \end{array}$$

All conditions are satisfied.

Distribution	Notation	Type	$E[X]$	$\text{Var}(X)$
Bernoulli	$\text{Bernoulli}(p)$	Discrete	p	$p(1-p)$
Binomial	$\text{Binomial}(n, p)$	Discrete	np	$np(1-p)$
Sample prop.	$\hat{p} = X/n$	—	p	$p(1-p)/n$
Diff. of props.	$\hat{p}_1 - \hat{p}_2$	—	$p_1 - p_2$	$\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}$

Summary: discrete distributions and proportions

6. Common Continuous Distributions

6.1 Uniform Distribution

- X is equally likely to take any value in the interval $[a, b]$
- We write $X \sim \text{Uniform}(a, b)$
- **PDF:**

$$f(x) = \frac{1}{b-a}, \quad a \leq x \leq b$$

- **CDF:**

$$F(x) = \frac{x-a}{b-a}, \quad a \leq x \leq b$$

- **Mean:** applying the definition $E[X] = \int_{-\infty}^{\infty} x f(x) dx$ with $f(x) = 1/(b-a)$:

$$E[X] = \int_a^b x \cdot \frac{1}{b-a} dx = \frac{1}{b-a} \cdot \frac{b^2 - a^2}{2} = \frac{(b-a)(b+a)}{2(b-a)} = \frac{a+b}{2}$$

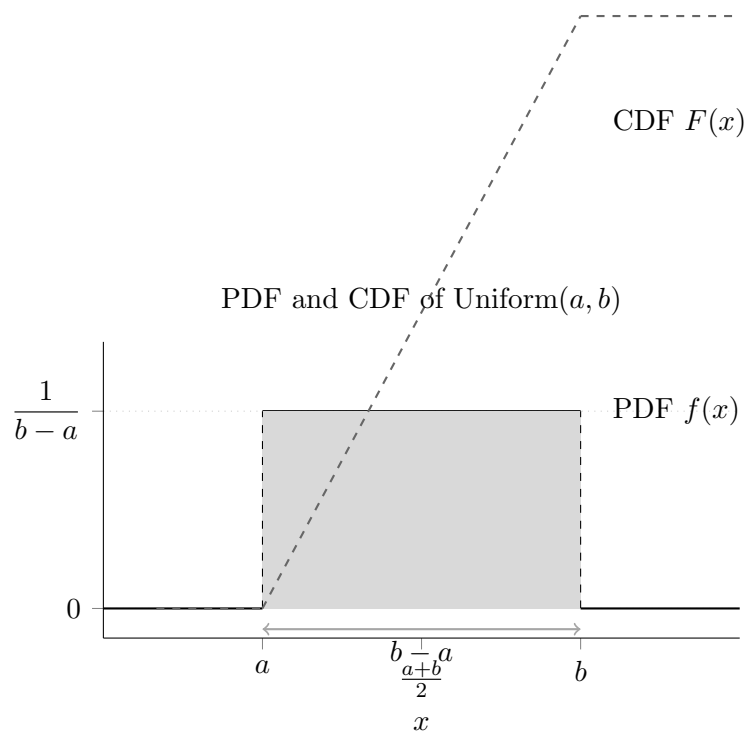
- **Variance:** applying the computational shortcut $\text{Var}(X) = E[X^2] - (E[X])^2$. First compute $E[X^2]$ via LOTUS:

$$E[X^2] = \int_a^b x^2 \cdot \frac{1}{b-a} dx = \frac{1}{b-a} \cdot \frac{b^3 - a^3}{3} = \frac{a^2 + ab + b^2}{3}$$

Then substitute into the shortcut:

$$\text{Var}(X) = E[X^2] - (E[X])^2 = \frac{a^2 + ab + b^2}{3} - \left(\frac{a+b}{2}\right)^2 = \frac{a^2 + ab + b^2}{3} - \frac{a^2 + 2ab + b^2}{4} = \frac{(b-a)^2}{12}$$

- The uniform distribution is the continuous analogue of a fair die: all values in the interval are equally likely, so the mean is simply the midpoint and the variance grows with the width of the interval.



The PDF (solid) is flat at height $1/(b-a)$ over $[a, b]$ and zero elsewhere. The CDF (dashed) rises linearly from 0 to 1 over $[a, b]$.

6.2 Normal Distribution

- The most important distribution in statistics, arising naturally whenever many small independent effects are added together
- Parameters: μ (mean), σ^2 (variance)
- We write $X \sim N(\mu, \sigma^2)$
- **PDF:**

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad -\infty < x < \infty$$

- The PDF is symmetric about μ , bell-shaped, and falls off rapidly in the tails.

$$E[X] = \mu, \quad \text{Var}(X) = \sigma^2$$

- **Key properties:**

- The normal is completely determined by its mean and variance
- Skewness = 0 (symmetric) and excess kurtosis = 0 (mesokurtic)
- Linear combinations of independent normals are normal: if $X \sim N(\mu_X, \sigma_X^2)$ and $Y \sim N(\mu_Y, \sigma_Y^2)$ independently, then

$$aX + bY \sim N(a\mu_X + b\mu_Y, a^2\sigma_X^2 + b^2\sigma_Y^2)$$

The mean and variance follow directly from the properties already established:

$$E[aX + bY] = a E[X] + b E[Y] = a\mu_X + b\mu_Y \quad (\text{linearity})$$

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) = a^2\sigma_X^2 + b^2\sigma_Y^2 \quad (\text{independence, scaling})$$

The additional fact that the result is *normal* (not just having these moments) follows from the MGF: $M_{aX+bY}(t) = M_X(at) \cdot M_Y(bt)$, which can be shown to equal the MGF of $N(a\mu_X + b\mu_Y, a^2\sigma_X^2 + b^2\sigma_Y^2)$.

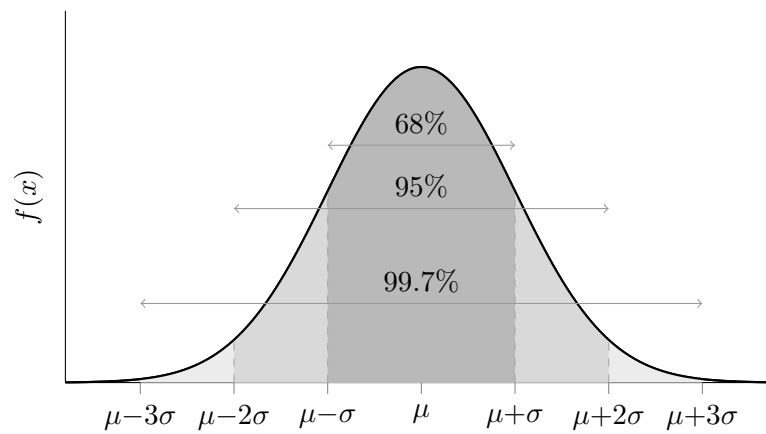
- **Empirical rule:** for any $X \sim N(\mu, \sigma^2)$,

$$P(\mu - \sigma \leq X \leq \mu + \sigma) \approx 0.68$$

$$P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 0.95$$

$$P(\mu - 3\sigma \leq X \leq \mu + 3\sigma) \approx 0.997$$

Normal distribution $N(\mu, \sigma^2)$ with empirical rule



The bell curve is symmetric about μ . Shading shows the three empirical rule bands: darker shading indicates higher probability density near the mean.

6.3 Standard Normal Distribution

- The **standard normal** is the special case $\mu = 0$, $\sigma^2 = 1$:

$$Z \sim N(0, 1)$$

- **PDF:**

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

- **CDF:** denoted $\Phi(z) = P(Z \leq z)$. There is no closed form; values are read from a table or computed numerically.
- **Standardization:** any normal random variable can be converted to a standard normal by subtracting the mean and dividing by the standard deviation:

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

Derivation: Since $E[X] = \mu$ and $\text{Var}(X) = \sigma^2$, by the linearity and scaling properties:

$$\begin{aligned} E\left[\frac{X - \mu}{\sigma}\right] &= \frac{E[X] - \mu}{\sigma} = \frac{\mu - \mu}{\sigma} = 0 \\ \text{Var}\left(\frac{X - \mu}{\sigma}\right) &= \frac{1}{\sigma^2} \text{Var}(X) = \frac{\sigma^2}{\sigma^2} = 1 \end{aligned}$$

- **Symmetry:** $\Phi(-z) = 1 - \Phi(z)$. This follows directly from the symmetry of the PDF about 0.
- **Computing probabilities:** for $X \sim N(\mu, \sigma^2)$,

$$P(a \leq X \leq b) = \Phi\left(\frac{b - \mu}{\sigma}\right) - \Phi\left(\frac{a - \mu}{\sigma}\right)$$

6.4 Sample Means

- Let $X_1, X_2, \dots, X_n$ be independent and identically distributed (i.i.d.) random variables, each with mean μ and variance σ^2 .
- The **sample mean** is:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

$\bar{X}$ is itself a random variable — it varies from sample to sample.

- **Mean of $\bar{X}$:**

$$E[\bar{X}] = E\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n E[X_i] = \frac{1}{n} \cdot n\mu = \mu$$

$\bar{X}$ is an **unbiased estimator** of μ , as shown in Section 4.

- **Variance of $\bar{X}$:** since the X_i are independent, variances add:

$$\text{Var}(\bar{X}) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2} \cdot n\sigma^2 = \frac{\sigma^2}{n}$$

- **Standard error of $\bar{X}$:**

$$\text{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

The standard error shrinks at rate $1/\sqrt{n}$: quadrupling the sample size halves the standard error.

- In practice σ is unknown and estimated by the sample standard deviation s :

$$\widehat{\text{SE}}(\bar{X}) = \frac{s}{\sqrt{n}}, \quad s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}$$

Central Limit Theorem

- For large n , the sample mean is approximately normally distributed *regardless of the distribution of the X_i* :

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right)$$

The standardized version is:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \approx N(0, 1)$$

- **Rule of thumb:** the approximation is generally reliable for $n \geq 30$, and for smaller n when the population distribution is roughly symmetric.
- This is the theoretical foundation for confidence intervals and hypothesis tests about means.

Confidence Interval for a Mean

- When σ is known, a 95% confidence interval for μ :

$$\bar{X} \pm 1.96 \cdot \frac{\sigma}{\sqrt{n}}$$

- When σ is unknown (the usual case), we use s and critical values from the t -distribution with $n - 1$ degrees of freedom:

$$\bar{X} \pm t_{n-1}^* \cdot \frac{s}{\sqrt{n}}$$

For large n , $t_{n-1}^* \approx 1.96$.

Practice Question 27-C: Sample Means

A quality control engineer measures the fill weight (in grams) of 36 randomly selected bottles. The sample mean is $\bar{X} = 503.2$ g and the sample standard deviation is $s = 4.8$ g. The target fill weight is 500 g.

1. Compute $E[\bar{X}]$ and $\widehat{SE}(\bar{X})$.
2. Construct a 95% confidence interval for the true mean fill weight μ .
3. Conduct a hypothesis test of $H_0 : \mu = 500$ against $H_a : \mu \neq 500$ at the 5% significance level.
4. Interpret the result in plain language.

Answer

1. **Mean and standard error:**

$$E[\bar{X}] = \mu \quad (\text{unbiased})$$

$$\widehat{SE}(\bar{X}) = \frac{s}{\sqrt{n}} = \frac{4.8}{\sqrt{36}} = \frac{4.8}{6} = 0.8 \text{ g}$$

2. **95% confidence interval** (using $t_{35}^* \approx 2.03$):

$$503.2 \pm 2.03 \times 0.8 = 503.2 \pm 1.624 = (501.6, 504.8) \text{ g}$$

3. **Hypothesis test:**

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} = \frac{503.2 - 500}{0.8} = \frac{3.2}{0.8} = 4.0$$

Since $|t| = 4.0 > 2.03 = t_{35}^*$, we **reject** H_0 at the 5% level.

4. **Interpretation:** There is strong statistical evidence that the true mean fill weight differs from the 500 g target. The sample mean of 503.2 g is 4 standard errors above the target, which is highly unlikely by chance alone. The engineer should investigate whether the filling machine is systematically overfilling.

6.5 Comparing Two Means

- We often want to compare the means of two independent groups.
- Let group 1 have n_1 observations with sample mean $\bar{X}_1$ and sample variance s_1^2 , drawn from a population with mean μ_1 .
- Let group 2 have n_2 observations with sample mean $\bar{X}_2$ and sample variance s_2^2 , drawn from a population with mean μ_2 .
- We are interested in the difference $\bar{X}_1 - \bar{X}_2$.

Mean and Variance of the Difference

- By linearity of expectation:

$$E[\bar{X}_1 - \bar{X}_2] = \mu_1 - \mu_2$$

- Since the two groups are independent, variances add:

$$\text{Var}(\bar{X}_1 - \bar{X}_2) = \text{Var}(\bar{X}_1) + \text{Var}(\bar{X}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$$

- **Standard error of the difference**, estimated by plugging in s_1^2 and s_2^2 :

$$\widehat{\text{SE}}(\bar{X}_1 - \bar{X}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Confidence Interval for $\mu_1 - \mu_2$

- A 95% confidence interval for $\mu_1 - \mu_2$:

$$(\bar{X}_1 - \bar{X}_2) \pm t^* \cdot \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

where t^* uses the Welch–Satterthwaite approximation for degrees of freedom:

$$df \approx \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$$

For large samples, $t^* \approx 1.96$.

Hypothesis Test for $H_0 : \mu_1 = \mu_2$

- The test statistic is:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Under H_0 , this follows approximately a t -distribution with degrees of freedom given by the Welch–Satterthwaite formula above.

- We reject H_0 at the 5% level when $|t| > t^*$.
- **Note on pooled variance:** if we are willing to assume $\sigma_1^2 = \sigma_2^2 = \sigma^2$, we can use a pooled estimate:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

giving the pooled t -statistic:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

with $n_1 + n_2 - 2$ degrees of freedom. In practice, the Welch test (unpooled) is preferred unless equal variances can be justified.

Practice Question 27-D: Comparing Two Means

A researcher compares exam scores between two teaching methods. Group A (traditional lecture) has $n_1 = 40$ students with $\bar{X}_1 = 74.3$ and $s_1 = 9.2$. Group B (flipped classroom) has $n_2 = 35$ students with $\bar{X}_2 = 79.1$ and $s_2 = 8.5$.

1. Compute the estimated standard error of $\bar{X}_1 - \bar{X}_2$.
2. Construct a 95% confidence interval for $\mu_1 - \mu_2$. Use $t^* \approx 1.99$ (Welch df ≈ 73).
3. Conduct a hypothesis test of $H_0 : \mu_1 = \mu_2$ against $H_a : \mu_1 \neq \mu_2$ at the 5% level.
4. Interpret the result in plain language.

Answer

1. **Standard error of the difference:**

$$\widehat{\text{SE}}(\bar{X}_1 - \bar{X}_2) = \sqrt{\frac{9.2^2}{40} + \frac{8.5^2}{35}} = \sqrt{\frac{84.64}{40} + \frac{72.25}{35}} = \sqrt{2.116 + 2.064} = \sqrt{4.180} \approx 2.045$$

2. **95% confidence interval:**

$$(74.3 - 79.1) \pm 1.99 \times 2.045 = -4.8 \pm 4.07 = (-8.87, -0.73)$$

3. **Hypothesis test:**

$$t = \frac{74.3 - 79.1}{2.045} = \frac{-4.8}{2.045} \approx -2.35$$

Since $|t| = 2.35 > 1.99 = t^*$, we **reject** H_0 at the 5% level.

4. **Interpretation:** There is statistically significant evidence that the two teaching methods produce different mean exam scores. Students in the flipped classroom scored an estimated 4.8 points higher on average, and the 95% confidence interval $(-8.87, -0.73)$ excludes zero, confirming the difference is unlikely to be due to chance. The researcher has evidence in favor of the flipped classroom method.

Distribution	Notation	$E[X]$	$\text{Var}(X)$	Key use
Uniform	$\text{Uniform}(a, b)$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	Equal-probability intervals
Normal	$N(\mu, \sigma^2)$	μ	σ^2	Natural phenomena, CLT limit
Standard normal	$N(0, 1)$	0	1	Probability tables, z -scores
Sample mean	$\bar{X}$	μ	σ^2/n	Estimating population mean
Diff. of means	$\bar{X}_1 - \bar{X}_2$	$\mu_1 - \mu_2$	$\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$	Two-sample comparison

Summary: continuous distributions and sample means

Reference

- 1.